

Number Toolkit

The words of number (factor, multiple, prime, square, cube), negatives and BIDMAS, and money problems with a clear decision at the end.



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THE TOPIC ON ONE MAP

Number Toolkit

01 Number families

a multiple of 8

a prime between 50 and 60

factors of 36

a cube number

02 Place value, negatives and BIDMAS

write in figures / words

7° below -3°

$-12 \div -2$

insert brackets

03 Money problems: decide

electricity bill

enough money?

which shop is cheaper?

cheapest packs

THE FIVE FAMILIES Know each word exactly:

ZONE 1 Factor

divides in:
factors of
12: 1, 2, 3, 4,
6, 12

ZONE 2 Multiple

in the times
table:
8, 16, 24, 32
...

ZONE 3 Prime

exactly two
factors:
2, 3, 5, 7, 11 ...

ZONE 4 Square

$n \times n$:
1, 4, 9, 16, 25
...

ZONE 5 Cube

$n \times n \times n$:
1, 8, 27, 64 ...

01 Number Families

Most “write down” marks are free — if you know the families by heart. Test each number in the list against the word.

WHEN YOU SEE → YOU DO

“a multiple of 7”

→ in the 7 times table: divide by 7, no remainder

“a prime number”

→ only 1 and itself divide it (1 is **not** prime)

“a square / cube number”

→ n^2 : 1, 4, 9, 16, 25, 36, 49, 64 ... n^3 : 1, 8, 27, 64, 125

“complete the list of factors”

→ work in pairs: 1×36 , 2×18 , 3×12 ...

“the difference between”

→ bigger – smaller

WORKED MODEL • FACTOR PAIRS

Complete the list of factors of 36: 1, 2, ..., 36.

1 1×36 , 2×18 , 3×12 , 4×9 , 6×6

factor pairs

ANSWER 1, 2, 3, 4, 6, 9, 12, 18, 36

× **TRAP** 1 is not a prime number (it has only one factor). 2 is the only even prime.

IDEA 01

Practise: Number Families

My questions first, then selected practice.

Q1 DR ESLAM Factor pairs

CH 1 • TOPIC 1 2 marks

Write down all the factors of 24.

Q2 DR ESLAM Which family?

CH 1 • TOPIC 1 3 marks

From 21, 23, 25, 27 write down (a) a prime number (b) a square number (c) a cube number.

SELECTED PRACTICE**Q3** PRACTICE Practice • Easy

CH 1 • TOPIC 1 1 mark

3 Give one reason why 0 is an even number.

(1 mark)

Q4

PRACTICE Practice · Easy

CH 1 · TOPIC 1 **3 marks**

7

32 33 34 35 36 37 38 39

From the list of numbers, write down

i) a multiple of 8,

[1]

ii) a square number,

[1]

- iii) a prime number.

[1]

(3 marks)

Q5

PRACTICE Practice · Easy

CH 1 • TOPIC 1 **2 marks**

15 Complete the list of factors of 36.

1, 2, , 36

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 1 mark

6 Write down a square number greater than 10.

(1 mark)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 2 marks

11

22

17

25

41

39

4

Work out the difference between the two prime numbers in the list above.

(2 marks)

Q8**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 1 mark

2 1, 2, 3, 5 and 7 are all common factors of two numbers.

Write down the digit that the two numbers must end in.

(1 mark)

02 Place Value, Negatives and BIDMAS

Temperatures are a number line: “below / fell / lower” move left, “above / rose / higher” move right. For a calculation, BIDMAS decides the order.

WHEN YOU SEE → YOU DO

“7°C below -3°C”, “fell by”

→ subtract: $-3 - 7 = -10$

“11 °C higher than”

→ add — or work back by subtracting

$-12 \div -2$

→ same signs → positive: 6

“write in figures”

→ use the place-value columns; fill empty places with 0

“insert one pair of brackets”

→ try each position; brackets first

Brackets

Indices

Divide / Multiply

Add / Subtract

WORKED MODEL • NUMBER LINE

The temperature at 07:00 is -3°C . This is 11°C higher than at 01:00. Find the temperature at 01:00.

1 01:00 is 11 degrees **lower**

reverse the direction

2 $-3 - 11$

move 11 left

ANSWER -14°C

× **TRAP** “Five million, two hundred and seven” is 5 000 207 — every empty column needs its 0.

IDEA 02

Practise: Place Value, Negatives and BIDMAS

My questions first, then selected practice.

Q9 DR ESAM Negatives

CH 1 • TOPIC 1 2 marks

Work out (a) $-4 - 7$ (b) $3 \times (-5) + 20$

SELECTED PRACTICE

Q10 PRACTICE Practice • Easy

CH 1 • TOPIC 1 1 mark

2 Circle the list of **all** the integers that satisfy $-2 < x \leq 4$

-2, -1, 0, 1, 2, 3	-1, 0, 1, 2, 3
-2, -1, 0, 1, 2, 3, 4	-1, 0, 1, 2, 3, 4

(1 mark)

Q11**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1**1 mark**

- 6 At noon the temperature in Maseru was 21°C .
At midnight the temperature had fallen by 26°C .

Work out the temperature at midnight.

..... $^{\circ}\text{C}$

(1 mark)

Q12**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1**1 mark**

- 9 Write down the temperature that is 7°C below -3°C .

..... $^{\circ}\text{C}$

(1 mark)

Q13**PRACTICE**

Practice · Easy

CH 1 • TOPIC 1 1 mark

13 Write the number five million, two hundred and seven in figures.

(1 mark)

Q14**PRACTICE**

Practice · Medium

CH 1 • TOPIC 1 1 mark

7 The lowest temperature recorded at Scott Base in Antarctica is -57.0°C .
The highest temperature recorded at Scott Base is 63.8°C more than this.

What is the highest temperature recorded at Scott Base?

..... $^{\circ}\text{C}$

(1 mark)

Q15**PRACTICE**

Practice · Hard

CH 1 • TOPIC 1 1 mark

5 Insert one pair of brackets to make this calculation correct.

$$7 - 5 - 3 + 4 = 9$$

(1 mark)

03 Money Problems: Decide

Work out each option separately, write money with 2 decimal places, then **answer the question** in a sentence: yes / no / which one, with the numbers.

WHEN YOU SEE → YOU DO

“Does he have enough money?”

→ total cost, compare with the money, say yes/no

“Which is cheaper? You must show all your working”

→ cost of each option, then choose

“plus VAT at 20%”

→ $\times 1.2$

“at least 100 tiles”, packs

→ try the combinations; the cheapest wins

price in pence, answer in pounds

→ $900\text{p} = \text{£}9.00$

Each option

Same units

Compare

Sentence answer

WORKED MODEL • WHICH SHOP?

25 plants: £2.39 each, or a pack of 25 for £52.50 plus VAT at 20%.

1 $25 \times 2.39 = \text{£}59.75$

option 1

2 $52.50 \times 1.2 = \text{£}63.00$

option 2

ANSWER Kirsty's Plants: $\text{£}59.75 < \text{£}63.00$

× **TRAP** A number alone is not an answer to “should he ...?”. Write the decision and the two amounts.

IDEA 03

Practise: Money Problems: Decide

My questions first, then selected practice.

Q16**DR ESLAM**

Better value

CH 1 • TOPIC 1

3 marks

Water: one bottle £1.35, or a pack of 6 bottles for £7.50. Which is better value? Show your working.

SELECTED PRACTICE

Q17**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1 4 marks

- 1 Here is part of Gary's electricity bill.

Electricity bill	
New reading	7155 units
Old reading	7095 units
Price per unit	15p

Work out, in pounds, how much Gary has to pay for the units of electricity he used.

(4 marks)

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 6 marks

- 1 (a) The diagram shows a patio in the shape of a rectangle.

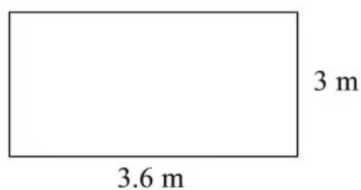


Diagram **NOT**
accurately drawn

The patio is 3.6m long and 3 m wide.

Matthew is going to cover the patio with paving slabs.
Each paving slab is a square of side 60 cm.

Matthew buys 32 of the paving slabs.

Does Matthew buy enough paving slabs to cover the patio?
You must show all your working.

(3 marks)

- (b) The paving slabs cost £8.63 each.

Work out the total cost of the 32 paving slabs.

(3 marks)

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 4 marks

2 Steve wants to put a hedge along one side of his garden.

He needs to buy 27 plants for the hedge. Each plant costs £5.54.

Steve has £150 to spend on plants for the hedge.

Does Steve have enough money to buy all the plants he needs?

(4 marks)

Q20**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 5 marks

4 Tom is going to buy 25 plants to make a hedge.

Here is information about the cost of buying the plants.

Kirsty's Plants

£2.39 each

Hedge World

Pack of 25

£52.50 plus VAT at 20%

Tom wants to buy the 25 plants as cheaply as possible.

Should Tom buy the plants from Kirsty's Plants or from Hedge World?

You must show all your working.

(5 marks)

Q21**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 4 marks**3 (a)** Tom is tiling a wall.

He needs to buy at least 100 tiles.

The tiles are sold in large packs and small packs.

Large pack 40 tiles	£18
Small pack 28 tiles	£14
Special offer	
25% reduction when you buy 3 or more large packs	

Work out the cheapest cost for Tom to buy the packs of tiles he needs.

£.....

(3 marks)**(b)** Tom is also tiling a floor.

The floor is a rectangle with length 600 cm and width 240 cm

Each tile is a square with side 40 cm

Tom uses this method to work out the number of tiles he needs.

Number of tiles that will fit along the length = $600 \div 40$
 $= 15$
 Number of tiles that will fit along the width = $240 \div 40$
 $= 6$
 Total number of tiles needed = $15 + 6$
 $= 21$

Give a reason why Tom's method is wrong.

(1 mark)

Q22**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 4 marks

- 4 The table shows the children nominated to win the subject prize in Mathematics and the subject prize in English.

Mathematics	English
Alice	Alice
Ben	Claire
Emma	Gabi
Paddy	Simon

The winner of each subject prize is picked at random.
It is possible for Alice to win both prizes.

In what percentage of the combinations of prize winners does Alice win **at least** one prize?

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 1 Factor divides in	ZONE 2 Multiple times table	ZONE 3 Prime two factors	ZONE 4 Square n^2	ZONE 5 Cube n^3
--	---	--	---	---------------------------------------

- 01 Families** Factor pairs; 1 is not prime; know squares to 225 and cubes to 125.
- 02 Negatives** Below/fell → left; above/rose → right. Same signs $\times \div$ → positive. BIDMAS.
- 03 Money** Each option, same units, 2 d.p., then a sentence with the decision.

BOTTOM LINE Know the words → draw the number line → money: compare, then decide in words.

Answers

Number Toolkit • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • NUMBER FAMILIES

Q1	Dr Eslam • Factor pairs	1, 2, 3, 4, 6, 8, 12, 24
Q2	Dr Eslam • Which family?	(a) 23 (b) 25 (c) 27
Q3	Practice • Easy	it is a multiple of 2 (divides by 2 exactly)
Q4	Practice • Easy	(i) 32 (ii) 36 (iii) 37
Q5	Practice • Easy	3, 4, 6, 9, 12, 18
Q6	Practice • Medium	16 (or 25, 36, ...)
Q7	Practice • Medium	$41 - 17 = 24$
Q8	Practice • Hard	0

IDEA 02 • PLACE VALUE, NEGATIVES AND BIDMAS

Q9	Dr Eslam • Negatives	(a) -11 (b) 5
Q10	Practice • Easy	-1, 0, 1, 2, 3, 4
Q11	Practice • Easy	-5
Q12	Practice • Easy	-10
Q13	Practice • Easy	5 000 207
Q14	Practice • Medium	6.8
Q15	Practice • Hard	$7 - (5 - 3) + 4 = 9$

IDEA 03 • MONEY PROBLEMS: DECIDE

Q16 Dr Eslam • Better value

pack: $7.50 \div 6 = £1.25 < £1.35$ — the pack

Q17 Practice • Easy

$$60 \times 15 = 900\text{p} = £9.00$$

Q18 Practice • Medium

(a) $6 \times 5 = 30$ slabs needed: yes (b) £276.16

Q19 Practice • Medium

$$27 \times 5.54 = £149.58: \text{yes, } 42\text{p left}$$

Q20 Practice • Medium

Kirsty £59.75, Hedge World £63.00: Kirsty's

Q21 Practice • Hard

(a) 3 large packs: £40.50 (b) multiply, not add: $15 \times 6 = 90$

Q22 Practice • Hard

$$\frac{7}{16} = 43.75\%$$



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Set Notation and Venn Diagrams

A set is a list in curly brackets. Four symbols ($\cap$, $\cup$, $'$, $\in$) and one rule for Venn diagrams: fill from the middle outwards.

THE TOPIC ON ONE MAP

Set Notation and Venn Diagrams

01 Set notation: list the members

$A \cap B$, $A \cup B$, A'

"is $24 \in A$?"

"one possible set C "

02 Read a Venn diagram

list the members

$n(A \cup B)$

$n(A' \cap B')$

03 Shade a region

shade $A \cap C'$

describe the shaded region

04 Fill it in: two sets

50 people: cat, dog

$n(A)$, $n(A \cap B)$ given

05 Three sets with x

dog, cat, rabbit

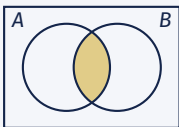
total = 80: find x

$3x$, $x + 2$ in regions

THE FOUR SYMBOLS Each one is a region — see it before you list it:

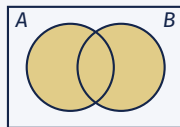
ZONE $\cap$

$A \cap B$: in both



ZONE $\cup$

$A \cup B$: in either



ZONE $'$

A' : not in A



ZONE $\cap'$

$A \cap B'$: A only



01 Set Notation: List the Members

Write the universal set E out in full first. Then test every member against the rule. Each member is listed **once**.

WHEN YOU SEE → YOU DO

$A \cap B$ (“and”)

→ members in **both**

$A \cup B$ (“or”)

→ members in either — list each once

A'

→ everything in E that is **not** in A

$\emptyset$, “ $B \cap K = \emptyset$ ”

→ the empty set: nothing in common

“is it true that $24 \in A$?”

→ check E first — 24 may not even be in it

“list the members of one possible set C ”

→ build it from the conditions, then check each one

Write out E

Write out A and B

Apply the symbol

Check each member

WORKED MODEL • JANUARY 2021 P1H

$E = \{20, 21, \dots, 29\}$, $A = \{\text{odd numbers}\}$, $B = \{\text{multiples of 3}\}$. List $A \cap B$ and $A \cup B$.

1 $A = \{21, 23, 25, 27, 29\}$, $B = \{21, 24, 27\}$

write them out

2 $A \cap B = \{21, 27\}$

in both

ANSWER $A \cup B = \{21, 23, 24, 25, 27, 29\}$

× **TRAP** Writing 21 twice in $A \cup B$, or forgetting the curly brackets $\{ \}$.

IDEA 01

Practise: Set Notation: List the Members

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Both and either

CH 1 • TOPIC 2 2 marks

$E = \{1, 2, \dots, 12\}$, $A = \{\text{factors of } 12\}$, $B = \{\text{even numbers}\}$. List (a) $A \cap B$ (b) A' .

SELECTED PRACTICE

Q2**PRACTICE** Practice • Easy

CH 1 • TOPIC 2 3 marks

1 (a) $A = \{p, r, a, g, u, e\}$

$B = \{p, a, r, i, s\}$

$C = \{b, u, d, a, p, e, s, t\}$

List the members of the set

i) $A \cap B$,

[1]

ii) $B \cup C$.

[1]

(2 marks)

(b) $D = \{r, o, m, e\}$

$E = \{l, i, s, b, o, n\}$

$F = \{b, e, r, l, i, n\}$

Put one of the letters D , E or F in the box below to make the statement correct.

$$A \cap \square = \emptyset$$

Explain your answer.

(1 mark)

Q3**PRACTICE** Practice • Easy**CH 1 • TOPIC 2** 4 marks

11 (a) $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 $A = \{2, 3, 5, 7\}$
 $B = \{4, 6, 8, 10\}$

Explain why $A \cap B = \emptyset$

(1 mark)

(b) $x \in \mathcal{E}$ and $x \notin A \cup B$

Write down the **two** possible values of x .

(1 mark)

(c) Set C is such that

$$A \cup B \cup C = \mathcal{E}$$

$$A \cap C = \{2\}$$

$$B \cap C = \{4, 6, 10\}$$

List all the members of set C .

(2 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2**2 marks**

7 $\xi = \{\text{whole numbers}\}$

$A = \{\text{factors of 100}\}$

$B = \{\text{multiples of 5}\}$

List the members of the set $A \cap B$.

(2 marks)

Q5**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 **3 marks**

4 (a) $\xi = \{\text{positive whole numbers less than } 13\}$

$A = \{\text{even numbers}\}$

$B = \{\text{multiples of } 3\}$

$C = \{\text{prime numbers}\}$

List the members of the set

i) $A' \cap B'$,

ii) $B \cup C$.

(2 marks)

(b) Is it true that $14 \in A$?

Tick (✓) the appropriate box.

☐ Yes ☐ No

Explain your answer.

(1 mark)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q4

CH 1 • TOPIC 2 3 marks

4 $B = \{b, l, u, e\}$

$G = \{g, r, e, y\}$

$W = \{w, h, i, t, e\}$

(a) List all the members of the set

(i) $B \cup G$

.....

(ii) $W \cap G'$

.....

Serena writes down the statement $B \cap G \cap W = \emptyset$

(b) Is Serena's statement correct?

You must give a reason for your answer.

.....

.....

.....

(Total for Question 4 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q4

CH 1 • TOPIC 2 2 marks

4 $\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$

$$A = \{\text{odd numbers}\}$$

$$B = \{\text{multiples of 3}\}$$

List the members of the set

(i) $A \cap B$

.....

(ii) $A \cup B$

.....

(Total for Question 4 is 2 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q3

CH 1 • TOPIC 2 3 marks

3 $\mathcal{E} = \{21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$

$$A = \{22, 24, 26, 28, 30\}$$

$$B = \{21, 24, 27, 30\}$$

(a) List the members of the set

(i) $A \cap B$

.....

(ii) A'

.....

$$C = \{23, 25, 29\}$$

(b) Using set notation, find an expression for C in terms of A and B .

.....

(Total for Question 3 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q4

CH 1 • TOPIC 2 3 marks**4** $\mathcal{E} = \{\text{letters of the alphabet}\}$

$$B = \{\text{b, r, a, z, i, l}\}$$

$$I = \{\text{i, r, e, l, a, n, d}\}$$

(a) List the members of the set

(i) $B \cup I$

.....

(ii) $B \cap I'$

.....

$$K = \{\text{k, e, n, y, a}\}$$

Cody writes down the statement $B \cap K = \emptyset$

Cody's statement is wrong.

(b) Explain why.

.....

(Total for Question 4 is 3 marks)

.....

Q10**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q4

CH 1 • TOPIC 2 5 marks

- 4 $\mathcal{E} = \{9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$
 $A = \{\text{multiples of 3}\}$
 $B = \{\text{odd numbers}\}$

(a) List the members of the set

(i) $A \cap B$

.....

(ii) $A \cup B$

.....

(b) Is it true that $24 \in A$?

Tick one of the boxes below.

Yes

No

☐
☐

Give a reason for your answer.

.....

Set C has 4 members such that $C \cap B' = \{10, 18\}$

(c) List the members of one possible set C

.....

(Total for Question 4 is 5 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q4

CH 1 • TOPIC 2 5 marks

4 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$A = \{\text{factors of } 6\}$

$B = \{\text{prime numbers}\}$

(a) List the members of the set

(i) $A \cup B$

(ii) A'

Harpreet states that $A \cap B = \emptyset$

Harpreet is incorrect.

(b) Explain why.

C is a set with 4 members such that

the set $A \cap C$ has 2 members

the set $B \cap C$ has 2 members

Set $A \cap C$ and set $B \cap C$ have no members in common.

(c) List the 4 members of set C

(Total for Question 4 is 5 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q7

CH 1 • TOPIC 2 5 marks

7 $\mathcal{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$

$A = \{\text{even numbers}\}$

$B = \{23, 29, 31\}$

$C = \{\text{multiples of 3}\}$

(a) List the members of the set

(i) $B \cup C$

.....

(ii) $A' \cap C$

.....

(b) Is it true that $B \cap C = \emptyset$?

Tick (✓) one of the boxes below.

Yes

No

☐
☐

Give a reason for your answer.

.....

The set D has 4 members and is such that $D \cap (A \cup C) = \emptyset$

(c) List the members of set D

.....

(Total for Question 7 is 5 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q5

CH 1 • TOPIC 2 4 marks

5 $\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

(a) List the members of the set

(i) $A \cup B$

.....

(ii) B'

.....

$$\mathcal{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$A = \{2, 4, 6, 8, 10, 12\}$$

$$B = \{3, 6, 9, 12\}$$

$$C = \{1, 3, 5, 7, 9, 11\}$$

| $\mathcal{E}$ $\cap$ $\cup$ $\emptyset$ $\in$ $\notin$ |

(b) Write a symbol from the box on each dotted line to make each of the following a true statement.

(i) $A \cap C = \dots\dots\dots$

(ii) $13 \dots\dots\dots \mathcal{E}$

(Total for Question 5 is 4 marks)

02 Read a Venn Diagram

Numbers in a region can mean two things: the **members** themselves, or **how many** members (n). Read the question: “list” or “ $n(\dots)$ ”.

WHEN YOU SEE → YOU DO

“list the members of A ”

→ every number inside circle A — including the overlap

“ $n(A \cup B)$ ”

→ add every region inside A or B

“ $n(A' \cap B')$ ”

→ outside both circles

“represent the numbers of elements”

→ add the numbers — do not list them

WORKED MODEL • MAY 2021 P1H

Regions: A only 12, $A \cap B$ only 5, B only 9, all three 6, $A \cap C$ only 10, $B \cap C$ only 3, C only 4, outside 8. Find $n(A \cup B)$ and $n(A' \cap B')$.

1 $12 + 5 + 9 + 6 + 10 + 3 = 45$

every region in A or B

2 $4 + 8 = 12$

outside A and outside B

ANSWER $n(A \cup B) = 45$, $n(A' \cap B') = 12$

IDEA 02

Practise: Read a Venn Diagram

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Count, do not list

CH 1 • TOPIC 2 2 marks

A Venn diagram has A only 7, $A \cap B$ 4, B only 9, outside 5. Find $n(A \cup B)$ and $n(B')$.

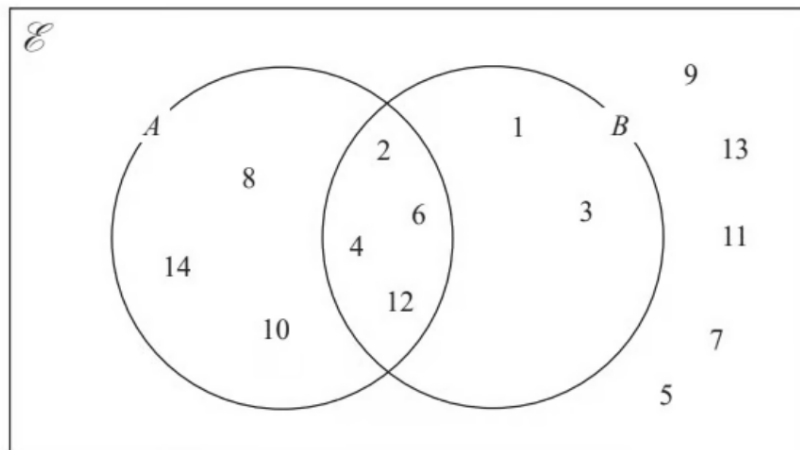
SELECTED PRACTICE

Q15**PRACTICE**

Practice • Easy

CH 1 • TOPIC 2 2 marks

10 The numbers from 1 to 14 are shown in the Venn diagram.



i) List the members of the set $A \cap B$

[1]

ii) List the members of the set B'

[1]

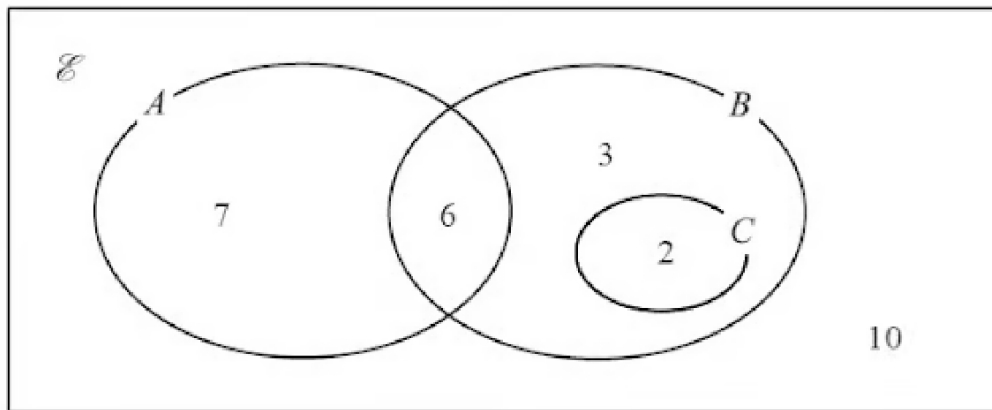
(2 marks)

Q16**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2 4 marks

6 (a) The Venn diagram shows a universal set ξ and three sets A , B and C .



7, 6, 3, 2 and 10 represent the **number** of elements in each set.

Find

$$n(A \cup B)$$

(1 mark)

(b) $n(A')$

(1 mark)

(c) $n(B \cap C)$

(1 mark)

(d) $n(A' \cup B')$

(1 mark)

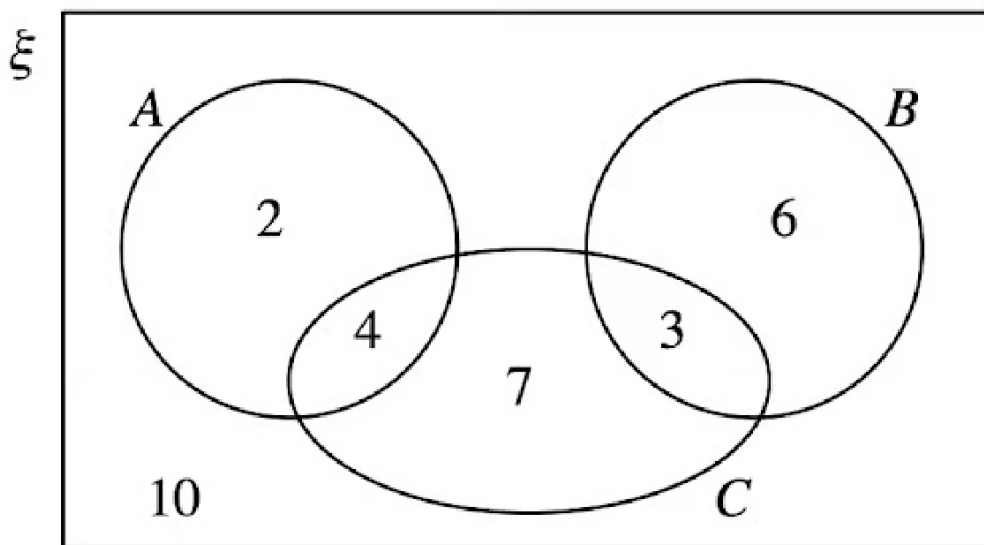
Q17**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2

4 marks

5 (a) The Venn diagram shows a universal set ξ and three sets A , B and C .



2, 4, 7, 3, 6 and 10 represent the **number** of elements in each set.

Find

$$n(A \cap B)$$

(1 mark)

(b) $n(B')$

(1 mark)

(c) $n(A \cap C)$

(1 mark)

(d) $n(B' \cap C)$

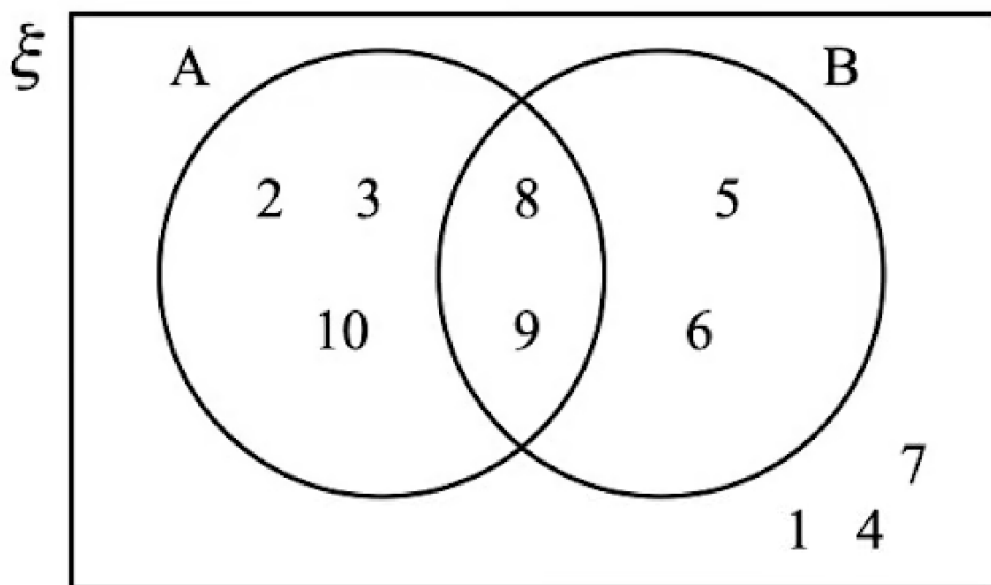
(1 mark)

Q18**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 4 marks

6 (a) The Venn diagram shows all of the elements in sets A , B and ξ .



Write down the elements in A' ,

(1 mark)

(b) Find $n(A \cap B)'$,

(1 mark)

(c) Find the elements in $(A \cap B) \cup (A \cup B)'$,

(1 mark)

(d) $A \cap C = \emptyset$

$$B \cup C = \{5, 6, 7, 8, 9\}$$

$$n(C) = 3$$

Write down the elements in C .

(1 mark)

PAST PAPERS

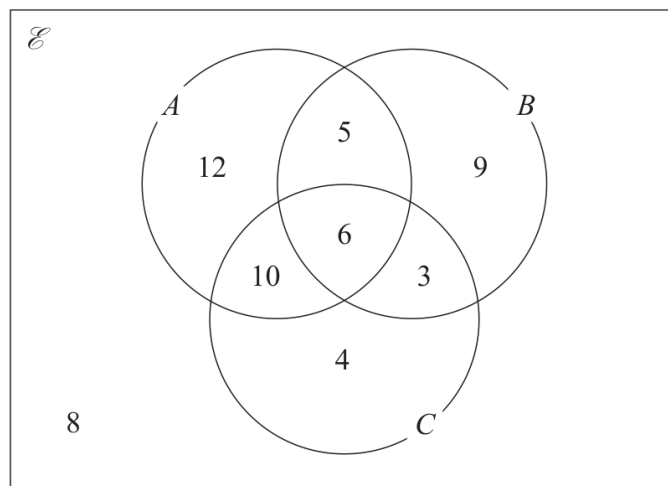
Q19

PAST PAPER

Edexcel 4MA1 • May 2021 • Paper 1H • Q19

CH 1 • TOPIC 2 3 marks

19 The Venn diagram shows a universal set, $\mathcal{E}$ and sets A , B and C .



12, 5, 9, 10, 6, 3, 4 and 8 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

(ii) $n(A' \cap B')$

.....

(iii) $n([A \cap B] \cup C)$

.....

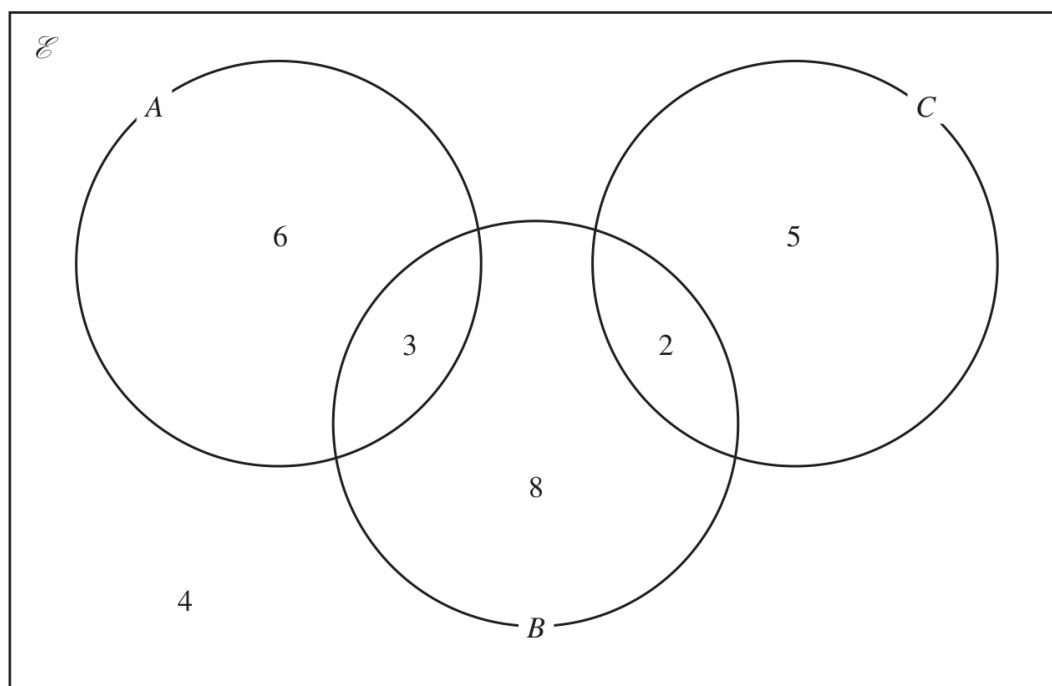
(Total for Question 19 is 3 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q17

CH 1 • TOPIC 2 4 marks

17 The Venn diagram shows a universal set $\mathcal{E}$ and three sets A , B and C .



6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

.....

(ii) $n(A \cap C)$

.....

(iii) $n(B \cap C')$

.....

(iv) $n(A' \cup B' \cup C')$

.....

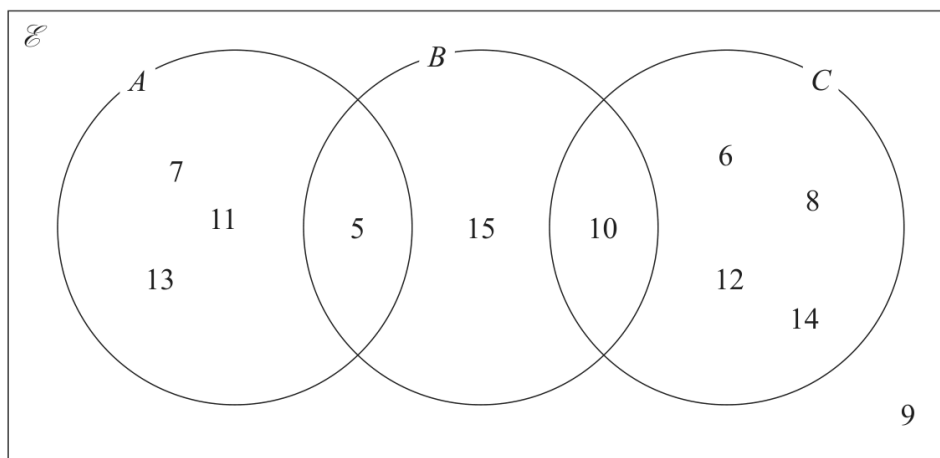
(Total for Question 17 is 4 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q2

CH 1 • TOPIC 2 3 marks

2 Here is a Venn diagram.



(a) Write down the numbers that are in the set

(i) A

.....

(ii) $B \cup C$

.....

Dominic writes down $9 \notin C$

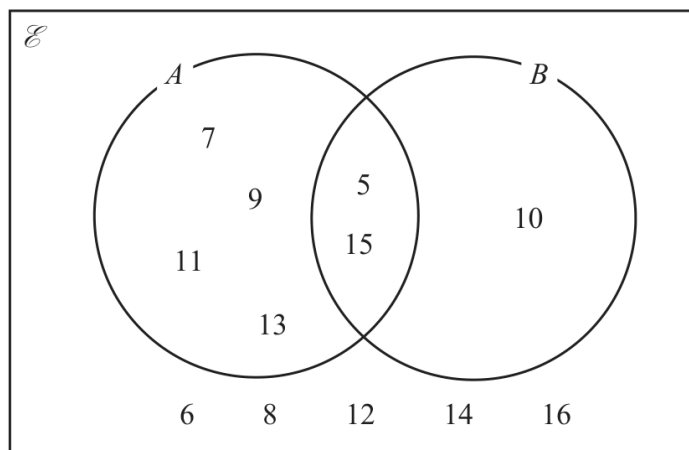
(b) Explain why Dominic is correct.

.....

(Total for Question 2 is 3 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q1

CH 1 • TOPIC 2 3 marks**You must write down all the stages in your working.****1** Here is a Venn diagram.

List the members of the set

(a) A

.....

(b) $A \cap B$

.....

(c) $(A \cup B)'$

.....

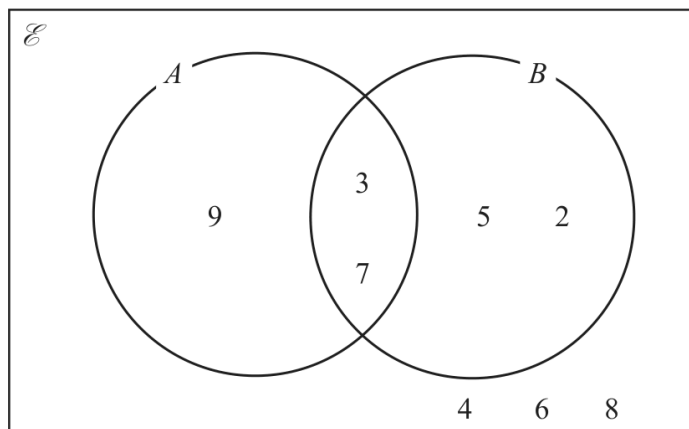
(Total for Question 1 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q4

CH 1 • TOPIC 2 3 marks

4 Here is a Venn diagram.

(a) List the members of the set B

.....

(b) List the members of the set $A \cap B$

.....

(c) List the members of the set A'

.....

(Total for Question 4 is 3 marks)

03 Shade a Region

Shade in stages: first the bracket, then the symbol outside it. Describe a shaded region by naming the circles it is **in** and **not in**.

WHEN YOU SEE → YOU DO

“shade $(A \cup B) \cap C$ ”

→ shade $A \cup B$, then remove C

“ $C \subset A$ ”

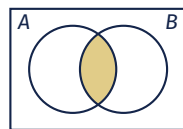
→ circle C drawn completely inside A

“ $A \cap B = \emptyset$ ”

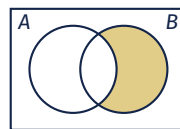
→ circles do not overlap

“use set notation to describe the shaded region”

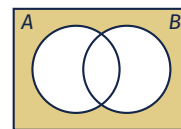
→ list what it is in, what it is not in



$A \cap B$



$A' \cap B$ (B only)



$A' \cap B' = (A \cup B)'$

TWO SYMBOLS $\subset$ = “is inside” (a subset). $\in$ = “is a member of”, $\notin$ = “is not a member of”.

IDEA 03

Practise: Shade a Region

My questions first, then selected practice.

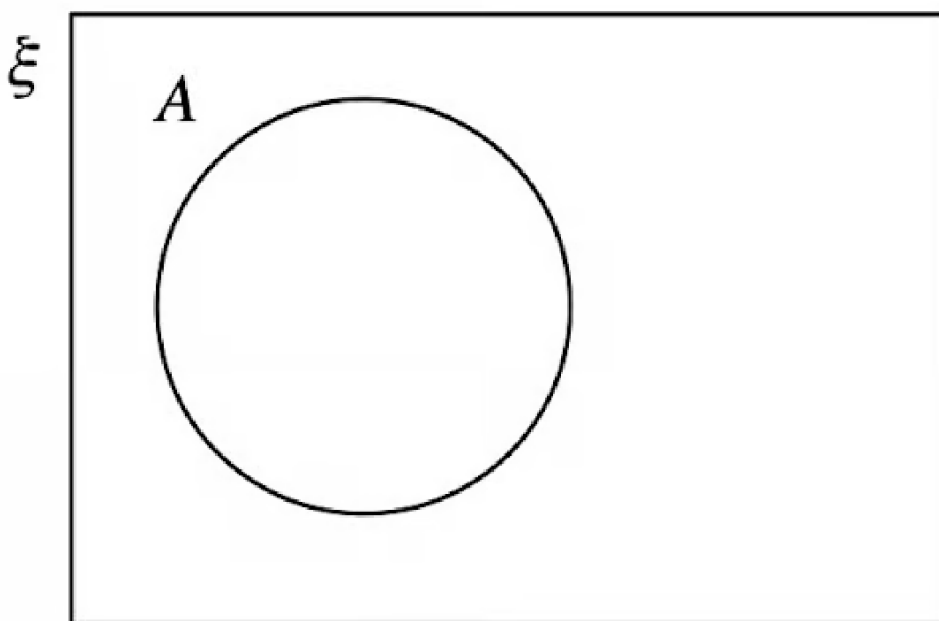
SELECTED PRACTICE**Q24****PRACTICE**

Practice • Medium

CH 1 • TOPIC 2 3 marks

5 (a) A , B and C are three sets.

$$A \cap B = \emptyset \text{ and } C \subset A$$



Complete the Venn diagram to show the sets B and C .

(2 marks)

(b) On the Venn diagram, shade the region that represents $A \cap C$.

(1 mark)

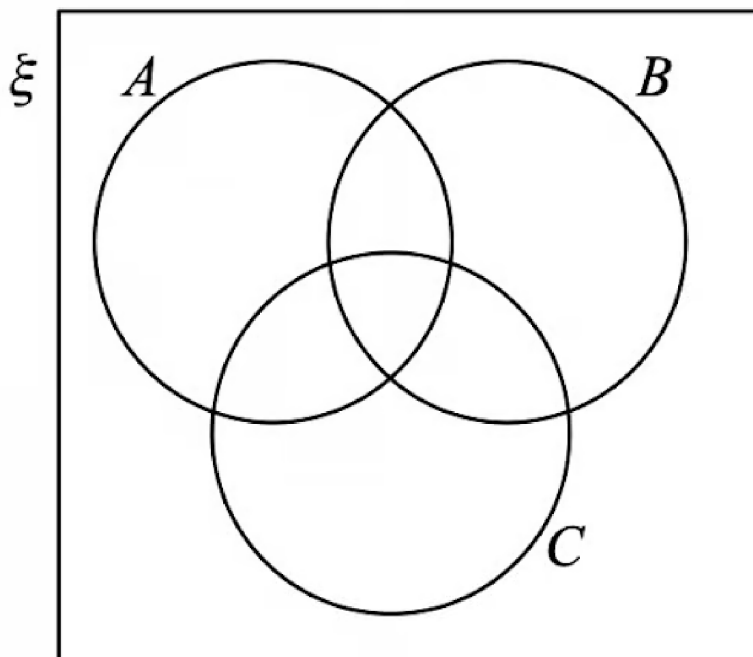
Q25

PRACTICE

Practice • Hard

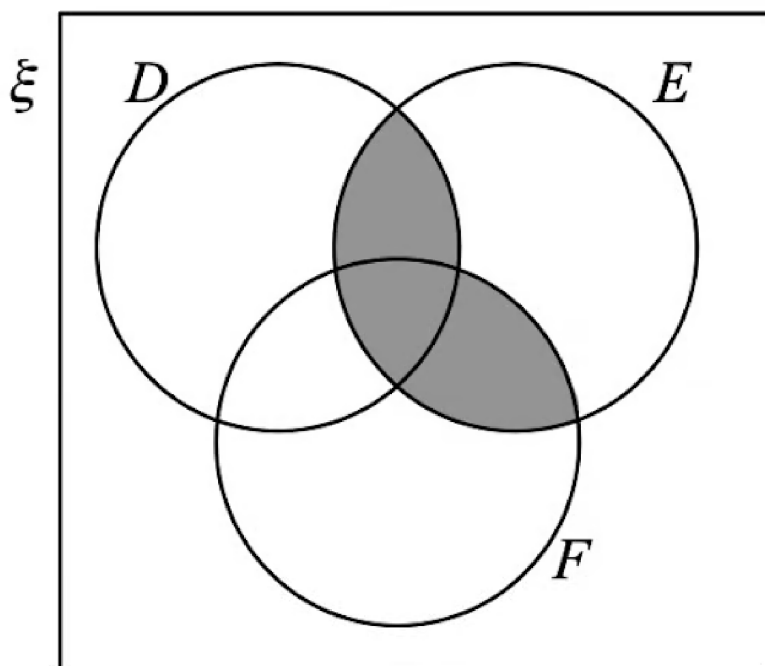
CH 1 • TOPIC 2 4 marks

7 (a) On the Venn diagram, shade the set $(A \cup B) \cap C'$.



(2 marks)

(b) Use set notation to describe the shaded region in the Venn diagram below.



(2 marks)

04 Fill It In: Work from the Middle

Always fill the **overlap first**, then the “only” parts, then the outside. Check that everything adds to the total.

WHEN YOU SEE → YOU DO

“ $n(A \cap B) = 12$ ”

→ write 12 in the overlap first

“ $n(A) = 22$ ”

→ $A \text{ only} = 22 - 12$

“6 have neither”

→ goes outside the circles, inside the box

“how many study only one language?”

→ add the two “only” regions

given $A \cap B, A', B'$ as lists

→ place each member in its one region

Overlap

Only A and only B

Outside

Check the total

WORKED MODEL • TWO SETS

50 people: 30 own a cat, 25 own a dog, 6 own neither. Complete the Venn diagram.

1 $30 + 25 - x + 6 = 50 \Rightarrow x = 11$

overlap x

2 cat only $30 - 11 = 19$, dog only $25 - 11 = 14$

“only” parts

ANSWER 19, 11, 14 and 6 outside

× **TRAP** Writing 30 in the “cat only” region. 30 is the whole circle — the overlap is part of it.

IDEA 04

Practise: Fill It In: Work from the Middle

My questions first, then selected practice, then real past papers.

Q26 DR ESLAM Middle first

CH 1 • TOPIC 2 3 marks

40 students: 22 play football, 18 play tennis, 7 play both. How many play neither?

SELECTED PRACTICE

Q27**PRACTICE**

Practice · Easy

CH 1 • TOPIC 2 **3 marks**

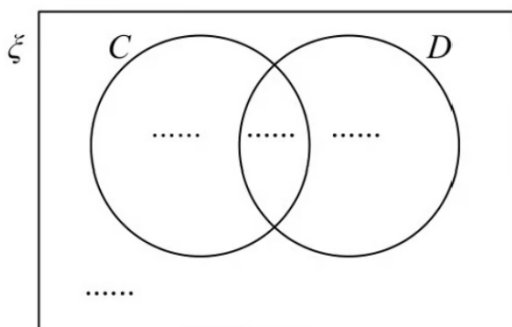
5 In a survey of 50 people,

30 owned a cat

25 owned a dog

6 owned neither a cat or a dog

Complete the Venn diagram to show this information.

**(3 marks)**

Q28**PRACTICE**

Practice • Medium

CH 1 • TOPIC 2

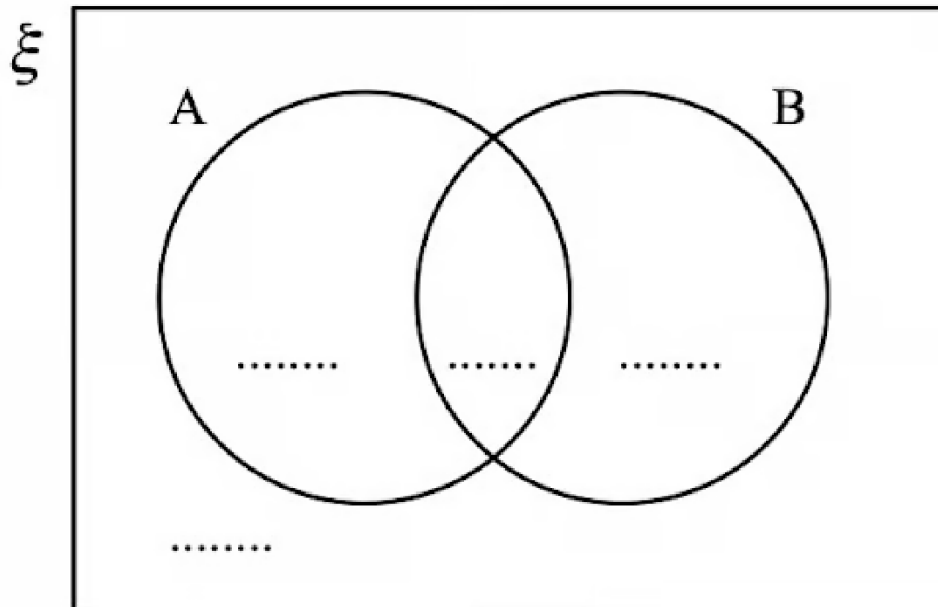
5 marks**11 (a)** A and B are two sets.

$$n(\xi) = 36$$

$$n(B) = 21$$

$$n(A \cap B) = 8$$

$$n(A') = 18$$

Complete the Venn diagram to show the **number of elements** in each region.**(3 marks)****(b)** Find

i) $n(A \cup B)$

ii) $n(A \cap B')$

(2 marks)

Q29**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 4 marks

2 There are 31 students in a class.

The only languages available for the class to study are French and Spanish.

17 students study French.

15 students study Spanish.

6 students study neither French nor Spanish.

Using a Venn diagram, or otherwise, work out how many students study only one language.

(4 marks)

Q30**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 5 marks

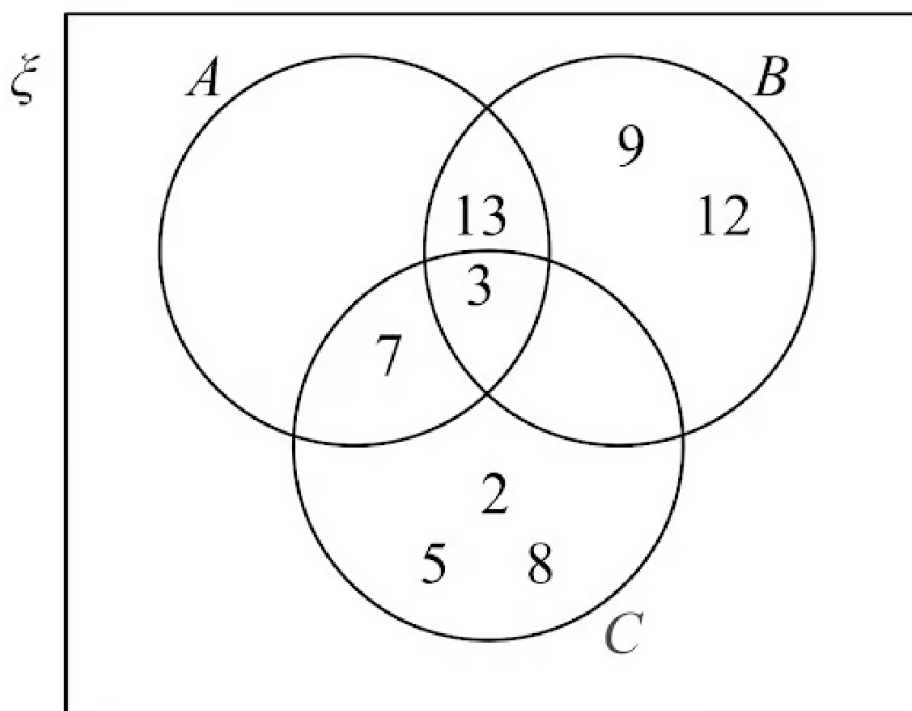
8 (a) $\xi = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\}$

$$A = \{3, 7, 11, 13\}$$

$$B = \{3, 6, 9, 12, 13\}$$

$$C = \{2, 3, 5, 6, 7, 8\}$$

Complete the Venn diagram



(2 marks)

(b) List the members of the set $B' \cap C$.

(1 mark)

(c) List the members of the set $(A \cup C)' \cap B$.

(1 mark)

(d) Find $n(A' \cap B')$

(1 mark)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1H • Q7

CH 1 • TOPIC 2 3 marks

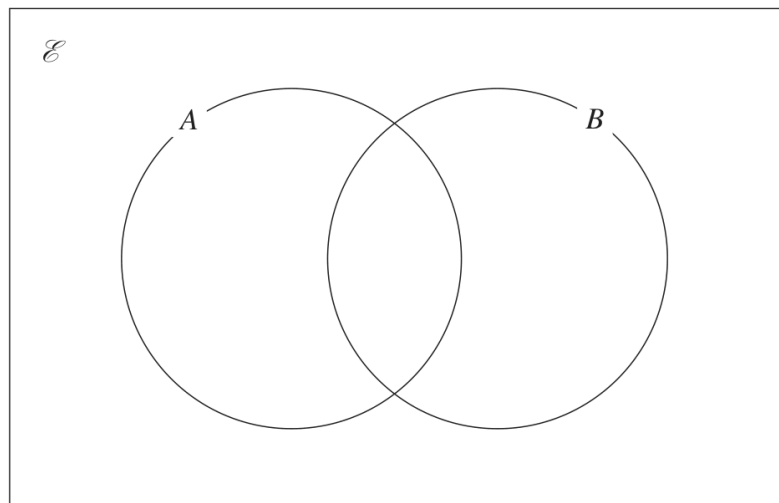
7 $\mathcal{E} = \{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15\}$

$$A \cap B = \{5, 10, 15\}$$

$$B' = \{7, 8, 9, 11, 12, 13, 14\}$$

$$A' = \{4, 6, 7, 8, 14\}$$

Complete the Venn diagram for this information.



(Total for Question 7 is 3 marks)

05 Three Sets with x

Same rule with three circles: the centre first, then the three pair-regions, then the “only” regions — in terms of x if needed. Then use the total.

WHEN YOU SEE → YOU DO

“6 have a dog, a cat and a rabbit”

→ centre

“9 have a dog and a rabbit”

→ pair region = 9 – centre

“Let x be the number who ...”

→ write every other region in terms of x

“given the total is 50, find x ”

→ add every region = 50, solve

Centre

Pairs minus centre

Only regions

Outside

Total = equation

WORKED MODEL • JANUARY 2020 P2HR

Dog 28, cat 18, rabbit 20; cat & rabbit 8, dog & rabbit 9, dog & cat x ; all three 6; none 5; total 50.

1 centre 6; C&R only 2, D&R only 3, D&C only $x - 6$

pairs

2 R only 9, D only $25 - x$, C only $16 - x$

only regions

3 $60 - x = 50$

total

ANSWER $x = 10$

WHERE YOU WILL USE IT Probability from Venn diagrams (Chapter 6): the same regions become probabilities.

IDEA 05

Practise: Three Sets with x

My questions first, then selected practice, then real past papers.

Q32 DR ESLAM Three sets

CH 1 • TOPIC 2 4 marks

60 people. All three sports: 4. $A \cap B$ only: 6, $B \cap C$ only: 5, $A \cap C$ only: 3. A only: x , B only: $2x$, C only: 10. None: 8. Find x .

$x =$ _____

SELECTED PRACTICE

Q33**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 4 marks

3 A garage tests cars for faults.

There are three types of fault – braking, steering and lighting.

A car fails the test if it has one or more of these three types of fault.

Last week, 11 cars had breaking faults,

9 cars had steering faults,

7 cars had lighting faults,

no car had both steering faults and lighting faults,

2 cars had both braking faults and steering faults,

3 cars had both braking faults and lighting faults.

By drawing a Venn diagram, or otherwise, find the number of cars which failed the test last week.

(4 marks)

Q34**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 **3 marks**

- 9 Each student in a group of 32 students was asked the following question.

“Do you have a desktop computer (D), a laptop (L) or a tablet (T)?”

Their answers showed that,

19 students have a desktop computer,

17 students have a laptop,

16 students have a tablet,

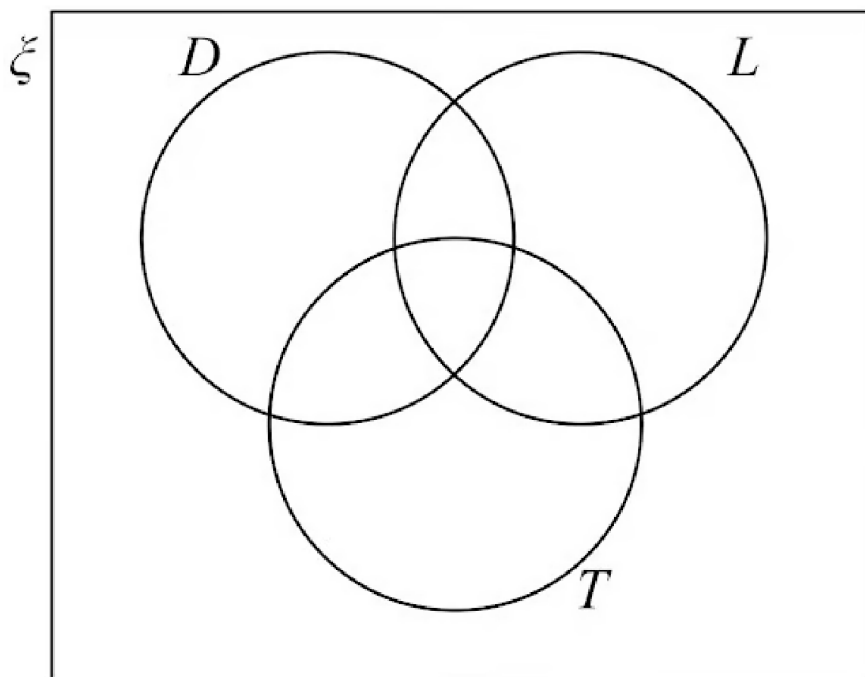
9 students have both a desktop computer and a laptop,

11 students have both a desktop computer and a tablet,

7 students have both a laptop and a tablet,

5 students have all three.

Using this information, complete the Venn diagram to show the number of students in each appropriate subset.

**(3 marks)**

Q35**PRACTICE**

Practice • Hard

CH 1 • TOPIC 2 **3 marks**

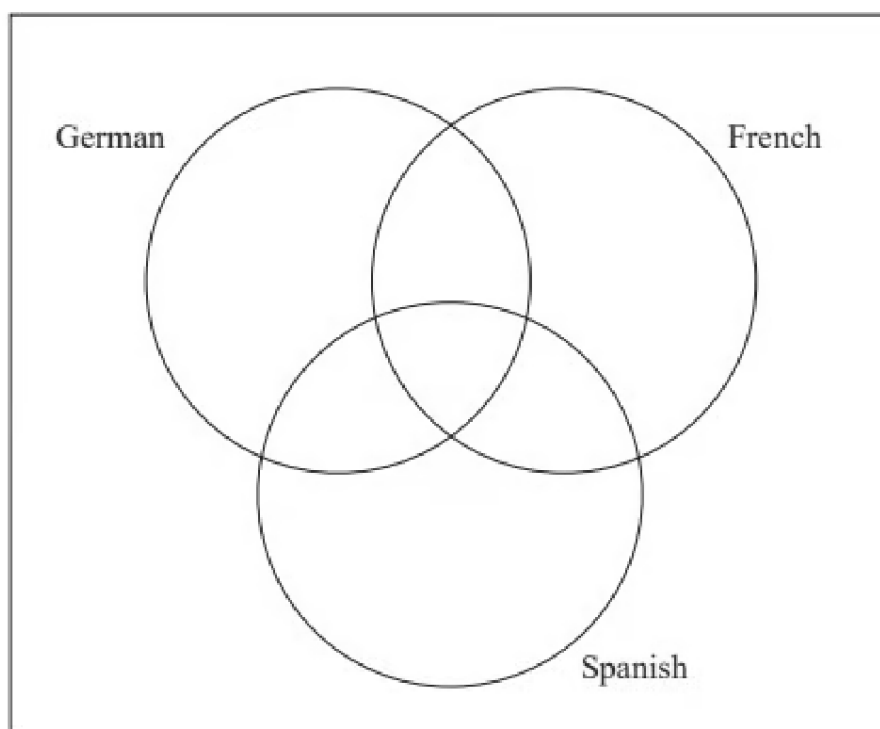
- 13** In a school, students must study at least one language from German, French and Spanish.

There are 30 students in a class of this school.

Of these students

- 7 study all three of the languages
- 10 study both German and French
- 12 study both Spanish and German
- 9 study both French and Spanish
- 16 study Spanish
- 18 study German

Work out the total number of the students in the class who study French.
You may use the Venn diagram to help with your calculations.

**(3 marks)**

PAST PAPERS

Q36 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2HR • Q17

CH 1 • TOPIC 2 6 marks

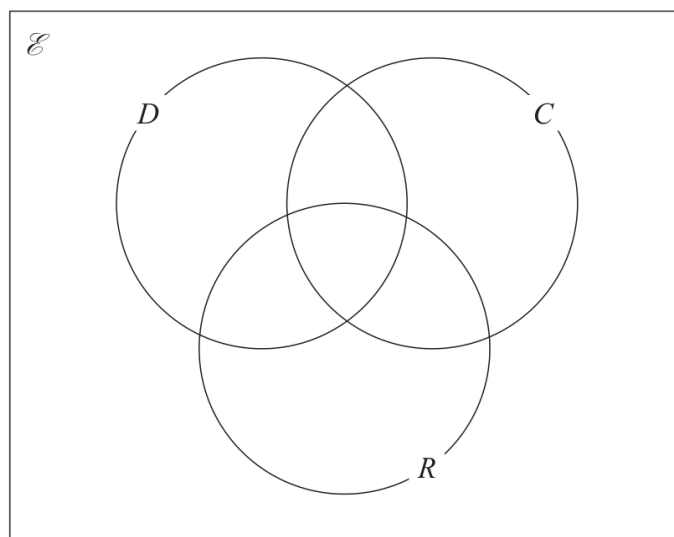
17 Some students in a school were asked the following question.

“Do you have a dog (D), a cat (C) or a rabbit (R)?”

Of these students

- 28 have a dog
- 18 have a cat
- 20 have a rabbit
- 8 have both a cat and a rabbit
- 9 have both a dog and a rabbit
- x have both a dog and a cat
- 6 have a dog, a cat and a rabbit
- 5 have not got a dog or a cat or a rabbit

- (a) Using this information, complete the Venn diagram to show the number of students in each appropriate subset.
Give the numbers in terms of x where necessary.



Given that a total of 50 students answered the question,

- (b) work out the value of x .

$x =$
(2)

- (c) Find $n(C' \cap D')$

.....

(Total for Question 17 is 6 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q16

CH 1 • TOPIC 2 6 marks

16 Some students were asked the following question.

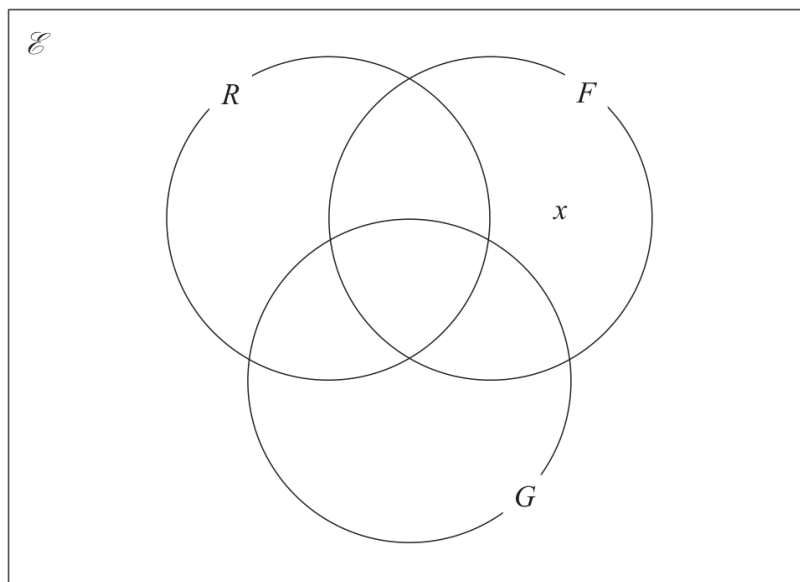
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

- (a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



Given that the number of students who were asked the question was 80

- (b) work out the number of these students that study Russian.

(Total for Question 16 is 6 marks)

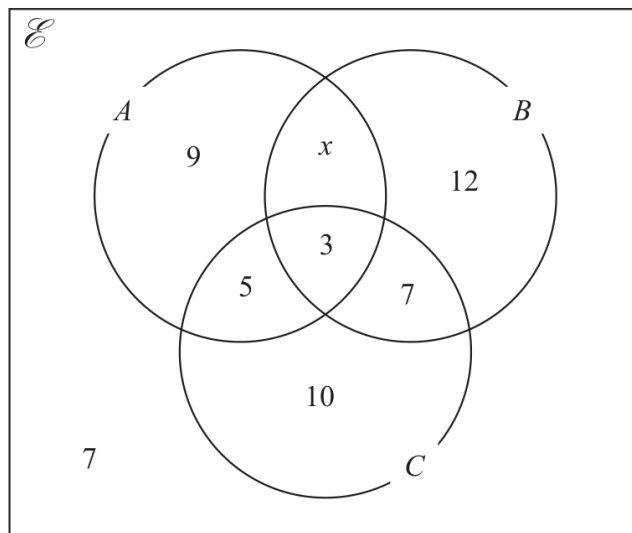
Q38**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q16

CH 1 • TOPIC 2 3 marks

16 The Venn diagram shows a universal set $\mathcal{E}$ and sets A , B and C

The numbers and the letter x represent **numbers** of elements.



Given that $n(A \cup B) = 42$

(a) find the value of x

$x = \dots\dots\dots$

(b) Find $n(A')$

$\dots\dots\dots$

(c) Find $n(B' \cap C)$

$\dots\dots\dots$

(Total for Question 16 is 3 marks)

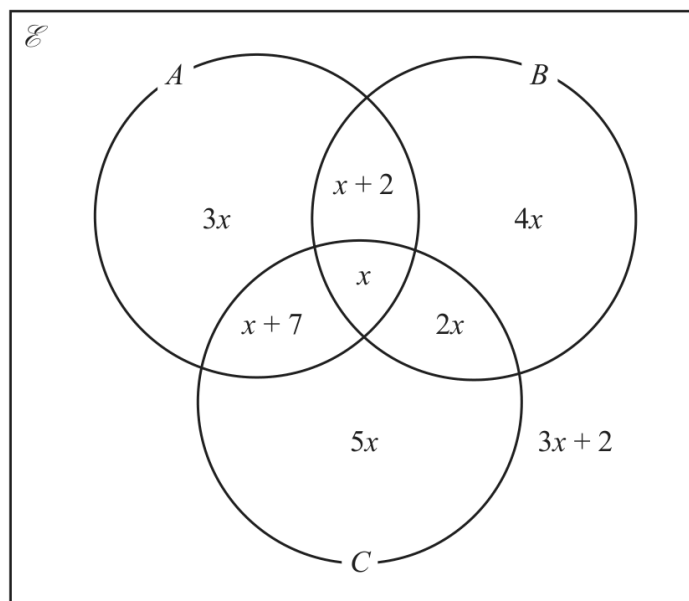
Q39

PAST PAPER

Edexcel 4MA1 • November 2024 • Paper 2H • Q21

CH 1 • TOPIC 2 4 marks

- 21 The Venn diagram shows information about the numbers of items in set A , set B and set C , where x is an integer.



Given that $n(A \cup B)' = 26$

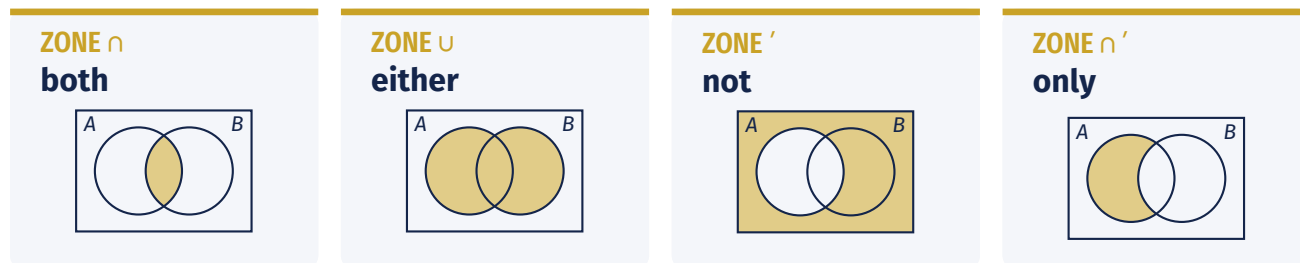
find $n(A' \cap C)$

$n(A' \cap C) = \dots\dots\dots$

(Total for Question 21 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.



01 Notation

Write E, A, B out; apply the symbol; each member once.

02 Read

“List” = the members; “ $n()$ ” = add the numbers.

03 Shade

Bracket first. Describe: in what, not in what.

04 Fill (2 sets)

Overlap $\rightarrow$ only $\rightarrow$ outside $\rightarrow$ check the total.

05 Fill (3 sets)

Centre $\rightarrow$ pairs $\rightarrow$ only $\rightarrow$ outside $\rightarrow$ total = equation in x .

BOTTOM LINE Always start in the middle of the Venn diagram $\rightarrow$ work outwards $\rightarrow$ check the total.

Answers

Set Notation and Venn Diagrams · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · SET NOTATION: LIST THE MEMBERS

- Q1** Dr Eslam • Both and either (a) $\{2, 4, 6, 12\}$ (b) $\{5, 7, 8, 9, 10, 11\}$
- Q2** Practice • Easy (a)(i) $\{p, r, a\}$ (ii) $\{p, a, r, i, s, b, u, d, e, t\}$ (b) E : no letter of A is in E
- Q3** Practice • Easy (a) no member of A is in B (b) 1, 9 (c) $\{1, 2, 8, 9\}$
- Q4** Practice • Medium $\{5, 10, 20, 25, 50, 100\}$
- Q5** Practice • Hard (a)(i) $\{1, 5, 7, 11\}$ (ii) $\{2, 3, 5, 6, 7, 9, 11, 12\}$ (b) No: 14 is not in E
- Q6** Edexcel 4MA1 • January 2020 • Paper 1HR • Q4 (a)(i) $\{b, l, u, e, g, r, y\}$ (ii) $\{w, h, i, t\}$ (b) No: e is in all three
- Q7** Edexcel 4MA1 • January 2021 • Paper 1H • Q4 (i) $\{21, 27\}$ (ii) $\{21, 23, 24, 25, 27, 29\}$
- Q8** Edexcel 4MA1 • January 2021 • Paper 2HR • Q3 (a)(i) $\{24, 30\}$ (ii) $\{21, 23, 25, 27, 29\}$ (b) $C = (A \cup B)'$
- Q9** Edexcel 4MA1 • May 2021 • Paper 2H • Q4 (a)(i) $\{b, r, a, z, i, l, e, n, d\}$ (ii) $\{b, z\}$ (b) a is in both B and K
- Q10** Edexcel 4MA1 • June 2022 • Paper 2HR • Q4 (a)(i) $\{9, 15\}$ (ii) $\{9, 11, 12, 13, 15, 17, 18, 19\}$ (b) No: 24 is not in E (c) e.g. $\{9, 10, 11, 18\}$
- Q11** Edexcel 4MA1 • June 2024 • Paper 2HR • Q4 (a)(i) $\{1, 2, 3, 5, 6, 7\}$ (ii) $\{4, 5, 7, 8, 9, 10\}$ (b) 2 and 3 are in both (c) $\{1, 5, 6, 7\}$
- Q12** Edexcel 4MA1 • May 2024 • Paper 1H • Q7 (a)(i) $\{23, 24, 27, 29, 30, 31, 33\}$ (ii) $\{27, 33\}$ (b) Yes: none of 23, 29, 31 is a multiple of 3 (c) $\{23, 25, 29, 31\}$
- Q13** Edexcel 4MA1 • June 2025 • Paper 1HR • Q5 (a)(i) $\{2, 3, 4, 6, 8, 9, 10, 12\}$ (ii) $\{1, 2, 4, 5, 7, 8, 10, 11\}$ (b)(i) $\emptyset$ (ii) $\notin$

IDEA 02 • READ A VENN DIAGRAM

- Q14** Dr Eslam • Count, do not list
 $n(A \cup B) = 20, n(B') = 12$
- Q15** Practice • Easy
 (i) $\{2, 4, 6, 12\}$ (ii) $\{5, 7, 8, 9, 10, 11, 13, 14\}$
- Q16** Practice • Medium
 (a) 18 (b) 15 (c) 9 (d) 22
- Q17** Practice • Hard
 (a) 0 (b) 23 (c) 2 (d) 12
- Q18** Practice • Hard
 (a) $\{1, 4, 5, 6, 7\}$ (b) 8 (c) $\{1, 4, 7, 8, 9\}$ (d) $\{5, 6, 7\}$
- Q19** Edexcel 4MA1 • May 2021 • Paper 1H • Q19
 (i) 45 (ii) 12 (iii) 28
- Q20** Edexcel 4MA1 • January 2022 • Paper 1HR • Q17
 (i) 19 (ii) 0 (iii) 11 (iv) 28
- Q21** Edexcel 4MA1 • May 2023 • Paper 1HR • Q2
 (a)(i) $\{5, 7, 11, 13\}$ (ii) $\{5, 6, 8, 10, 12, 14, 15\}$ (b) 9 is outside circle C
- Q22** Edexcel 4MA1 • November 2023 • Paper 1H • Q1
 (a) $\{5, 7, 9, 11, 13, 15\}$ (b) $\{5, 15\}$ (c) $\{6, 8, 12, 14, 16\}$
- Q23** Edexcel 4MA1 • November 2024 • Paper 1H • Q4
 (a) $\{2, 3, 5, 7\}$ (b) $\{3, 7\}$ (c) $\{2, 4, 5, 6, 8\}$

IDEA 03 • SHADE A REGION

- Q24** Practice • Medium
 (a) B separate from A; C inside A (b) A outside C shaded
- Q25** Practice • Hard
 (a) $A \cup B$ outside C shaded (b) $E \cap (D \cup F)$

IDEA 04 • FILL IT IN: WORK FROM THE MIDDLE

- Q26** Dr Eslam • Middle first $15 + 7 + 11 = 33$; neither = 7
- Q27** Practice • Easy cat only 19, both 11, dog only 14, outside 6
- Q28** Practice • Medium (a) 10, 8, 13, outside 5 (b)(i) 31 (ii) 10
- Q29** Practice • Hard both = $17 + 15 - 25 = 7$; only one = 18
- Q30** Practice • Hard (a) 11 in A only, 6 in $B \cap C$ only, 1, 4, 10 outside
(b) {2, 5, 7, 8} (c) {9, 12} (d) 6
- Q31** Edexcel 4MA1 • January 2022 • Paper 1H • Q7
A only 9, 11, 12, 13; $A \cap B$ 5, 10, 15; B only 4, 6;
outside 7, 8, 14

IDEA 05 • THREE SETS WITH x

- Q32** Dr Eslam • Three sets $3x + 36 = 60$; $x = 8$
- Q33** Practice • Hard braking only 6, B&S 2, B&L 3, steering only 7,
lighting only 4: 22 cars
- Q34** Practice • Hard centre 5; D&L 4, D&T 6, L&T 2; D only 4, L only
6, T only 3; outside 2
- Q35** Practice • Hard French only 8: French total 20
- Q36** Edexcel 4MA1 • January 2020 • Paper 2HR • Q17
(b) $x = 10$ (c) 14
- Q37** Edexcel 4MA1 • January 2021 • Paper 2H • Q16
(a) R only $2x$, F only x , R&F only 6, all 4, R&G
only 2, F&G only 9, G only 9, none 11 (b) $x =$
13: 38 study Russian
- Q38** Edexcel 4MA1 • May 2023 • Paper 1H • Q16
(a) $x = 6$ (b) 36 (c) 15
- Q39** Edexcel 4MA1 • November 2024 • Paper 2H • Q21
 $8x + 2 = 26$, $x = 3$: $n(A' \cap C) = 7x = 21$

Prime Factors, HCF and LCM

Break every number into primes. HCF = what they share (lowest powers). LCM = everything (highest powers).



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Prime Factors, HCF and LCM

01 Product of prime factors

1400 as powers of primes

3.6×10^3

smallest multiplier for a square

02 HCF and LCM of numbers

HCF of 56 and 84

LCM of 28, 42, 63

LCM of $5A$ and $7B$

03 HCF and LCM in index form

$A = 2^8 \times 3^5 \times 11^4$

unknown powers m, n, p

$P^3 \times Q$

04 LCM problems: “together again”

buses, trams, lamps

packs of pens and pencils

next date both

WHICH ONE? Read the story first:

ZONE HCF

Biggest that divides both

shares out / splits into equal groups:
lowest powers of the **common** primes

ZONE LCM

Smallest both divide into

happens together again / same number
of each: highest power of **every** prime

01 Product of Prime Factors

Split the number into any two factors, keep splitting until every branch ends in a prime. Then write the primes in order, as powers.

WHEN YOU SEE → YOU DO

“as a product of its prime factors”

→ factor tree (or ladder), then multiply the primes

“as a product of powers of its prime factors”

→ index form: $2^3 \times 5^2 \times 7$

“Show your working clearly”

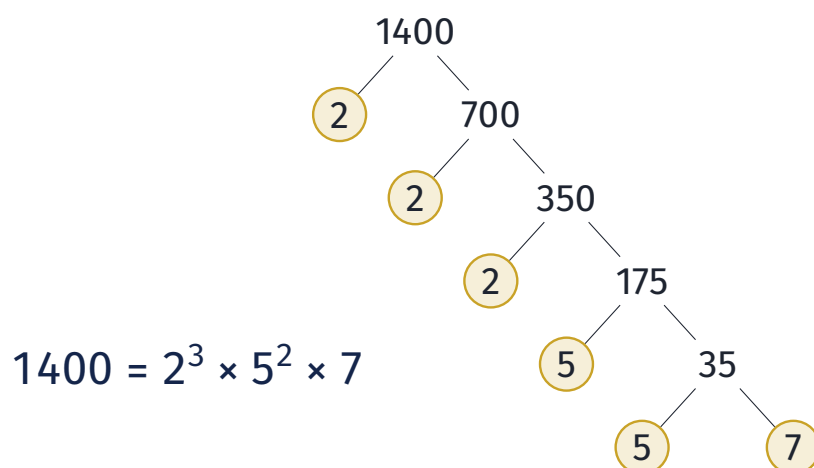
→ the tree / ladder is the method mark

3.6×10^3 , 480×10^9

→ split: 36×100 — then $10 = 2 \times 5$

“smallest number to multiply by to make a square”

→ make every power even



× **TRAP** Stopping at 35 or 25: every end of the tree must be a **prime** (circled).

IDEA 01

Practise: Product of Prime Factors

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Factor tree

CH 1 • TOPIC 3 2 marks

Write 90 as a product of powers of its prime factors.

Q2 DR ESLAM Index form

CH 1 • TOPIC 3 2 marks

Write 360 as a product of powers of its prime factors.

SELECTED PRACTICE

Q3

PRACTICE

Practice • Easy

CH 1 • TOPIC 3

3 marks

- 1 Write 525 as a product of its prime factors.

(3 marks)

Q4

PRACTICE

Practice • Easy

CH 1 • TOPIC 3

2 marks

6 $A = 2^n \times 3 \times 5^m$

Write $8A$ as a product of powers of its prime factors.

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 3 3 marks

14 Write 504 as the product of its prime factors.

(3 marks)

Q6**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 3 marks

10 N is a number.

As a product of prime factors in index form $N = 2 \times 3^4 \times y^3$

Work out $3N^2$ as a product of prime factors in index form.

Give your answer in terms of y .

(3 marks)

PAST PAPERS

Q7**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 (a) Write $5^{17} \times 5^2$ as a single power of 5

.....

- (b) Write 800 as a product of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 Write 600 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q7

CH 1 • TOPIC 3 3 marks

- 7 Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.
-

(Total for Question 7 is 3 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q2

CH 1 • TOPIC 3 3 marks

- 2 Write 1200 as a product of powers of its prime factors.
Show your working clearly.
-

(Total for Question 2 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 Write 2250 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q1

CH 1 • TOPIC 3 3 marks

You must write down all the stages in your working.

- 1 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 1 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q2

CH 1 • TOPIC 3 3 marks

- 2 Write 1400 as a product of powers of its prime factors.
Show your working clearly.

.....

(Total for Question 2 is 3 marks)

02 HCF and LCM of Numbers: the Venn Method

Write both numbers as prime factors. Shared primes go in the middle. HCF = multiply the middle. LCM = multiply **everything** in the diagram.

WHEN YOU SEE → YOU DO

“highest common factor (HCF)”

→ multiply the overlap

“lowest common multiple (LCM)”

→ multiply every number in the Venn diagram

three numbers: “LCM of 28, 42 and 63”

→ highest power of each prime from any of them

“circle the LCM”

→ check it is divisible by every number

Prime factors

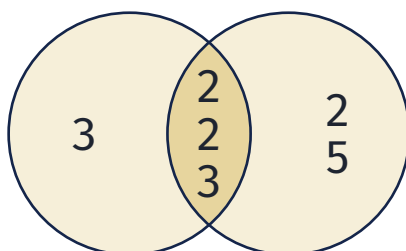
Venn diagram

HCF = middle

LCM = all

$$36 = 2^2 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$



$$\text{HCF} = 2 \times 2 \times 3 = 12$$

$$\text{LCM} = 3 \times 2 \times 2 \times 3 \times 2 \times 5 = 360$$

CALCULATOR CHECK The calculator has FACT (prime factors) — use it to check your tree, never instead of the working.

IDEA 02

Practise: HCF and LCM of Numbers

My questions first, then selected practice, then real past papers.

Q14 DR ESLAM Venn method

CH 1 • TOPIC 3 3 marks

Find (a) the HCF (b) the LCM of 48 and 180.

SELECTED PRACTICE**Q15** PRACTICE Practice • Easy

CH 1 • TOPIC 3 2 marks

11 Find the Highest Common Factor (HCF) of 140 and 245

(2 marks)

Q16 **PRACTICE** Practice • Medium

CH 1 • TOPIC 3 2 marks

- 5 Find the highest common factor (HCF) of 72 and 90.
(2 marks)

Q17 **PRACTICE** Practice • Medium

CH 1 • TOPIC 3 3 marks

- 14 (a) Written as the product of its prime factors

$$672 = 2^5 \times 3 \times 7$$

Write 252 as the product of its prime factors.

(2 marks)

- (b) Work out the value of the highest common factor of 672 and 252

(1 mark)

Q18**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 4 marks

13 Here are some properties of a number.

- It is a common factor of 288 and 360.
- It is a common multiple of 4 and 6.
- It is larger than 25.

Find the **two** possible numbers with these properties.

(4 marks)**PAST PAPERS****Q19****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q2

CH 1 • TOPIC 3 2 marks

2 Find the lowest common multiple (LCM) of 28 and 105

(Total for Question 2 is 2 marks)

Q20**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q7

CH 1 • TOPIC 3 3 marks

- 7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.
-

(Total for Question 7 is 3 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q3

CH 1 • TOPIC 3 4 marks

- 3 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.
-
- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.
-

(Total for Question 3 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q8

CH 1 • TOPIC 3 4 marks

- 8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

$$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

$$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$$

- (b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

(Total for Question 8 is 4 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q3

CH 1 • TOPIC 3 2 marks

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

(Total for Question 3 is 2 marks)

Q24**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q1

CH 1 • TOPIC 3 2 marks

You must write down all the stages in your working.

- 1** Find the lowest common multiple (LCM) of 45 and 70

(Total for Question 1 is 2 marks)

03 HCF and LCM in Index Form

No calculation needed. Line the primes up. HCF: **lowest** power of each **shared** prime. LCM: **highest** power of **every** prime.

WHEN YOU SEE → YOU DO

$$A = 2^8 \times 3^5 \times 11^4, B = 2^6 \times 3 \times 11^8$$

$$\longrightarrow \text{HCF } 2^6 \times 3 \times 11^4 \cdot \text{LCM } 2^8 \times 3^5 \times 11^8$$

a prime in only one number

→ in the LCM, never in the HCF

“HCF of 5A and 2B”

→ first update the powers: 5A adds 1 to the power of 5

unknown powers with $m > 2$

→ use the inequality to see which power is smaller

“find p, q, r ” from a given HCF/LCM

→ match each prime's power one at a time

	2	3	5	7
$A = 3^2 \times 5^4 \times 7$	—	3^2	5^4	7
$B = 3^4 \times 5^3 \times 7 \times 11$	—	3^4	5^3	7
HCF (lowest)		3^2	5^3	7
LCM (highest, + 11)		3^4	5^4	7×11

× **TRAP** Adding the powers (that is multiplying $A \times B$), not choosing the lowest or highest one.

IDEA 03

Practise: HCF and LCM in Index Form

My questions first, then selected practice, then real past papers.

Q25 DR ESLAM Lowest, highest

CH 1 • TOPIC 3 2 marks

$A = 2^4 \times 3 \times 5^2$, $B = 2^2 \times 3^3 \times 7$. Find the HCF and the LCM of A and B .

SELECTED PRACTICE**Q26** PRACTICE Practice • Medium

CH 1 • TOPIC 3 4 marks

7 (a) $A = 3^2 \times 5^4 \times 7$ $B = 3^4 \times 5^3 \times 7 \times 11$

Find the highest common factor (HCF) of A and B .

(2 marks)

(b) Find the lowest common multiple (LCM) of A and B .

(2 marks)

Q27**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3 7 marks

17 (a) i) Write 120 as a product of its prime factors.

[3]

ii) The lowest common multiple (LCM) of x and 120 is 360. Find the smallest possible value of x .

[2]

(5 marks)

(b) Two numbers, A and B , are written as a product of prime factors.

$$A = 2^4 \times 3^2 \times 7^2 \quad B = 2^3 \times 3 \times 5 \times 7$$

Find the highest common factor (HCF) of A and B .

(2 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 2 marks

11 You are given that

$$90 = 3^2 \times 2 \times 5 \text{ and } 177147 = 3^{11}$$

Find the lowest common multiple (LCM) of 270 and 177147. Give your answer using power notation and as an ordinary number.

using power notation

as an ordinary number

(2 marks)**Q29****PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 1 mark

14 Here is a list of five numbers.

$$98^{53} \quad 98^{64} \quad 98^{73} \quad 98^{88} \quad 98^{91}$$

Find the lowest common multiple of these five numbers.

(1 mark)

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q10

CH 1 • TOPIC 3 3 marks

10 $A = 2 \times 3^{43}$
 $B = 16 \times 3^{37}$

(a) Find the highest common factor (HCF) of A and B .

.....

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

.....

(Total for Question 10 is 3 marks)**Q31****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q9

CH 1 • TOPIC 3 4 marks

9 $A = 2^8 \times 3^5 \times 11^4$ $B = 2^6 \times 3 \times 11^8$

(a) Find the highest common factor (HCF) of A and B .

.....

(b) Find the lowest common multiple (LCM) of $2A$ and $3B$.
Give the LCM as a product of powers of its prime factors.

.....

(Total for Question 9 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q9

CH 1 • TOPIC 3 2 marks

9

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B .

Give your answer as a product of powers of prime numbers.

(Total for Question 9 is 2 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q6

CH 1 • TOPIC 3 4 marks

6 (a) Work out the lowest common multiple (LCM) of 36 and 120

$$A = 5^2 \times 7^4 \times 11^p$$

$$B = 5^m \times 7^{n-5} \times 11$$

m , n and p are integers such that

$$m > 2$$

$$n > 10$$

$$p > 1$$

(b) Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

(Total for Question 6 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1HR • Q12

CH 1 • TOPIC 3 3 marks

$$12 \quad P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

(a) Write down the highest common factor (HCF) of P and Q

$$P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

(b) Work out the value of $P^3 \times Q$ Give your answer in the form $3^x \times 5^y \times 7^z$ where x , y and z are positive integers.

(Total for Question 12 is 3 marks)

Q35 PAST PAPER Edexcel 4MA1 • November 2023 • Paper 1H • Q7

CH 1 • TOPIC 3 2 marks

$$7 \quad A = 5^3 \times 7^3 \times 11^6 \quad \text{and} \quad B = 5^6 \times 7^2 \times 11^4$$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

(Total for Question 7 is 2 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q11

CH 1 • TOPIC 3 4 marks

11 $A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (a) Write down the highest common factor (HCF) of
- $5A$
- and
- $2B$

Give your answer as a product of prime factors.

.....
(2)

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

- (b) Work out the value of
- $(AB)^2$

Give your answer as a product of prime factors.

.....
(Total for Question 11 is 4 marks)

Q37**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q6

CH 1 • TOPIC 3 2 marks

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

Find the highest common factor (HCF) of A , B and C
Write your answer as a product of prime factors.

.....
(Total for Question 6 is 2 marks)

04 LCM Problems: “Together Again”

Two things repeat at different intervals and start together — they meet again after the LCM. Buy “the same number of each” — the LCM again.

WHEN YOU SEE → YOU DO

“every 24 minutes ... every 20 minutes ... next leave together”

→ $\text{LCM}(24, 20) = 120$ min, add to the start time

“exactly the same number of pens as pencils”

→ LCM of the pack sizes, then count the packs

“more than 60 of each”

→ first multiple of the LCM above 60

“how many times in one hour”

→ $60 \text{ min} \div \text{LCM}$

“state an assumption”

→ they keep to the timetable exactly

Find the LCM

Answer the real question

WORKED MODEL • JUNE 2023 P2HR

Pens: packets of 9 at £7.60. Pencils: boxes of 12 at £4.80. Same number of pens as pencils. Minimum cost?

1 $\text{LCM}(9, 12) = 36$

same number of each

2 $36 \div 9 = 4$ packets, $36 \div 12 = 3$ boxes

count the packs

3 $4 \times 7.60 + 3 \times 4.80 = 30.40 + 14.40$

cost

ANSWER £44.80

IDEA 04

Practise: LCM Problems: Together Again

My questions first, then selected practice, then real past papers.

Q38 **DR ESLAM** Together again

CH 1 • TOPIC 3 3 marks

Two lights flash every 15 seconds and every 20 seconds. They flash together at 9:00:00. When do they next flash together?

SELECTED PRACTICE**Q39** **PRACTICE** Practice • Easy

CH 1 • TOPIC 3 3 marks

4 Trams leave Piccadilly

to Eccles every 9 minutes
to Didsbury every 12 minutes

A tram to Eccles and a tram to Didsbury both leave Piccadilly at 9 am.
At what time will a tram to Eccles and a tram to Didsbury next leave Piccadilly at the same time?

(3 marks)

Q40**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3**3 marks**

3 Matt and Dan cycle around a cycle track.

Each lap Matt cycles takes him 50 seconds.

Each lap Dan cycles takes him 80 seconds.

Dan and Matt start cycling at the same time at the start line.

Work out how many laps they will each have cycled when they are next at the start line together.

(3 marks)

Q41**PRACTICE**

Practice • Medium

CH 1 • TOPIC 3 5 marks

16 (a) A bus timetable shows the following information.

- A bus following route T leaves for the train station every 20 minutes.
- A bus following route A leaves for the airport every 18 minutes.
- A bus following route T and a bus following route A both leave at 8.37 am.

When is the next time one of each bus is timetabled to leave at the same time?

(4 marks)

(b) Write down one assumption that was necessary to solve this problem.

(1 mark)

Q42**PRACTICE**

Practice • Hard

CH 1 • TOPIC 3 3 marks

5 Liz buys packets of coloured buttons.

There are 8 red buttons in each packet of red buttons.

There are 6 silver buttons in each packet of silver buttons.

There are 5 gold buttons in each packet of gold buttons.

Liz buys equal numbers of red buttons, silver buttons and gold buttons.

How many packets of each colour of buttons did Liz buy?

..... packets of red buttons

..... packets of silver buttons

..... packets of gold buttons

(3 marks)

PAST PAPERS

Q43**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q2

CH 1 • TOPIC 3**4 marks**

- 2 Sandeep wants to buy some packets of pens and some boxes of pencils for his stationery shop.

Each packet of pens contains 9 pens.

Each box of pencils contains 12 pencils.

Each packet of pens costs £7.60

Each box of pencils costs £4.80

Sandeep can only buy full packets of pens and full boxes of pencils.

He wants to buy exactly the same number of pens as pencils.

Work out the minimum amount Sandeep needs to pay.

£.....

(Total for Question 2 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE HCF

shared primes

lowest powers — “share into equal groups”

ZONE LCM

every prime

highest powers — “together again”,
“same number of each”

- | | | |
|-----------|----------------------|--|
| 01 | Prime factors | Tree until every end is prime; write as powers in order. |
| 02 | Venn method | Middle = HCF; everything = LCM. |
| 03 | Index form | HCF lowest shared powers; LCM highest powers of all primes. |
| 04 | Problems | “Together again” / “same number of each” = LCM, then answer the story. |

BOTTOM LINE Primes first → HCF = shared, lowest → LCM = all, highest.

Answers

Prime Factors, HCF and LCM · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PRODUCT OF PRIME FACTORS

Q1	Dr Eslam · Factor tree	$2 \times 3^2 \times 5$
Q2	Dr Eslam · Index form	$2^3 \times 3^2 \times 5$
Q3	Practice · Easy	$3 \times 5^2 \times 7$
Q4	Practice · Easy	$2^{n+3} \times 3 \times 5^m$
Q5	Practice · Easy	$2^3 \times 3^2 \times 7$
Q6	Practice · Hard	$2^2 \times 3^9 \times y^6$
Q7	Edexcel 4MA1 · January 2020 · Paper 2HR · Q1	(a) 5^{19} (b) $2^5 \times 5^2$
Q8	Edexcel 4MA1 · May 2021 · Paper 2H · Q1	$2^3 \times 3 \times 5^2$
Q9	Edexcel 4MA1 · November 2021 · Paper 1H · Q7	$2^4 \times 3^2 \times 5^2$
Q10	Edexcel 4MA1 · June 2022 · Paper 2HR · Q2	$2^4 \times 3 \times 5^2$
Q11	Edexcel 4MA1 · May 2023 · Paper 1HR · Q1	$2 \times 3^2 \times 5^3$
Q12	Edexcel 4MA1 · June 2024 · Paper 2HR · Q1	$2^3 \times 5^2 \times 7$
Q13	Edexcel 4MA1 · June 2025 · Paper 2HR · Q2	$2^3 \times 5^2 \times 7$

IDEA 02 • HCF AND LCM OF NUMBERS

- Q14** Dr Eslam • Venn method (a) 12 (b) 720
- Q15** Practice • Easy 35
- Q16** Practice • Medium 18
- Q17** Practice • Medium (a) $2^2 \times 3^2 \times 7$ (b) 84
- Q18** Practice • Hard 36 and 72
- Q19** Edexcel 4MA1 • January 2020 • Paper 1H • Q2 420
- Q20** Edexcel 4MA1 • January 2022 • Paper 2H • Q7 252
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q3 (a) 28 (b) 360
- Q22** Edexcel 4MA1 • May 2023 • Paper 1H • Q8 (a) $2^2 \times 3 \times 5^2$ (b) $2^3 \times 3^3 \times 5^2 \times 7 = 37\,800$
- Q23** Edexcel 4MA1 • May 2024 • Paper 1H • Q3 36
- Q24** Edexcel 4MA1 • June 2025 • Paper 1HR • Q1 630

IDEA 03 • HCF AND LCM IN INDEX FORM

- Q25** Dr Eslam • Lowest, highest HCF $2^2 \times 3$; LCM $2^4 \times 3^3 \times 5^2 \times 7$
- Q26** Practice • Medium (a) $3^2 \times 5^3 \times 7$ (b) $3^4 \times 5^4 \times 7 \times 11$
- Q27** Practice • Medium (a)(i) $2^3 \times 3 \times 5$ (ii) 9 (b) $2^3 \times 3 \times 7 = 168$
- Q28** Practice • Hard $2 \times 3^{11} \times 5 = 1\,771\,470$
- Q29** Practice • Hard 98^{91}
- Q30** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q10 (a) 2×3^{37} (b) $2^5 \times 3^{80}$
- Q31** Edexcel 4MA1 • January 2021 • Paper 1HR • Q9 (a) $2^6 \times 3 \times 11^4$ (b) $2^9 \times 3^5 \times 11^8$
- Q32** Edexcel 4MA1 • January 2021 • Paper 2H • Q9 $2^4 \times 3^2 \times 5^4 \times 11 \times 13$
- Q33** Edexcel 4MA1 • January 2022 • Paper 1HR • Q6 (a) 360 (b) $5^2 \times 7^4 \times 11$
- Q34** Edexcel 4MA1 • June 2022 • Paper 1HR • Q12 (a) $3^2 \times 5 \times 7$ (b) $3^{11} \times 5^7 \times 7^5$
- Q35** Edexcel 4MA1 • November 2023 • Paper 1H • Q7 $5^3 \times 7^2 \times 11^4$
- Q36** Edexcel 4MA1 • May 2024 • Paper 1HR • Q11 (a) $2^4 \times 5^2 \times 7^2$ (b) $2^{16} \times 5^8 \times 7^{12}$
- Q37** Edexcel 4MA1 • November 2024 • Paper 1H • Q6 $2^2 \times 5^2 \times 7$

IDEA 04 • LCM PROBLEMS: TOGETHER AGAIN

- Q38** Dr Eslam • Together again LCM = 60 s: 9:01:00
- Q39** Practice • Easy LCM 36: 9:36 am
- Q40** Practice • Medium LCM 400 s: Matt 8 laps, Dan 5 laps
- Q41** Practice • Medium (a) LCM 180 min: 11:37 am (b) the buses leave exactly on time
- Q42** Practice • Hard 15 red, 20 silver, 24 gold
- Q43** Edexcel 4MA1 • June 2023 • Paper 2HR • Q2 £44.80

Powers, Roots and Standard Form



DR ESLAM AHMED
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Index laws for numbers, then standard form:
 $A \times 10^n$ with $1 \leq A < 10$. Convert, calculate, and
 compare big and small numbers.

THE TOPIC ON ONE MAP

Powers, Roots and Standard Form

01 Powers and roots of numbers

$$5^0, 3^{-2}$$

$$125^{\frac{2}{3}}, 27^{-\frac{2}{3}}$$

$$\sqrt{2} \div \frac{8^3}{16^{3/2}} = 2^n$$

02 Standard form: convert

76 000 000

$$5.4 \times 10^{-4}$$

which is in standard form?

order them

03 Standard form: calculate

$$(5 \times 10^{42}) \times (7 \times 10^{-180})$$

$$\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$$

$$4\% \text{ of } 4.5 \times 10^{157}$$

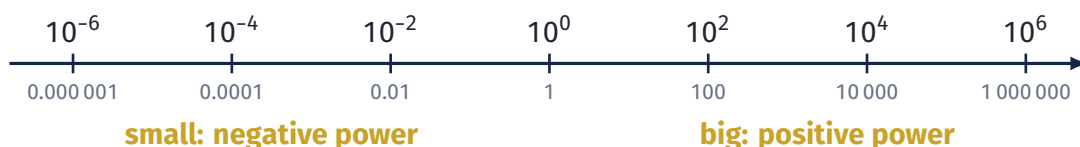
04 Tables and real data

difference of two oceans

"k times as big"

per person

THE SLIDING SCALE The power of 10 counts how far the decimal point moves:



01 Powers and Roots of Numbers

Three rules turn any strange power into a normal number: zero $\rightarrow$ 1, negative $\rightarrow$ flip, fraction $\rightarrow$ root then power.

WHEN YOU SEE $\rightarrow$ YOU DO

a^0 (anything to the power 0)

$\rightarrow = 1$

3^{-2} (negative power)

$\rightarrow$ flip: $\frac{1}{3^2} = \frac{1}{9}$

$125^{\frac{2}{3}}$ (fraction power)

$\rightarrow$ bottom = root, top = power: $(\sqrt[3]{125})^2 = 25$

“write as a power of 2 / find n ”

$\rightarrow$ write every number as a power of 2, then use the laws

“show that”

$\rightarrow$ show the root step and the power step

Negative? Flip

Root (bottom)

Power (top)

WORKED MODEL • NEGATIVE FRACTION POWER

Work out $27^{-\frac{2}{3}}$.

1 $\left(\frac{1}{27}\right)^{\frac{2}{3}}$

flip

2 $\left(\frac{1}{3}\right)^2$

cube root

ANSWER $\frac{1}{9}$

× **TRAP** $64^{\frac{1}{4}}$ is **not** $\frac{1}{4}$ of 64. It is the fourth root: $\sqrt[4]{64} = 2\sqrt{2}$.

IDEA 01

Practise: Powers and Roots of Numbers

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Three rules

CH 1 • TOPIC 4 3 marks

Work out (a) 2^{-3} (b) $27^{\frac{2}{3}}$ (c) $16^{-\frac{1}{2}}$

SELECTED PRACTICE

Q2 PRACTICE Practice • Easy

CH 1 • TOPIC 4 2 marks

1 Write these numbers in order of size. Start with the smallest number.

5^{-1}	0.5	-5	5^0
----------	-----	----	-------

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 4 1 mark

14 Circle the expression that has the **largest** value when $a < -1$

$\frac{1}{2}a$	a	a^2	a^3
----------------	-----	-------	-------

(1 mark)

Q4**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 2 marks

1 (a) Show that $2^{-3} = \frac{1}{8}$

(1 mark)

(b) $5\sqrt{5}$ can be written in the form 5^k .

Find the value of k .

(1 mark)

Q5**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4

3 marks

14 Show that $\frac{\sqrt[3]{81}}{3}$ can be written as $3^{\frac{1}{3}}$

(3 marks)**PAST PAPERS****Q6****PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q15

CH 1 • TOPIC 4

6 marks

15 (a) $\sqrt{2} \div \frac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

 $n = \dots\dots\dots$

(b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

 $\dots\dots\dots$ **(Total for Question 15 is 6 marks)**

02 Standard Form: Convert

Standard form is $A \times 10^n$ with $1 \leq A < 10$. Move the point until one non-zero digit is in front of it; count the moves.

WHEN YOU SEE → YOU DO

“write in standard form”, a big number

→ power = number of places moved left (positive)

0.000068 (small number)

→ negative power: 6.8×10^{-5}

“as an ordinary number”

→ positive power → bigger; negative power → smaller

“circle the number in standard form”

→ check $1 \leq A < 10$: 0.9×10^{-3} and 12×10^7 are not

“write in order of size”

→ convert all to ordinary numbers (or the same power), then order

ZONE BIG

$$76\,000\,000 = 7.6 \times 10^7$$

the point moved 7 places left

ZONE SMALL

$$0.000\,54 = 5.4 \times 10^{-4}$$

the point moved 4 places right

× **TRAP** 0.38×10^{-1} and 3800×10^{-4} are not in standard form — convert before you compare.

IDEA 02

Practise: Standard Form: Convert

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Both directions

CH 1 • TOPIC 4 2 marks

(a) Write 0.003 05 in standard form. (b) Write 4.6×10^5 as an ordinary number.

SELECTED PRACTICE**Q8****PRACTICE**

Practice • Easy

CH 1 • TOPIC 4 2 marks

4 (a) Write 7.8×10^{-4} as an ordinary number.

(1 mark)

(b) Write 95 600 000 as a number in standard form.

(1 mark)

Q9**PRACTICE**

Practice • Easy

CH 1 • TOPIC 4 2 marks

8 (a) Write 0.00562 in standard form.

(1 mark)

(b) Write 1.452×10^3 as an ordinary number.

(1 mark)

Q10**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

3 (a) Write down the value of 10^0 .

(1 mark)

(b) Write down the value of 10^{-2} .

(1 mark)

(c) Write these numbers in order of size.
Start with the smallest number.

2.73×10^3	27.3×10^{-3}	273×10^2	0.00273
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(2 marks)

Q11**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 4 marks

10 (a) Write 8.2×10^5 as an ordinary number.

(1 mark)

(b) Write 0.000 376 in standard form.

(1 mark)

(c) Work out the value of $(2.3 \times 10^{12}) \div (4.6 \times 10^3)$
Give your answer in standard form.

(2 marks)

PAST PAPERS

Q12 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q5**CH 1 • TOPIC 4** **4 marks**

5 (a) Write 5.7×10^{-3} as an ordinary number.

.....

(b) Write 800 000 in standard form.

.....

(c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....

(Total for Question 5 is 4 marks)**Q13** **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1H • Q6**CH 1 • TOPIC 4** **2 marks**

6 (a) Write 2 840 000 000 in standard form.

.....

(b) Write 2.5×10^{-4} as an ordinary number.

.....

(Total for Question 6 is 2 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q8

CH 1 • TOPIC 4 4 marks

8 (a) Write 5×10^4 as an ordinary number.

.....

(b) Write 0.00006 in standard form.

.....

(c) Work out $(4 \times 10^{512}) \div (1.6 \times 10^{700})$

Give your answer in standard form.

.....

(Total for Question 8 is 4 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q8

CH 1 • TOPIC 4 2 marks

8 (a) Write 0.000089 in standard form.

.....

(b) Write 8.34×10^4 as an ordinary number.

.....

(Total for Question 8 is 2 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q6

CH 1 • TOPIC 4 2 marks

6 (a) Write 76 000 000 in standard form.

.....

(b) Write 5.4×10^{-4} as an ordinary number.

.....

(Total for Question 6 is 2 marks)**Q17****PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q9

CH 1 • TOPIC 4 4 marks

9 (a) Write 5.87×10^{-4} as an ordinary number.

.....

(b) Write 84 000 000 in standard form.

.....

The number of neurons in a human brain is 8.5×10^{10}

The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

(c) Work out the value of K

Give your answer correct to one decimal place.

 $K =$ **(Total for Question 9 is 4 marks)**

Q18**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q9

CH 1 • TOPIC 4 4 marks

9 (a) Write 6.04×10^5 as an ordinary number.

.....

(b) Write 0.000 07 in standard form.

.....

(1)

(c) Work out $\frac{7.6 \times 10^{10}}{4 \times 10^5 - 2 \times 10^4}$

Give your answer in standard form.

.....

(Total for Question 9 is 4 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q8

CH 1 • TOPIC 4 4 marks

8 (a) Write 520 million in standard form.

.....

(b) Write 8.79×10^{-5} as an ordinary number.

.....

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....

(Total for Question 8 is 4 marks)

03 Standard Form: Calculate

Multiply or divide: numbers with numbers, powers with powers — then fix A. Add or subtract: use the calculator (or make the powers the same).

WHEN YOU SEE → YOU DO

$$(5 \times 10^{42}) \times (7 \times 10^{-180})$$

$$\longrightarrow 35 \times 10^{-138} \rightarrow \text{fix to } 3.5 \times 10^{-137}$$

÷ **in standard form**

→ divide the numbers, subtract the powers

“+” or “-”

→ calculator ($\times 10^x$ key), or same power first

$$(6 \times 10^{40})^3$$

$$\longrightarrow 6^3 \times 10^{120} = 216 \times 10^{120} \rightarrow 2.16 \times 10^{122}$$

“give your answer in standard form”

→ last line must have $1 \leq A < 10$

Numbers together

Powers together

Fix A

Check $1 \leq A < 10$

WORKED MODEL • JUNE 2025 P1H

Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$.

1 $(5 \times 7) \times 10^{42-180}$

numbers · powers

2 35×10^{-138}

not yet standard form

ANSWER 3.5×10^{-137}

CALCULATOR CHECK Use the $\times 10^x$ key, never “ $\times 10^{\wedge}$ ”. Put a fraction’s top and bottom in brackets.

IDEA 03

Practise: Standard Form: Calculate

My questions first, then selected practice, then real past papers.

Q20 DR ESLAM Fix A

CH 1 • TOPIC 4 2 marks

Work out $(3 \times 10^4) \times (4 \times 10^5)$. Give your answer in standard form.

Q21 DR ESLAM Divide

CH 1 • TOPIC 4 2 marks

Work out $(6 \times 10^8) \div (2 \times 10^{-3})$ in standard form.

SELECTED PRACTICE**Q22** PRACTICE Practice • Easy

CH 1 • TOPIC 4 2 marks

- 5 Work out the value of $(7.5 \times 10^4) \times (2.5 \times 10^3)$
Give your answer in standard form.
-

(2 marks)

Q23**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4**3 marks**

- 1 One sheet of paper is 9×10^{-3} cm thick.

Mark wants to put 500 sheets of paper into the paper tray of his printer.
The paper tray is 4 cm deep.

Is the paper tray deep enough for 500 sheets of paper?
You must explain your answer.

(3 marks)

Q24**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 9 marks

- 16 (a)** A company makes sweets.
The sweets are put into packets.

Here are some facts.

1.47×10^7	3.5×10^5
sweets are made every day	packets of sweets are produced every day

Calculate the mean number of sweets in one packet.

(2 marks)

- (b)** Sweets are made on 288 days each year.

Calculate the number of sweets made each year.
Give your answer in standard form.

(3 marks)

- (c)** The company has 152 machines making the sweets. Each machine operates for 15 hours each day.

- i) Calculate the number of sweets made by one machine each hour.
Give your answer as an ordinary number correct to the nearest 10.

[3]

- ii) State one assumption you have made in part **(c)(i)**.

[1]

(4 marks)

Q25**PRACTICE**

Practice • Hard

CH 1 • TOPIC 4 2 marks

- 11 Work out the value of $(3.5 \times 10^6) \div (5 \times 10^{-3})$.
Give your answer in standard form.

(2 marks)

PAST PAPERS**Q26****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q6

CH 1 • TOPIC 4 3 marks

- 6 (a) Write 7.8×10^{-4} as an ordinary number.

(b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

(Total for Question 6 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q6

CH 1 • TOPIC 4 4 marks

- 6 (a) Write down the value of y^0

.....
(1)

- (b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

.....
(Total for Question 6 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q8

CH 1 • TOPIC 4 2 marks

8

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

.....
(Total for Question 8 is 2 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q12

CH 1 • TOPIC 4 3 marks

12 $a = 6 \times 10^{40}$

Work out the value of a^3
Give your answer in standard form.

.....

(Total for Question 12 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q8

CH 1 • TOPIC 4 3 marks**8** (a) Write 5.6×10^{-3} as an ordinary number.

(b) Work out
$$\frac{6 \times 10^3}{2.1 \times 10^{-4} + 9 \times 10^{-5}}$$

Give your answer in standard form.

.....

(Total for Question 8 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q9

CH 1 • TOPIC 4 3 marks

9 (a) Write 8.4×10^{-5} as an ordinary number.

.....

(b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....

(Total for Question 9 is 3 marks)**Q32****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q14

CH 1 • TOPIC 4 2 marks

14 Work out $6.7 \times 10^{135} + 3 \times 10^{134}$
Give your answer in standard form.

.....

(Total for Question 14 is 2 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q8

CH 1 • TOPIC 4 3 marks

8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(Total for Question 8 is 3 marks)

04 Tables and Real Data

The numbers are in a table: read the right row, then do exactly the operation the words ask for.

WHEN YOU SEE → YOU DO

“the difference between”

→ bigger – smaller

“ k times the surface area”, “how many times further”

→ divide

“per person”, “bacteria per raindrop”

→ divide by the number of people / drops

“correct to the nearest whole number / 2 s.f.”

→ round only at the end

WORKED MODEL • JANUARY 2020 P2HR

Pacific $1.56 \times 10^8 \text{ km}^2$, Arctic $1.41 \times 10^7 \text{ km}^2$. Pacific = $k \times$ Arctic. Find k to the nearest whole number.

$$1 \quad k = \frac{1.56 \times 10^8}{1.41 \times 10^7}$$

“times” → divide

$$2 \quad k = 11.06 \dots$$

calculator

ANSWER $k = 11$

IDEA 04

Practise: Tables and Real Data

My questions first, then selected practice, then real past papers.

Q34 **DR ESLAM** How many times

CH 1 • TOPIC 4 **2 marks**

The Earth's mass is 5.97×10^{24} kg; the Moon's is 7.35×10^{22} kg. How many times heavier is the Earth? (1 d.p.)

SELECTED PRACTICE

Q35**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

6 (a) The table shows some information about eight planets.

Planet	Distance from Earth (km)	Mass (kg)
Earth	0	5.97×10^{24}
Jupiter	6.29×10^8	1.898×10^{27}
Mars	7.83×10^7	6.42×10^{23}
Mercury	9.17×10^7	3.302×10^{23}
Neptune	4.35×10^9	1.024×10^{26}
Saturn	1.28×10^9	5.68×10^{26}
Uranus	2.72×10^9	8.683×10^{25}
Venus	4.14×10^7	4.869×10^{24}

Write down the name of the planet with the greatest mass.

(1 mark)

(b) Find the difference between the mass of Venus and the mass of Mercury.

(1 mark)

(c) Nishat says that Neptune is over a hundred times further away from Earth than Venus is.

Is Nishat right?

You must show how you get your answer.

(2 marks)

Q36**PRACTICE**

Practice • Medium

CH 1 • TOPIC 4 4 marks

- 17 (a)** The table shows the population, correct to two significant figures, of each of six countries in April 2016.

Country	Population (April 2016)
Hungary	9.8×10^6
Mexico	1.3×10^8
Thailand	6.8×10^7
Nigeria	1.9×10^8
Singapore	5.7×10^6
Egypt	9.3×10^7

Write 9.3×10^7 as an ordinary number.

(1 mark)

- (b)** Which of these countries had the least population?

(1 mark)

- (c)** The population of China was 1.382×10^9 in April 2016.
The population of India was 1.327×10^9 in April 2016.

Work out the difference between the population of China and the population of India in April 2016.

Give your answer in standard form.

(2 marks)

PAST PAPERS

Q37 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 2HR • Q10

CH 1 • TOPIC 4 3 marks

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

- (a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean.
Give your answer in standard form.

..... square kilometres

The surface area of the Pacific Ocean is k times the surface area of the Arctic Ocean.

- (b) Work out the value of k .
Give your answer correct to the nearest whole number.

$k =$

(Total for Question 10 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q8

CH 1 • TOPIC 4 3 marks

- 8 The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

- (a) Which ocean has the greatest length of coastline?

.....

- (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline.
Give your answer in standard form.

..... km

(Total for Question 8 is 3 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q8

CH 1 • TOPIC 4 4 marks

8 The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

.....

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

- (b) work out the value of k .
Give your answer correct to the nearest whole number.

 $k =$

(Total for Question 8 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q9

CH 1 • TOPIC 4 5 marks

- 9 A rainwater tank contains 2.4×10^7 raindrops.

The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.

Give your answer in standard form correct to 2 significant figures.

.....

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.

Give your answer in standard form correct to one significant figure.

..... molecules

(Total for Question 9 is 5 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q10

CH 1 • TOPIC 4 3 marks

- 10 The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

Country	Total population	Total spent on healthcare (\$)
Austria	8.7×10^6	4.2×10^{10}
Luxembourg	6.3×10^5	3.7×10^9

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.

Give your answer correct to the nearest whole number.

..... dollars

(Total for Question 10 is 3 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q9

CH 1 • TOPIC 4 4 marks

- 9 The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

- (a) Write 8.8×10^6 as an ordinary number.
.....
- (b) Which of these cities had the least population in 2018?
.....
- (c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018
Give your answer in standard form correct to 2 significant figures.
.....

(Total for Question 9 is 4 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q9

CH 1 • TOPIC 4 3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes

(Total for Question 9 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 0

$$a^0 = 1$$

always

ZONE -

$$a^{-n} = \frac{1}{a^n}$$

flip

ZONE $\frac{m}{n}$

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m$$

root, then power

- 01 Powers of numbers** Zero $\rightarrow$ 1, negative $\rightarrow$ flip, fraction $\rightarrow$ root then power.
- 02 Convert** $A \times 10^n$, $1 \leq A < 10$; big $\rightarrow +n$, small $\rightarrow -n$.
- 03 Calculate** Numbers with numbers, powers with powers, then fix A.
- 04 Tables** Difference = subtract; “times” / “per” = divide; round last.

BOTTOM LINE Standard form always ends with $1 \leq A < 10 \rightarrow$ check it on every last line.

Answers

Powers, Roots and Standard Form • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • POWERS AND ROOTS OF NUMBERS

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam • Three rules | (a) $\frac{1}{8}$ (b) 9 (c) $\frac{1}{4}$ |
| Q2 | Practice • Easy | $-5, 5^{-1}, 0.5, 5^0$ |
| Q3 | Practice • Easy | a^2 |
| Q4 | Practice • Hard | (a) $\frac{1}{2^3}$ (b) $k = \frac{3}{2}$ |
| Q5 | Practice • Hard | $\frac{3^{\frac{4}{3}}}{3} = 3^{\frac{1}{3}}$ |
| Q6 | Edexcel 4MA1 • May 2023 • Paper 1H • Q15 | (a) $n = -\frac{5}{2}$ (b) 1.8×10^{156} |

IDEA 02 • STANDARD FORM: CONVERT

- Q7** Dr Eslam • Both directions (a) 3.05×10^{-3} (b) 460 000
- Q8** Practice • Easy (a) 0.000 78 (b) 9.56×10^7
- Q9** Practice • Easy (a) 5.62×10^{-3} (b) 1452
- Q10** Practice • Medium (a) 1 (b) 0.01 (c) 0.00273, 27.3×10^{-3} , 2.73×10^3 , 273×10^2
- Q11** Practice • Hard (a) 820 000 (b) 3.76×10^{-4} (c) 5×10^8
- Q12** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q5 (a) 0.0057 (b) 8×10^5 (c) 4.55×10^6
- Q13** Edexcel 4MA1 • January 2021 • Paper 1H • Q6 (a) 2.84×10^9 (b) 0.000 25
- Q14** Edexcel 4MA1 • January 2022 • Paper 1HR • Q8 (a) 50 000 (b) 6×10^{-5} (c) 2.5×10^{12}
- Q15** Edexcel 4MA1 • June 2022 • Paper 1H • Q8 (a) 8.9×10^{-5} (b) 83 400
- Q16** Edexcel 4MA1 • June 2023 • Paper 2H • Q6 (a) 7.6×10^7 (b) 0.000 54
- Q17** Edexcel 4MA1 • November 2023 • Paper 1H • Q9 (a) 0.000 587 (b) 8.4×10^7 (c) $K = 57.8$
- Q18** Edexcel 4MA1 • June 2024 • Paper 2HR • Q9 (a) 604 000 (b) 7×10^{-5} (c) 2×10^5
- Q19** Edexcel 4MA1 • June 2025 • Paper 1H • Q8 (a) 5.2×10^8 (b) 0.000 087 9 (c) 3.5×10^{-137}

IDEA 03 • STANDARD FORM: CALCULATE

Q20	Dr Eslam • Fix A	$12 \times 10^9 = 1.2 \times 10^{10}$
Q21	Dr Eslam • Divide	3×10^{11}
Q22	Practice • Easy	1.875×10^8
Q23	Practice • Medium	$500 \times 0.009 = 4.5 \text{ cm} > 4 \text{ cm}$: No
Q24	Practice • Medium	(a) 42 (b) 4.2336×10^9 (c)(i) 6450 (ii) every machine works at the same rate all day
Q25	Practice • Hard	7×10^8
Q26	Edexcel 4MA1 • January 2020 • Paper 1H • Q6	(a) 0.000 78 (b) 2.25×10^7
Q27	Edexcel 4MA1 • January 2021 • Paper 2H • Q6	(a) 1 (b) 3.2×10^{125}
Q28	Edexcel 4MA1 • January 2022 • Paper 1H • Q8	1.26×10^{122}
Q29	Edexcel 4MA1 • January 2022 • Paper 2H • Q12	2.16×10^{122}
Q30	Edexcel 4MA1 • May 2023 • Paper 1HR • Q8	(a) 0.0056 (b) 2×10^9
Q31	Edexcel 4MA1 • November 2024 • Paper 1H • Q9	(a) 0.000 084 (b) 5.2×10^{146}
Q32	Edexcel 4MA1 • June 2025 • Paper 1HR • Q14	7×10^{135}
Q33	Edexcel 4MA1 • November 2025 • Paper 2H • Q8	(a) 57 600 (b) 7.7×10^6

IDEA 04 • TABLES AND REAL DATA

- Q34** Dr Eslam • How many times 81.2
- Q35** Practice • Medium (a) Jupiter (b) 4.5388×10^{24} kg (c) $4.35 \times 10^9 \div 4.14 \times 10^7 = 105$: yes
- Q36** Practice • Medium (a) 93 000 000 (b) Singapore (c) 5.5×10^7
- Q37** Edexcel 4MA1 • January 2020 • Paper 2HR • Q10 (a) 3.74×10^7 (b) $k = 11$
- Q38** Edexcel 4MA1 • January 2021 • Paper 2HR • Q8 (a) Pacific (b) 9.393×10^4
- Q39** Edexcel 4MA1 • May 2021 • Paper 1H • Q8 (a) 1.318×10^9 (b) $k = 11$
- Q40** Edexcel 4MA1 • November 2021 • Paper 2H • Q9 (a) 0.073 (b) 4×10^{28}
- Q41** Edexcel 4MA1 • January 2022 • Paper 2HR • Q10 \$1045
- Q42** Edexcel 4MA1 • June 2022 • Paper 2HR • Q9 (a) 8 800 000 (b) Barcelona (c) 2.9×10^7
- Q43** Edexcel 4MA1 • May 2024 • Paper 1H • Q9 (a) 35 000 000 (b) 7.6×10^6 tonnes

Fractions

Mixed numbers to top-heavy fractions first — then add with a common denominator, multiply straight across, and divide with KFC.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Fractions

01 — Fraction basics

$\frac{7}{11}$ of 198 kg

£1.50 as a fraction of 60p

lowest terms

half way between

02 — Add and subtract

$4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

$5\frac{1}{3} - 2\frac{6}{7}$

“as a mixed number”

03 — Multiply and divide

$3\frac{1}{5} \times 2\frac{5}{8}$

$6\frac{3}{4} \div 2\frac{4}{7}$

“show that”

04 — Fraction problems

rent, travel, \$420 left

$\frac{7}{20}$ of the bricks

ratio of students

WHICH MOVE? Every calculation starts the same way — mixed numbers become top-heavy:

ZONE + −

Common denominator

same bottom, then add / subtract the tops

ZONE ×

Straight across

top × top, bottom × bottom (cancel first)

ZONE ÷

KFC

Keep – Flip – Change to ×

01 Fraction Basics

“Of” means multiply. “As a fraction of” means the first amount on top — in the **same units**. Then cancel to lowest terms.

WHEN YOU SEE → YOU DO

“ $\frac{7}{11}$ of 198 kg”

→ $198 \div 11 \times 7$

“£1.50 as a fraction of 60p”

→ same units: $\frac{150}{60} = \frac{5}{2}$

“in its lowest terms / simplest form”

→ divide top and bottom by the HCF

“half way between two fractions”

→ add them, divide by 2

WORKED MODEL • FRACTION OF AN AMOUNT

Work out $\frac{7}{11}$ of 198 kg.

1 $198 \div 11 = 18$

one eleventh

2 18×7

seven elevenths

ANSWER 126 kg

× **TRAP** £1.50 and 60p are different units: $\frac{1.50}{60}$ is wrong. Change both to pence first.

IDEA 01

Practise: Fraction Basics

My questions first, then selected practice.

Q1 **DR ESLAM** Of an amount

CH 1 • TOPIC 5 2 marks

Work out $\frac{3}{8}$ of 240 kg.**SELECTED PRACTICE****Q2** **PRACTICE** Practice • Easy

CH 1 • TOPIC 5 1 mark

3 Work out £1.50 as a fraction of 60p

Circle your answer.

$\frac{2}{5}$	$\frac{1}{4}$	$\frac{4}{1}$	$\frac{5}{2}$
---------------	---------------	---------------	---------------

(1 mark)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 5**1 mark**

4 Circle the fraction that is equivalent to 3.875

$\frac{15}{4}$	$\frac{29}{8}$	$\frac{31}{8}$	$\frac{15}{8}$
----------------	----------------	----------------	----------------

(1 mark)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 5**2 marks**

5 Write \$24.60 as a fraction of \$2870.

Give your answer in its lowest terms.

(2 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 1 mark

- 11 Circle the number half way between $\frac{7}{12}$ and $\frac{3}{4}$

$\frac{7}{32}$	$\frac{5}{8}$	$\frac{2}{3}$	$\frac{1}{2}$
----------------	---------------	---------------	---------------

(1 mark)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 3 marks

- 13** A unit fraction has a numerator equal to 1, for example $\frac{1}{3}$, $\frac{1}{7}$ and $\frac{1}{25}$.

Unit fractions can be written as the sum of two different unit fractions, for example

$$\frac{1}{2} = \frac{1}{3} + \frac{1}{6}$$

Write each of the following unit fractions as the sum of two **different** unit fractions.

$$\frac{1}{4} = \frac{1}{\square} + \frac{1}{\square}$$

$$\frac{1}{5} = \frac{1}{\square} + \frac{1}{\square}$$

$$\frac{1}{6} = \frac{1}{\square} + \frac{1}{\square}$$

(3 marks)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5**1 mark**

17 Giulio's reaction times are measured in two games.

In the first game his reaction time is $\frac{1}{3}$ of a second.

In the second game his reaction time is $\frac{1}{8}$ of a second.

Find the difference between the two reaction times.

..... S

(1 mark)

02 Add and Subtract

Top-heavy first, then one common denominator. Multiply each top by what its bottom was multiplied by.

WHEN YOU SEE → YOU DO

“Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$ ”

→ write both top-heavy, common bottom, convert back

“without a calculator ... simplest form”

→ show every line; cancel at the end

“as a mixed number”

→ divide: top ÷ bottom, remainder over the bottom

subtraction where the first fraction is smaller

→ top-heavy avoids the borrowing problem

Top-heavy

Common denominator

Add / subtract tops

Back to mixed

WORKED MODEL • JUNE 2022 P2H

Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$.

1 $\frac{16}{3} - \frac{20}{7}$

top-heavy

2 $\frac{112}{21} - \frac{60}{21}$

common denominator 21

3 $\frac{52}{21}$

subtract the tops

ANSWER $2\frac{10}{21}$

× **TRAP** Adding the bottoms: $\frac{2}{3} + \frac{4}{5} \neq \frac{6}{8}$. Only the tops add, once the bottoms are the same.

IDEA 02

Practise: Add and Subtract

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Add mixed numbers

CH 1 • TOPIC 5 3 marks

Show that $2\frac{1}{3} + 1\frac{3}{4} = 4\frac{1}{12}$.

Q9 DR ESLAM Subtract

CH 1 • TOPIC 5 3 marks

Work out $3\frac{2}{5} - 1\frac{3}{4}$. Give your answer as a mixed number.

SELECTED PRACTICE

Q10

PRACTICE

Practice • Easy

CH 1 • TOPIC 5 3 marks

1 (a) Show that $\frac{1}{7} \times \frac{2}{3}$ is equal to $\frac{2}{21}$

(1 mark)

(b) Show that $\frac{3}{5} - \frac{1}{3}$ is equal to $\frac{4}{15}$

(2 marks)

Q11

PRACTICE

Practice • Medium

CH 1 • TOPIC 5 3 marks

16 Without using a calculator, work out $\frac{5}{6} + \frac{2}{3}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

Q12**PRACTICE**

Practice · Hard

CH 1 • TOPIC 5**3 marks**

4 Show that $5\frac{2}{3} - 2\frac{3}{4} = 2\frac{11}{12}$

(3 marks)**Q13****PRACTICE**

Practice · Hard

CH 1 • TOPIC 5**3 marks**

15 **Without using a calculator**, workout $3\frac{5}{8} - 1\frac{2}{3}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

PAST PAPERS

Q14 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 2H • Q3**CH 1 • TOPIC 5** **3 marks**

3 Show that $4\frac{2}{3} + 3\frac{4}{5} = 8\frac{7}{15}$

(Total for Question 3 is 3 marks)

Q15 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 1HR • Q6**CH 1 • TOPIC 5** **3 marks**

6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

(Total for Question 6 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q4

CH 1 • TOPIC 5 3 marks

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 4 is 3 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q3

CH 1 • TOPIC 5 3 marks

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q5

CH 1 • TOPIC 5 3 marks

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

03 Multiply and Divide

Top-heavy first. Multiply: straight across (cancel first to keep numbers small). Divide: **KFC** — Keep the first, Flip the second, Change $\div$ to $\times$.

WHEN YOU SEE → YOU DO

" $3\frac{1}{5} \times 2\frac{5}{8}$ "

→ $\frac{16}{5} \times \frac{21}{8}$, cancel 16 with 8

" $\div$ "

→ Keep – Flip – Change

"You must write down all the stages"

→ top-heavy line, KFC line, answer line

a whole number in the answer

→ $\frac{56}{7} = 8$

Top-heavy

KFC if $\div$

Cancel

Multiply across

Mixed number

WORKED MODEL • JANUARY 2022 P2H

Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$.

1 $\frac{27}{4} \div \frac{18}{7}$

top-heavy

2 $\frac{27}{4} \times \frac{7}{18}$

KFC

3 $\frac{3 \times 7}{4 \times 2} = \frac{21}{8}$

cancel 27 and 18 by 9

ANSWER $2\frac{5}{8}$

× **TRAP** $3\frac{1}{5} \times 2\frac{5}{8}$ is not $6 + \frac{1}{5} \times \frac{5}{8}$. Never multiply the whole numbers and the fractions separately.

IDEA 03

Practise: Multiply and Divide

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM KFC

CH 1 • TOPIC 5 3 marks

Show that $2\frac{1}{4} \div 1\frac{1}{2} = 1\frac{1}{2}$.

SELECTED PRACTICE**Q20** PRACTICE Practice • Medium

CH 1 • TOPIC 5 6 marks

1 (a) Show that $1\frac{1}{5} \times 2\frac{1}{3} = 2\frac{4}{5}$

(3 marks)

(b) Show that $2\frac{7}{15} - 1\frac{2}{3} = \frac{4}{5}$

(3 marks)

Q21 **PRACTICE** Practice • Medium

CH 1 • TOPIC 5 2 marks

20 Without using a calculator, work out $\frac{1}{4} \div \frac{2}{3}$.

You must show all your working and give your answer as a fraction.

(2 marks)

Q22 **PRACTICE** Practice • Hard

CH 1 • TOPIC 5 2 marks

14 Without using a calculator, work out $\frac{5}{16} \times 1\frac{1}{7}$.

You must show all your working and give your answer as a fraction in its simplest form.

(2 marks)

Q23**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5**3 marks**

20 Without using a calculator, work out $\frac{2}{3} \div 1\frac{1}{5}$.

You must show all your working and give your answer as a fraction in its simplest form.

(3 marks)

PAST PAPERS

Q24**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q2

CH 1 • TOPIC 5**3 marks**

2 Show that $3\frac{1}{5} \times 2\frac{5}{8} = 8\frac{2}{5}$

(Total for Question 2 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $3\frac{3}{4} \times \frac{7}{9} = 2\frac{11}{12}$

(Total for Question 1 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

(Total for Question 1 is 3 marks)

Q27 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 2H • Q2**CH 1 • TOPIC 5** **3 marks**

2 Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

(Total for Question 2 is 3 marks)

Q28 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 2H • Q2**CH 1 • TOPIC 5** **3 marks**

2 Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

(Total for Question 2 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

(Total for Question 1 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q6

CH 1 • TOPIC 5 3 marks

6 Show that $3\frac{3}{7} \div 2\frac{2}{3} = 1\frac{2}{7}$

(Total for Question 6 is 3 marks)

Q31 **PAST PAPER** Edexcel 4MA1 • June 2024 • Paper 2H • Q4**CH 1 • TOPIC 5** **3 marks**

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

(Total for Question 4 is 3 marks)

Q32 **PAST PAPER** Edexcel 4MA1 • May 2024 • Paper 1HR • Q4**CH 1 • TOPIC 5** **3 marks**

4 Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(Total for Question 4 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q1

CH 1 • TOPIC 5 3 marks

You must write down all the stages in your working.

1 Show that $1\frac{5}{7} \times 2\frac{3}{16} = 3\frac{3}{4}$

(Total for Question 1 is 3 marks)

Q34**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q3

CH 1 • TOPIC 5 3 marks

3 Show that $2\frac{1}{4} \times 1\frac{5}{7} = 3\frac{6}{7}$

(Total for Question 3 is 3 marks)

04 Fraction Problems

Fractions of a whole: add the fractions you know, the rest is 1 – that. Then match the rest to the amount you are given.

WHEN YOU SEE → YOU DO

“she spends $\frac{1}{3}$... $\frac{1}{5}$... and \$420”

→ \$420 is the fraction left: $1 - \frac{1}{3} - \frac{1}{5}$

“ $\frac{7}{20}$ of the bricks are yellow”

→ the others are $\frac{13}{20}$ – that matches the 52 you know

“ $\frac{2}{3}$ of the students ... $\frac{2}{5}$ of those”

→ “of those” → multiply

“ratio of ... to ...”

→ find both fractions, then write as a ratio

WORKED MODEL • NOVEMBER 2025 P2H

9 red, 43 blue, and yellow bricks; $\frac{7}{20}$ are yellow; each brick 35 g. Weight of the yellow bricks?

1 red + blue = 52 = $\frac{13}{20}$ of the bricks

the known part

2 $\frac{1}{20} = 4$ bricks, yellow = $7 \times 4 = 28$

one part

ANSWER $28 \times 35 = 980$ g

IDEA 04

Practise: Fraction Problems

My questions first, then selected practice, then real past papers.

Q35 DR ESLAM The rest

CH 1 • TOPIC 5 3 marks

Sam spends $\frac{1}{4}$ of his money on food and $\frac{2}{5}$ on rent. He has \$70 left. How much did he start with?

SELECTED PRACTICE**Q36** PRACTICE Practice • Medium

CH 1 • TOPIC 5 3 marks

12 In a school, $\frac{2}{3}$ of the students study a language.

Of those students who study a language, $\frac{2}{5}$ study Spanish.

Find the ratio of students who study Spanish to students who do not study Spanish.

(3 marks)

Q37**PRACTICE**

Practice • Medium

CH 1 • TOPIC 5 3 marks

- 21** Collette has \$136 and spends half of her money on clothes and $\frac{1}{5}$ of her money on books.

Calculate the amount she has left.

\$

(3 marks)

Q38**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5**4 marks**

5 Each month Edna spends all her income on rent, on travel and on other living expenses.

She spends $\frac{1}{3}$ of her income on rent.

She spends $\frac{1}{5}$ of her income on travel.

She spends \$420 of her income on other living expenses.

Work out her income each month.

(4 marks)

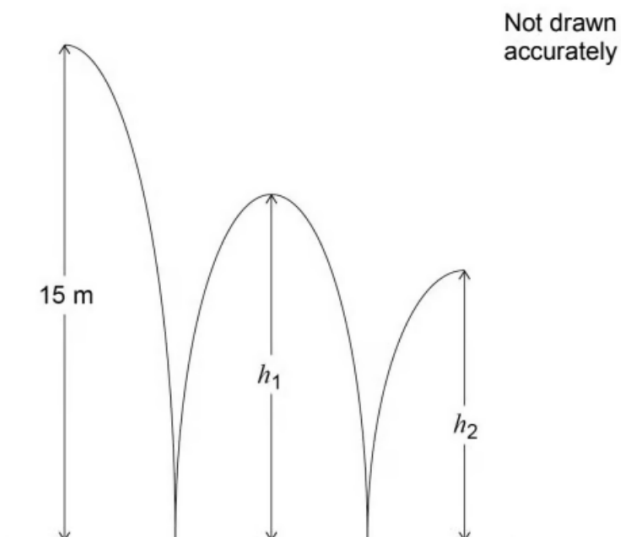
Q39**PRACTICE**

Practice • Hard

CH 1 • TOPIC 5 4 marks

6 (a) A ball is thrown from a height of 15 metres.

It bounces to height h_1 , then to height h_2 as shown.



h_1 is three quarters of the original height.

Jack expects h_2 to be three quarters of h_1

Work out the value of h_2 that he expects.

.....metres

(2 marks)

(b) In fact, h_2 is two thirds of h_1

How does this affect the answer to part (a)? Tick a box.

☐ The ball bounced higher than he expected

☐ The ball bounced lower than he expected

Show working to support your answer.

(2 marks)

PAST PAPERS

Q40**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q3

CH 1 • TOPIC 5 4 marks

3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE + –
same bottom
then the tops

ZONE ×
across
cancel first

ZONE ÷
KFC
Keep–Flip–Change

- | | |
|-----------------------------|---|
| 01 Basics | “Of” = $\times$. “As a fraction of”: same units, then cancel. |
| 02 Add / subtract | Top-heavy $\rightarrow$ common denominator $\rightarrow$ tops $\rightarrow$ mixed number. |
| 03 Multiply / divide | Top-heavy; $\times$ across; $\div$ KFC; cancel first. |
| 04 Problems | Known part = 1 – fractions; find one part, then the rest. |

BOTTOM LINE Mixed numbers $\rightarrow$ top-heavy, always $\rightarrow$ then the right move $\rightarrow$ back to a mixed number.

Answers

Fractions • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • FRACTION BASICS

- | | | |
|-----------|-------------------------|--|
| Q1 | Dr Eslam • Of an amount | $240 \div 8 \times 3 = 90 \text{ kg}$ |
| Q2 | Practice • Easy | $\frac{5}{2}$ |
| Q3 | Practice • Easy | $\frac{31}{8}$ |
| Q4 | Practice • Easy | $\frac{3}{350}$ |
| Q5 | Practice • Medium | $\frac{2}{3}$ |
| Q6 | Practice • Medium | $\frac{1}{4} = \frac{1}{5} + \frac{1}{20}, \frac{1}{5} = \frac{1}{6} + \frac{1}{30}, \frac{1}{6} = \frac{1}{7} + \frac{1}{42}$ |
| Q7 | Practice • Medium | $\frac{5}{24} \text{ s}$ |

IDEA 02 • ADD AND SUBTRACT

- Q8** Dr Eslam • Add mixed numbers $\frac{7}{3} + \frac{7}{4} = \frac{28}{12} + \frac{21}{12} = \frac{49}{12}$
- Q9** Dr Eslam • Subtract $\frac{68}{20} - \frac{35}{20} = \frac{33}{20} = 1\frac{13}{20}$
- Q10** Practice • Easy (a) $\frac{1 \times 2}{7 \times 3}$ (b) $\frac{9}{15} - \frac{5}{15}$
- Q11** Practice • Medium $1\frac{1}{2}$
- Q12** Practice • Hard $\frac{68}{12} - \frac{33}{12} = \frac{35}{12}$
- Q13** Practice • Hard $1\frac{23}{24}$
- Q14** Edexcel 4MA1 • January 2020 • Paper 2H • Q3 $\frac{70}{15} + \frac{57}{15} = \frac{127}{15}$
- Q15** Edexcel 4MA1 • June 2022 • Paper 1HR • Q6 $\frac{32}{12} + \frac{45}{12} = \frac{77}{12}$
- Q16** Edexcel 4MA1 • June 2022 • Paper 2H • Q4 $\frac{112}{21} - \frac{60}{21} = \frac{52}{21}$
- Q17** Edexcel 4MA1 • June 2025 • Paper 2H • Q3 $\frac{154}{21} - \frac{75}{21} = \frac{79}{21}$
- Q18** Edexcel 4MA1 • November 2025 • Paper 1H • Q5 $\frac{78}{21} + \frac{35}{21} = \frac{113}{21}$

IDEA 03 • MULTIPLY AND DIVIDE

Q19	Dr Eslam • KFC	$\frac{9}{4} \times \frac{2}{3} = \frac{18}{12} = \frac{3}{2}$
Q20	Practice • Medium	(a) $\frac{6}{5} \times \frac{7}{3} = \frac{14}{5}$ (b) $\frac{37}{15} - \frac{25}{15} = \frac{12}{15}$
Q21	Practice • Medium	$\frac{3}{8}$
Q22	Practice • Hard	$\frac{5}{14}$
Q23	Practice • Hard	$\frac{5}{9}$
Q24	Edexcel 4MA1 • January 2020 • Paper 1HR • Q2	$\frac{16}{5} \times \frac{21}{8} = \frac{42}{5}$
Q25	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q1	$\frac{15}{4} \times \frac{7}{9} = \frac{35}{12}$
Q26	Edexcel 4MA1 • January 2021 • Paper 1HR • Q1	$\frac{16}{5} \times \frac{11}{6} = \frac{88}{15}$
Q27	Edexcel 4MA1 • May 2021 • Paper 2H • Q2	$\frac{18}{7} \times \frac{8}{9} = \frac{16}{7}$
Q28	Edexcel 4MA1 • January 2022 • Paper 2H • Q2	$\frac{27}{4} \times \frac{7}{18} = \frac{21}{8}$
Q29	Edexcel 4MA1 • June 2023 • Paper 2H • Q1	$\frac{14}{3} \times \frac{5}{6} = \frac{35}{9}$
Q30	Edexcel 4MA1 • November 2023 • Paper 2H • Q6	$\frac{24}{7} \times \frac{3}{8} = \frac{9}{7}$
Q31	Edexcel 4MA1 • June 2024 • Paper 2H • Q4	$\frac{18}{7} \times \frac{28}{9} = 8$
Q32	Edexcel 4MA1 • May 2024 • Paper 1HR • Q4	$\frac{7}{3} \times \frac{4}{21} = \frac{4}{9}$
Q33	Edexcel 4MA1 • November 2024 • Paper 2H • Q1	$\frac{12}{7} \times \frac{35}{16} = \frac{15}{4}$
Q34	Edexcel 4MA1 • June 2025 • Paper 1HR • Q3	$\frac{9}{4} \times \frac{12}{7} = \frac{27}{7}$

IDEA 04 • FRACTION PROBLEMS

Q35 Dr Eslam • The rest

$$\text{left} = \frac{7}{20}: 70 \div 7 \times 20 = \$200$$

Q36 Practice • Medium

$$\frac{4}{15} : \frac{11}{15} = 4 : 11$$

Q37 Practice • Medium

\$40.80

Q38 Practice • Hard

$$\$420 \text{ is } \frac{7}{15}: \$900$$

Q39 Practice • Hard

(a) $8\frac{7}{16}$ m (b) $h_2 = 7.5$ m < $8\frac{7}{16}$: lower than expected

Q40 Edexcel 4MA1 • November 2025
• Paper 2H • Q3

80 bricks, 28 yellow: 980 g

Percentages

Turn every percentage into one multiplier. Forwards you multiply, backwards you divide, twice you multiply twice.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Percentages

01 — Percentage of an amount

145% of 1.80 m VAT at 20% commission, discount
is the target met?

02 — Percentage change and profit

% increase in population % profit which coupon saves more?

03 — Reverse percentages

sale price 20.40 after 15% off 15% more than March 17% of N is 357

04 — Repeated and combined changes

+15% then -8% "4 + 2.5 = 6.5" — is he right? $\times k$ each hour

05 — Percentages with fractions and ratio

mugs: $\frac{2}{5}$ and 32% bulbs 9:4:2 combined savings

THE MULTIPLIER Write the multiplier first — every question then becomes one calculation:

ZONE UP 12%

$\times 1.12$

100% + 12%

ZONE DOWN 15%

$\times 0.85$

100% - 15%

ZONE FORWARDS

old $\times m$

new value

ZONE BACKWARDS

new $\div m$

original value

01 Percentage of an Amount

Change the percentage to a decimal and multiply. Several parts? Work each one out, then add.

WHEN YOU SEE → YOU DO

“145% of 1.80 m”

→ 1.80×1.45

“plus VAT at 20%”

→ $\times 1.2$

“5% of the cost of all the car insurance he sells”

→ commission: $0.05 \times \text{total}$

“a target of 94%”

→ work out the actual %, then compare

“is it cheaper from ...”

→ work out both totals, then decide in words

WORKED MODEL • VAT AND PAYMENTS

A caravan costs £7000 plus VAT at 20%. Deposit £3000, the rest in 6 equal payments.

1 $7000 \times 1.2 = £8400$

add VAT

2 $8400 - 3000 = £5400$

after deposit

ANSWER $5400 \div 6 = £900$ a month

IDEA 01

Practise: Percentage of an Amount

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Multiplier

CH 1 • TOPIC 6 2 marks

A coat costs £85 plus VAT at 20%. Work out the total cost.

SELECTED PRACTICE

Q2 **PRACTICE** Practice • Easy

CH 1 • TOPIC 6 4 marks

2 Ria is going to buy a caravan.

The total cost of the caravan is £7000 **plus** VAT at 20%.

Ria pays a deposit of £3000 She pays the rest of the total cost in 6 equal monthly payments.

Work out the amount of each monthly payment.

(4 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 6 1 mark**11** Work out 40 as a percentage of 10

Circle your answer.

4%	25%	300%	400%
----	-----	------	------

(1 mark)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

4 Mr Watkins needs to buy some oil for his central heating.

Mr Watkins can put up to 1500 litres of oil in his oil tank.

There are already 850 litres of oil in the tank.

Mr Watkins is going to fill the tank with oil.

The price of oil is 67.2p per litre.

Mr Watkins gets 5% off the price of the oil.

How much does Mr Watkins pay for the oil he needs to buy?

(5 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

8 Bill buys and sells laptops.

Last month Bill bought 50 laptops.

He paid £400 for each laptop.

He sold

40 of these laptops at a profit of 30% on each laptop

10 of these laptops at a profit of 15% on each laptop

Bill's target last month was to sell all 50 laptops for a total of at least £25 000

Did Bill reach this target?

(5 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 • May 2024 • Paper 1HR • Q3

CH 1 • TOPIC 6 5 marks

- 3 Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.

He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.

60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£.....

(Total for Question 3 is 5 marks)

02 Percentage Change and Profit

Change over the **original**, times 100. Profit is the change from cost to selling price.

WHEN YOU SEE → YOU DO

“percentage increase / decrease”

$$\rightarrow \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

“percentage profit”

$$\rightarrow \frac{\text{selling} - \text{cost}}{\text{cost}} \times 100$$

“is Brigitte correct?” (two increases)

→ work out both %, compare

“which coupon saves more?”

→ work out each saving in money

Change = new – old

Divide by the ORIGINAL

× 100

ZONE %
change
original × 100

divide by where you **started**

ZONE PROFIT
total income – total cost

count **all** costs: delivery, materials, adhesive ...

WORKED MODEL • NOVEMBER 2021 P2H

Victor buys 12 bottles for \$21 and sells all 12 at \$2.45 each. Percentage profit?

1 $12 \times 2.45 = \$29.40$

income

2 $29.40 - 21 = \$8.40$

profit

ANSWER $\frac{8.40}{21} \times 100 = 40\%$

× **TRAP** Dividing by the new value: $\frac{8.40}{29.40}$ is wrong. The original (cost) always goes on the bottom.

IDEA 02

Practise: Percentage Change and Profit

My questions first, then selected practice, then real past papers.

Q7**DR ESLAM**

Percentage change

CH 1 • TOPIC 6

3 marks

A car's value fell from £12 500 to £9 800. Work out the percentage decrease.

SELECTED PRACTICE**Q8****PRACTICE**

Practice • Easy

CH 1 • TOPIC 6

3 marks

- 1 Bill's weight decreases from 64.8 kg to 59.3 kg.

Calculate the percentage decrease in Bill's weight.

Give your answer correct to 3 significant figures.

(3 marks)

Q9**PRACTICE**

Practice • Easy

CH 1 • TOPIC 6 3 marks

4 George wants to watch all 23 games that a football team will play at home next season.

He can buy

a season ticket costing £425

or 23 separate tickets costing £24 each ticket.

What percentage of the total cost of 23 separate tickets does George save by buying a season ticket?

(3 marks)**Q10****PRACTICE**

Practice • Easy

CH 1 • TOPIC 6 3 marks

13 In 1999 the minimum wage for adults was £3.60 per hour.

In 2013 it was £6.31 per hour.

Work out the percentage increase in the minimum wage.

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6

4 marks

- 9 Renee buys 5 kg of sweets to sell.
She pays £10 for the sweets.

Renee puts all the sweets into bags.
She puts 250 g of sweets into each bag.
She sells each bag of sweets for 65p.

Renee sells all the bags of sweets.

Work out her percentage profit.

(4 marks)

PAST PAPERS

Q12**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q8

CH 1 • TOPIC 6 7 marks

- 8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says,

“The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017”

- (a) Is Brigitte correct?

You must show how you get your answer.

Henri wants to insure his car.

He gets a discount of 15% off the normal price.

Henri pays 952 euros for his car insurance after the discount.

- (b) Work out the discount that Henri gets.

..... euros

(Total for Question 8 is 7 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q5

CH 1 • TOPIC 6 6 marks

- 5 In 2017, the population of a village was 7500
In 2019, the population of the village was 8265

(a) Work out the percentage increase in the population of the village from 2017 to 2019

..... %

In a sale, normal prices are reduced by 30%
The sale price of a T-shirt was 31.50 euros.

(b) Work out the normal price of the T-shirt.

..... euros

(Total for Question 5 is 6 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q3

CH 1 • TOPIC 6 6 marks

- 3 Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

.....%

Gladys has a sale in her shop.

She decreases all the normal prices by 12%

The normal price of an armchair was \$550

- (b) Work out the sale price of the armchair.

\$.....

(Total for Question 3 is 6 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q6

CH 1 • TOPIC 6**3 marks**

- 6 Victor buys 12 bottles of apple juice for a total cost of \$21
Victor sells all 12 bottles at \$2.45 each bottle.

Work out Victor's percentage profit.

..... %

(Total for Question 6 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q4

CH 1 • TOPIC 6**5 marks**

- 4 Sarah makes and sells mugs.

One day she makes 150 mugs.

Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	£8.50
Medium	£11.20
Large	£14.20

Sarah sells all 150 mugs.

Work out her percentage profit.

Give your answer correct to the nearest whole number.

.....%

(Total for Question 4 is 5 marks)

Q17**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

- 8 The table gives information about the average house price in England in 2018 and in 2019

Year	2017	2018	2019
Average house price (£)		228 314	231 776

- (a) Work out the percentage increase in the average house price from 2018 to 2019
Give your answer correct to one decimal place.

..... %

The average house price in 2019 was 7.7% greater than the average house price in 2017

- (b) Work out the average house price in 2017
Give your answer correct to 3 significant figures.

£

(Total for Question 8 is 5 marks)

Q18**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q2

CH 1 • TOPIC 6 7 marks

- 2 Mary saves for a holiday each year.

In 2020 she saved a total of \$720

In 2021, each month she saved \$78

The total amount Mary saved in 2021 was $P\%$ more than the total she saved in 2020

- (a) Work out the value of P

.....

Roberto is going to go on holiday.

He has two coupons that will save him money on his holiday.

Coupon A

18% off the cost of the
accommodation

Coupon B

12.5% off the total cost of the
accommodation **and** the flights

For Roberto's holiday

the cost of the accommodation is \$1600

the cost of the flights is \$800

Roberto can only use one of the coupons.

He wants to save as much money as he can.

- (b) Which of the two coupons, **A** or **B**, should he use?

Show your working clearly.

(Total for Question 2 is 7 marks)

Q19**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q6

CH 1 • TOPIC 6 7 marks

- 6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week.

In his new job, Pierre is paid 14 euros per hour.

In his new job, Pierre works for 7 hours each day for a 5-day week.

- (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week.

..... %

Marie changes her job.

Her salary decreases by 6%

Her new salary is 23 030 euros.

- (b) Work out Marie's salary before she changes her job.

..... euros

(Total for Question 6 is 7 marks)

Q20 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1H • Q3**CH 1 • TOPIC 6** **4 marks**

- 3 Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.

Give your answer correct to 3 significant figures.

..... %

(Total for Question 3 is 4 marks)

Q21**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 1H • Q3

CH 1 • TOPIC 6 4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

..... %

(Total for Question 3 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q2

CH 1 • TOPIC 6 6 marks

2 Anna makes cups.

Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.

Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

(a) Work out the percentage profit Anna makes.

Show your working clearly.

..... %

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g

(Total for Question 2 is 6 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q4

CH 1 • TOPIC 6**5 marks**

- 4 Joshua is going to cover a floor with tiles for a customer.
The area of the floor is 45 m^2

Joshua buys one box of tiles for each 1.5 m^2 of floor area.
Each box of tiles costs £64

Joshua also buys 5 bags of tile adhesive.
Each bag of tile adhesive costs £12

Joshua charges the customer £3000

Work out his percentage profit.
Give your answer correct to one decimal place.

..... %

(Total for Question 4 is 5 marks)

03 Reverse Percentages

You are given the value **after** the change. Write the multiplier, then **divide**. Never take the percentage of the new price.

WHEN YOU SEE → YOU DO

“sale price ... reduced by 15% ... find the normal price”

→ sale $\div$ 0.85

“943 is 15% more than March”

→ $943 \div 1.15$

“17% of N is 357”

→ $357 \div 0.17$

“5.48 million was 22% of the population”

→ $5.48 \div 0.22$

“by how much was the price reduced?”

→ original – sale

Multiplier

New $\div$ multiplier

Check forwards

WORKED MODEL • JANUARY 2020 P1HR

A hat is reduced by 15%. The sale price is 20.40 euros. Find the normal price.

1 multiplier $1 - 0.15 = 0.85$

write the multiplier

2 $20.40 \div 0.85$

backwards: divide

ANSWER 24 euros

× **TRAP** $20.40 \times 1.15 = 23.46$ is wrong. 15% of the sale price is not 15% of the normal price.

CALCULATOR CHECK Check forwards: $24 \times 0.85 = 20.40$ — correct

IDEA 03

Practise: Reverse Percentages

My questions first, then selected practice, then real past papers.

Q24 **DR ESLAM** BackwardsCH 1 • TOPIC 6 **3 marks**

After a 12% increase, a train ticket costs £39.20. What did it cost before?

SELECTED PRACTICE**Q25** **PRACTICE** Practice • EasyCH 1 • TOPIC 6 **3 marks**

14 Ed has a card shop.

Ed's profit on "Good Luck" cards in 2018 was £360. This was a decrease of 20% on his profit in 2017.

Work out Ed's profit on "Good Luck" cards in 2017.

£

(3 marks)

Q26**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 5 marks

- 12 (a)** Each year Wenford Hospital records how long patients wait to be treated in the Accident and Emergency department.

In 2015 patients waited 11% less time than in 2014

In 2015 the average time patients waited was 68 minutes.

Work out the average time patients waited in 2014

Give your answer to the nearest minute.

(3 marks)

- (b)** The hospital has a target to reduce the average time patients wait to be treated in the Accident and Emergency department to 60 minutes in 2016.

Work out the percentage decrease from 68 minutes to 60 minutes.

(2 marks)

Q27**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6 6 marks

- 17 (a)** Lijuan's salary is 180000 Hong Kong Dollars (HK\$).
She gets a salary increase of 3%

Work out Lijuan's salary after this increase.

(3 marks)

- (b)** In a sale, all normal prices are reduced by 15%
The sale price of a camera is HK\$6630

Work out the normal price of the camera.

(3 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6

4 marks

4 A shop has a sale.

Microwave ovens

$\frac{1}{3}$ off normal price

Combination ovens

40% off normal price

A microwave oven has a sale price of £90

A combination oven has a sale price of £84

Which of these ovens has the greater normal price?

You must show all your working.

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q7

CH 1 • TOPIC 6 3 marks

- 7 In a sale, the normal price of a hat is reduced by 15%
The sale price of the hat is 20.40 euros.

Work out the normal price of the hat.

..... euros

(Total for Question 7 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q9

CH 1 • TOPIC 6 3 marks

- 9 In a sale, normal prices are reduced by 20%
A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q6

CH 1 • TOPIC 6**3 marks**

- 6 On Wednesday, the price of 1 litre of petrol was £1.26
The price of petrol on Wednesday was 5% more than the price of petrol on the previous Monday.

Calculate the price of 30 litres of petrol on the previous Monday.

£.....

(Total for Question 6 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q8

CH 1 • TOPIC 6**3 marks**

- 8 In 2018, the population of Sydney was 5.48 million.
This was 22% of the total population of Australia.

Work out the total population of Australia in 2018
Give your answer correct to 3 significant figures.

..... million

(Total for Question 8 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q8

CH 1 • TOPIC 6 3 marks

- 8 A cinema increased the cost of an adult ticket by 12%

After the increase, the cost of an adult ticket was £18.20

Work out the cost of an adult ticket before the increase.

£.....

(Total for Question 8 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 1H • Q6**CH 1 • TOPIC 6** **4 marks**

- 6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket A $\frac{1}{6}$ off normal price

ticket B 20% off normal price

The sale price of ticket **A** is 140 euros.

The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

..... euros

(Total for Question 6 is 4 marks)

Q35**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q8

CH 1 • TOPIC 6**3 marks**

- 8 In a sale, all normal prices are reduced by 17%

The sale price of a fridge is 6225 rupees.

Work out the normal price of the fridge.

..... rupees

(Total for Question 8 is 3 marks)

Q36**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q4

CH 1 • TOPIC 6**3 marks**

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

(Total for Question 4 is 3 marks)

7 Shop A and Shop B have offers for buying the same type of laptop.

The normal price of the laptop in Shop A is different to the normal price of the laptop in Shop B

Shop A	Shop B
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

(Total for Question 7 is 4 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q4

CH 1 • TOPIC 6 5 marks

- 4 N is a number.
17% of N is 357

(a) Work out the value of N

$N =$

In 2019, the population of a village was 650
In 2020, the population of the village was 806

(b) Work out the percentage increase in the population.

..... %

(Total for Question 4 is 5 marks)

Q39**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q5

CH 1 • TOPIC 6 3 marks

- 5 In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.

Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q9

CH 1 • TOPIC 6 3 marks

- 9 In a sale, normal prices are reduced by 15%

The sale price of a dishwasher is 612 Swiss francs.

Work out the normal price of the dishwasher.

..... Swiss francs

(Total for Question 9 is 3 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q9

CH 1 • TOPIC 6 3 marks

- 9 Hari scored 140 points playing a game on Tuesday.

This was 12% more points than he scored playing the game on Monday.

Work out the number of points that Hari scored playing the game on Monday.

.....

(Total for Question 9 is 3 marks)

04 Repeated and Combined Changes

Two changes one after the other: multiply the multipliers. Percentages do **not** add.

WHEN YOU SEE → YOU DO

“increased by 15% and then decreased by 8%”

→ $1.15 \times 0.92 = 1.058$: up 5.8%

“4 + 2.5 = 6.5 ... is he right?”

→ $0.96 \times 0.975 = 0.936$: 6.4%, not 6.5%

“reduced by 20%, then by 30%”

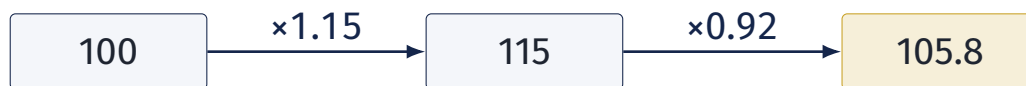
→ $0.8 \times 0.7 = 0.56$: 44% off, not 50%

“2% less every hour”, $V_{t+1} = kV_t$

→ $k = 0.98$

length -15%, width -10%

→ work out the new perimeter / area, then the %



overall multiplier $1.15 \times 0.92 = 1.058 \rightarrow$ up 5.8%, not 7%

× **TRAP** Adding or subtracting percentages ($15 - 8 = 7\%$). Each change acts on a different amount.

IDEA 04

Practise: Repeated and Combined Changes

My questions first, then selected practice.

Q42 DR ESLAM Twice

CH 1 • TOPIC 6 2 marks

A price rises by 10% and then falls by 10%. Find the overall percentage change.

SELECTED PRACTICE

Q43

PRACTICE

Practice • Easy

CH 1 • TOPIC 6 3 marks

12 The table shows information about the population of a city.

Population in 2001	Population in 2011
420 000	480 000

Liam claims,

“From 2011 to 2021 the population of the city will increase by the same percentage as from 2001 to 2011”

He works out,

$$\text{population increase from 2001 to 2011} = 480\,000 - 420\,000$$

$$= 60\,000$$

$$\text{population in 2021} = 480\,000 + 60\,000$$

$$= 540\,000$$

Does the population of 540 000 match his claim? You **must** show your working.

(3 marks)

Q44**PRACTICE**

Practice • Medium

CH 1 • TOPIC 6**2 marks**

15 In a shop, all normal prices are reduced by 20% to give the sale price.

The sale price of a TV set is then reduced by 30%.

Mary says,

" $30 + 20 = 50$, so this means that the normal price of the TV set has been reduced by 50% "

Is Mary right?

You must give a reason for your answer.

(2 marks)

Q45**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**5 marks**

2 (a) Ibrar bought a house for £145 000

The value of the house depreciated by 4% in the first year. The value of the house depreciated by 2.5% in the second year.

Ibrar says,

" $4 + 2.5 = 6.5$ so in two years the value of my house depreciated by 6.5%"

Is Ibrar right?

You must give a reason for your answer.

(2 marks)

(b) The value of Ibrar's house increases by $x\%$ in the third year.

At the end of the third year the value of Ibrar's house is £140 000.

Work out the value of x .

Give your answer correct to 3 significant figures.

(3 marks)

Q46**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 2 marks

- 9 During last season the cost of a ticket to watch Seapron United increased by 15% and then decreased by 8%.

Work out the overall percentage change in the cost of a ticket to watch Seapron United during last season.

(2 marks)**Q47****PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 2 marks

- 12 (a) Show that $a\%$ of $b = b\%$ of a

(1 mark)

- (b) Rosie says,

"160% of 40 = 140% of 60 because $a\%$ of $b = b\%$ of a "

Is she correct?

Tick a box.

☐

Yes

☐

No

Give a reason for your answer.

(1 mark)

Q48**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**6 marks**

- 18** In 2017, the value of a house increased by 4%.
In 2018, the value of the house then decreased by 3%.

Teresa says

Over the two years the value of the house increased by exactly 1% because $4 - 3 = 1$.

Show that Teresa is wrong.

(6 marks)

05 Percentages with Fractions and Ratio

Mixed questions: split the whole with the ratio or the fractions first, then take each percentage of its own part.

WHEN YOU SEE → YOU DO

“ratio 9 : 4 : 2 ... 45% of the tulips ...”

→ share the total first, then the % of each share

“ $\frac{3}{8}$... 45% ... and \$406”

→ the \$406 is what is left: $1 - \frac{3}{8} - 0.45$

“45% more than Abel”

→ Abel = a , Bahira = $1.45a$

“what fraction ... give in its simplest form”

→ add the parts, cancel

WORKED MODEL • JUNE 2024 P2H

600 bulbs, tulip : crocus : daffodil = 9 : 4 : 2. Yellow: 45% of tulips, $\frac{5}{8}$ of crocus, all daffodils.

1 $600 \div 15 = 40$: 360, 160, 80

share by the ratio

2 $0.45 \times 360 = 162$, $\frac{5}{8} \times 160 = 100$, 80

each part

ANSWER $162 + 100 + 80 = 342$ yellow

IDEA 05

Practise: Percentages with Fractions and Ratio

My questions first, then selected practice, then real past papers.

Q49**DR ESLAM**

Split first

CH 1 • TOPIC 6 3 marks

80 students: boys : girls = 3 : 5. 40% of the boys and 60% of the girls walk to school. How many walk?

SELECTED PRACTICE**Q50****PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 3 marks

- 6 Marie invests £8000 in an account for one year.
At the end of the year, interest is added to her account.

Marie pays tax on this interest at a rate of 20%
She pays £28.80 tax.

Work out the percentage interest rate for the account.

(3 marks)

Q51**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6**4 marks**

13 When you earn money you pay income tax.

The amount you pay depends on how much you earn that year. You pay

0% on the first £12500 you earn

20% on the next £37 500 you earn

40% on the next £112 500 you earn.

One year, Kim paid £9260 income tax.

Work out how much she earned that year.

£.....

(4 marks)

Q52

PRACTICE

Practice • Hard

CH 1 • TOPIC 66 marks

PRACTISE • 1.6

15 Lily buys and sells microwaves.

She buys each one for £32 and sells it for £60.
She also pays £7 for the delivery of each microwave she sells.

If she sells a microwave that is faulty then Lily must pay for its repair and redelivery.
This costs her another £25 for each faulty microwave.

Last month, 6 out of the 80 microwaves Lily sold were faulty.

This month she has orders for 133 microwaves.
Calculate her expected percentage profit on this month’s order. Showing your working
in the boxes below may help you present your work.

Expected number of faulty microwaves:	Expected costs:
Income from sales:	Expected percentage profit:

.....%

(6 marks)

Q53**PRACTICE**

Practice • Hard

CH 1 • TOPIC 6 6 marks

19 In a village the ratio of males to females is 2:1.

40% of the people in the village are right-handed males.

25% of the people in the village are right-handed females.

Show that the proportion of females who are right-handed is greater than the proportion of males who are right-handed.

(6 marks)

PAST PAPERS

Q54 **PAST PAPER** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q10**CH 1 • TOPIC 6** **5 marks****10** Alex makes 80 cakes to sell.

He makes chocolate cakes, lemon cakes and fruit cakes where

$$\begin{array}{ccccc} \text{number of} & : & \text{number of} & : & \text{number of} \\ \text{chocolate cakes} & & \text{lemon cakes} & & \text{fruit cakes} \end{array} = 3 : 2 : 5$$

Alex sells

all of the chocolate cakes

 $\frac{3}{4}$ of the lemon cakes $\frac{7}{8}$ of the fruit cakes

The profit he makes on each cake he sells is shown in the table.

Type of cake	Profit per cake he sells
chocolate	£2.00
lemon	£1.70
fruit	£2.40

Work out the total profit that Alex makes from the cakes he sells.

£.....

(Total for Question 10 is 5 marks)

Q55**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q10

CH 1 • TOPIC 6 3 marks

10 The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

.....
(Total for Question 10 is 3 marks)

Q56**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q6

CH 1 • TOPIC 6 4 marks

- 6 Mariana sells bags of bird food.

The bags that Mariana sold last week each contained 12 kg of seeds.

The bags that she is going to sell next week will each contain a mixture of nuts and seeds where for each bag

$$\text{weight of nuts : weight of seeds} = 4 : 5$$

The total weight of the nuts and the seeds in each bag will be 19.35 kg

The weight of seeds in each bag that Mariana sells next week will be less than the weight of seeds in each bag that Mariana sold last week.

Work out this decrease as a percentage of the weight of seeds in each bag that Mariana sold last week.

Give your answer correct to one decimal place.

..... %

(Total for Question 6 is 4 marks)

Q57**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q4

CH 1 • TOPIC 6**4 marks**

- 4 The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.

45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

.....%

(Total for Question 4 is 4 marks)

Q58**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q13

CH 1 • TOPIC 6**5 marks****13** Antoine is going on holiday.

He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	Amount paid
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	\$406

Work out how much Antoine has to pay for Payment 2

\$

(Total for Question 13 is 5 marks)

Q59**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

8 Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7:5:3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

.....
(Total for Question 8 is 5 marks)

Q60**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q8

CH 1 • TOPIC 6 5 marks

- 8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

(Total for Question 8 is 5 marks)

Q61**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q14

CH 1 • TOPIC 6**5 marks**

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

(Total for Question 14 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE UP

$$\times(1 + r)$$

ZONE DOWN

$$\times(1 - r)$$

ZONE FORWARDS

$$\times m$$

ZONE BACKWARDS

$$\div m$$

- | | | |
|-----------|------------------------|--|
| 01 | Of an amount | Decimal $\times$ amount; VAT $\times 1.2$. |
| 02 | Change / profit | $\frac{\text{change}}{\text{original}} \times 100$ — original on the bottom. |
| 03 | Reverse | New $\div$ multiplier. Never a % of the new value. |
| 04 | Repeated | Multiply the multipliers. Percentages do not add. |
| 05 | Mixed | Split the whole first, then each %. |

BOTTOM LINE Multiplier first $\rightarrow$ forwards $\times \rightarrow$ backwards $\div \rightarrow$ twice: multiply them.

Answers

Percentages · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PERCENTAGE OF AN AMOUNT

- | | | |
|-----------|--|--|
| Q1 | Dr Eslam · Multiplier | $85 \times 1.2 = \text{£}102$ |
| Q2 | Practice · Easy | $\text{£}900$ |
| Q3 | Practice · Easy | 400% |
| Q4 | Practice · Medium | $650 \text{ L: } \text{£}414.96$ |
| Q5 | Practice · Medium | $20\,800 + 4\,600 = \text{£}25\,400$: Yes |
| Q6 | Edexcel 4MA1 · May 2024 · Paper 1HR · Q3 | $120 \times 0.60 + 80 \times 1.05 = \text{£}156$ |

IDEA 02 • PERCENTAGE CHANGE AND PROFIT

Q7	Dr Eslam • Percentage change	$\frac{2700}{12500} \times 100 = 21.6\%$
Q8	Practice • Easy	8.49%
Q9	Practice • Easy	$552 - 425 = 127$: 23.0%
Q10	Practice • Easy	75.3%
Q11	Practice • Medium	30%
Q12	Edexcel 4MA1 • January 2020 • Paper 2H • Q8	(a) 9% then 8.6%: No (b) 168 euros
Q13	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q5	(a) 10.2% (b) 45 euros
Q14	Edexcel 4MA1 • January 2021 • Paper 2H • Q3	(a) 11.8% (b) \$484
Q15	Edexcel 4MA1 • November 2021 • Paper 2H • Q6	40%
Q16	Edexcel 4MA1 • January 2022 • Paper 1H • Q4	income £1644: 44%
Q17	Edexcel 4MA1 • January 2022 • Paper 2H • Q8	(a) 1.5% (b) £215 000
Q18	Edexcel 4MA1 • January 2022 • Paper 2HR • Q2	(a) $P = 30$ (b) A saves \$288, B saves \$300: B
Q19	Edexcel 4MA1 • June 2023 • Paper 2HR • Q6	(a) 22.5% (b) 24 500 euros
Q20	Edexcel 4MA1 • May 2023 • Paper 1H • Q3	62.3%
Q21	Edexcel 4MA1 • November 2023 • Paper 1H • Q3	41.6%
Q22	Edexcel 4MA1 • June 2025 • Paper 1H • Q2	(a) 12.5% (b) 8.5 cm (c) 125 g
Q23	Edexcel 4MA1 • June 2025 • Paper 2HR • Q4	cost £1980: 51.5%

IDEA 03 • REVERSE PERCENTAGES

		39.20 ÷ 1.12 = £35
Q24	Dr Eslam • Backwards	
		360 ÷ 0.8 = £450
Q25	Practice • Easy	
		(a) 76 minutes (b) 11.8%
Q26	Practice • Medium	
		(a) HK\$185 400 (b) HK\$7800
Q27	Practice • Medium	
		microwave £135, combination £140: the
Q28	Practice • Hard	combination oven
		24 euros
Q29	Edexcel 4MA1 • January 2020 • Paper 1HR • Q7	
		1350 rupees
Q30	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q9	
		£36
Q31	Edexcel 4MA1 • January 2021 • Paper 2HR • Q6	
		24.9 million
Q32	Edexcel 4MA1 • November 2021 • Paper 1H • Q8	
		£16.25
Q33	Edexcel 4MA1 • June 2022 • Pa- per 2HR • Q8	
		A 168, B 170: 2 euros
Q34	Edexcel 4MA1 • November 2023 • Paper 1H • Q6	
		7500 rupees
Q35	Edexcel 4MA1 • June 2024 • Pa- per 2HR • Q8	
		820 km
Q36	Edexcel 4MA1 • May 2024 • Pa- per 1H • Q4	
		A £76 off, B £72 off: Shop A
Q37	Edexcel 4MA1 • November 2024 • Paper 1H • Q7	
		(a) 2100 (b) 24%
Q38	Edexcel 4MA1 • November 2024 • Paper 2H • Q4	
		275 euros
Q39	Edexcel 4MA1 • June 2025 • Pa- per 1H • Q5	
		720 Swiss francs
Q40	Edexcel 4MA1 • June 2025 • Pa- per 2HR • Q9	

IDEA 04 • REPEATED AND COMBINED CHANGES**Q42** Dr Eslam • Twice

$$1.1 \times 0.9 = 0.99: 1\% \text{ decrease}$$

Q43 Practice • Easy

+14.3% gives about 548 571, not 540 000: No

Q44 Practice • Medium

No: $0.8 \times 0.7 = 0.56$, a 44% reduction

Q45 Practice • Hard

(a) No: $0.96 \times 0.975 = 0.936$, 6.4% (b) $x = 3.15$

Q46 Practice • Hard

5.8% increase

Q47 Practice • Hard

(a) $\frac{a}{100}b = \frac{b}{100}a$ (b) No: $64 \neq 84$

Q48 Practice • Hard

$$1.04 \times 0.97 = 1.0088: \text{up } 0.88\%, \text{ not } 1\%$$

IDEA 05 • PERCENTAGES WITH FRACTIONS AND RATIO

- Q49** Dr Eslam • Split first 30 boys, 50 girls: $12 + 30 = 42$
- Q50** Practice • Hard interest £144: 1.8%
- Q51** Practice • Hard £54 400
- Q52** Practice • Hard cost £5436.38, income £7980: 46.8%
- Q53** Practice • Hard females $\frac{0.25}{1/3} = 75\% > \text{males } \frac{0.40}{2/3} = 60\%$
- Q54** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q10 $48 + 20.40 + 84 = £152.40$
- Q55** Edexcel 4MA1 • January 2021 • Paper 2H • Q10 $\frac{12}{25}$
- Q56** Edexcel 4MA1 • May 2021 • Paper 2H • Q6 10.4%
- Q57** Edexcel 4MA1 • June 2022 • Paper 1HR • Q4 $\frac{84}{96} = 87.5\%$
- Q58** Edexcel 4MA1 • June 2022 • Paper 1H • Q13 total \$2320: \$1044
- Q59** Edexcel 4MA1 • June 2022 • Paper 2H • Q8 $\frac{23}{75}$
- Q60** Edexcel 4MA1 • June 2024 • Paper 2H • Q8 342
- Q61** Edexcel 4MA1 • May 2024 • Paper 1H • Q14 Abel 6300, Bahira 9135: Chanda 6090 dinars

Compound Interest and Depreciation



DR ESLAM AHMED
eliteigcse.com

One formula does it all: $\text{amount} = \text{start} \times \text{multiplier}^{\text{years}}$. Growth has a multiplier above 1, depreciation below 1.

THE TOPIC ON ONE MAP

Compound Interest and Depreciation

01 Compound growth

savings for 3 years

value increases 4% a year

"total interest"

02 Depreciation

car loses 15% a year

total % lost in 2 years

band changed?

03 Compare and changing rates

which bank?

1.8% then $x\%$

compound vs simple

04 Work backwards: rate, start or time

find x from the total

find P

least number of years

THE FORMULA Read the story, fill three boxes:

$$\begin{array}{c}
 \boxed{\text{amount}} = \boxed{\text{start}} \times \boxed{\text{multiplier}}^{\text{years}} \\
 \text{up } r\%: 1 + \frac{r}{100} \quad \text{down } r\%: 1 - \frac{r}{100}
 \end{array}$$

01 Compound Growth

Interest is added to the new amount each year, so the multiplier is used again and again: multiply once, with a power.

WHEN YOU SEE → YOU DO

“compound interest”, “increases by 4% each year”

→ $\text{start} \times 1.04^n$

“how much money will she have”

→ the amount (start + interest)

“total amount of interest”

→ amount – start

“to the nearest dollar / 3 s.f.”

→ round only at the end

Multiplier

Power = years

Start $\times$ multiplierⁿ

Round

WORKED MODEL • NOVEMBER 2024 P2H

Nisha invests 20 000 euros for 3 years at 3.5% per year compound interest. How much will she have?

1 multiplier 1.035

up 3.5%

2 $20\,000 \times 1.035^3 = 22\,174.36$

power 3

ANSWER 22 174 euros

× **TRAP** Simple interest: $20\,000 \times 0.035 \times 3$ only adds interest on the first amount. “Compound” uses the power.

IDEA 01

Practise: Compound Growth

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Growth

CH 1 • TOPIC 7 3 marks

£5000 is invested for 2 years at 3% per year compound interest. Work out the value at the end.

SELECTED PRACTICE**Q2** **PRACTICE** Practice • Medium

CH 1 • TOPIC 7 2 marks

- 1 Toby invested £7500 for 2 years in a savings account. He was paid 4% per annum compound interest.

How much money did Toby have in his savings account at the end of 2 years?

(2 marks)

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 6 marks

5 (a) Derek buys a house for £150 000

He sells the house for £154 500

Work out Derek's percentage profit.

(3 marks)

(b) Derek invests £154 500 for 2 years at 4% per year compound interest.

Work out the value of the investment at the end of 2 years.

(3 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 3 marks

7 Katy invests £200 000 in a savings account for 4 years.

The account pays compound interest at a rate of 1.5 % per annum.

Calculate the total amount of interest Katy will get at the end of 4 years.

(3 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 3 marks

12 Aayush invests 18 000 rupees for 3 years at a rate of 4% per year compound interest.

Work out the total amount of interest Aayush has received by the end of 3 years.

Give your answer correct to the nearest rupee.

(3 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q8

CH 1 • TOPIC 7 3 marks

8 Harold bought an antique clock for £1200
The clock increased in value by 8% per year.

Find the value of the clock exactly 3 years after Harold bought the clock.

Give your answer correct to the nearest £.

£.....

(Total for Question 8 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q7

CH 1 • TOPIC 7 3 marks

- 7 Kuro invests 50 000 yen for 3 years in a savings account.
She gets 2.4% per year compound interest.

Work out how much money Kuro will have in her savings account at the end of the 3 years.
Give your answer correct to the nearest yen.

..... yen

(Total for Question 7 is 3 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q7

CH 1 • TOPIC 7 6 marks

- 7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

- (a) Work out the value of Chen's investment at the end of 3 years.

Give your answer to the nearest yuan.

..... yuan

Wang invested P yuan.

The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

- (b) Work out the value of P .

$P =$

(Total for Question 7 is 6 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q11

CH 1 • TOPIC 7 6 marks

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

(a) Work out how much Chengbo paid for the house.

Give your answer correct to 3 significant figures.

..... yuan

Zhi bought a house on 1st January 2017

When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

(b) Work out the value of Zhi's house on 1st January 2020

Give your answer correct to 3 significant figures.

..... yuan

(Total for Question 11 is 6 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q5

CH 1 • TOPIC 7 3 marks

- 5 Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

..... dollars

(Total for Question 5 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Pasha invests 50000 dollars in a savings account for 4 years.
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.
Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 9 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q8

CH 1 • TOPIC 7 3 marks

- 8 Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of
4 years.

Give your answer correct to the nearest koruna.

..... koruna

(Total for Question 9 is 3 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q5

CH 1 • TOPIC 7 3 marks

- 5 Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account
at the end of 4 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 5 is 3 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 9 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q7

CH 1 • TOPIC 7 3 marks

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

02 Depreciation

Same formula, multiplier below 1. The value goes down by a percentage of the **current** value each year.

WHEN YOU SEE → YOU DO

“depreciates by 19% each year”

→ start $\times 0.81^n$

“the total percentage it had depreciated by”

→ $1 - 0.85^2 = 0.2775$: 27.75%

“lost more than half its value?”

→ compare the final value with half the start

different rates each year

→ multiply each year's multiplier

ZONE GROWTH

multiplier > 1

1.04, 1.025, 1.35

ZONE DEPRECIATION

multiplier < 1

0.81, 0.85, 0.8

WORKED MODEL • JANUARY 2020 P2HR

A car bought for \$20 000 depreciates by 19% each year. Value after 3 years?

1 multiplier $1 - 0.19 = 0.81$

down 19%

2 $20\,000 \times 0.81^3 = 10\,628.82$

power 3

ANSWER \$10 629

× **TRAP** $19\% \times 3 = 57\%$ off is wrong: after the first year you lose 19% of a smaller value.

IDEA 02

Practise: Depreciation

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Depreciation

CH 1 • TOPIC 7 3 marks

A machine worth £8000 depreciates by 10% each year. Find its value after 3 years.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 4 marks

- 4 This notice was in a car magazine.

Most new cars lose more than half of their value in the first three years

Paul bought a new car.

The value of the car was £15 000

In the first year, the value of the car depreciated by 23%.

After the first year, the value of the car depreciated by 18% each year.

Work out if Paul's car lost more than half of its value by the end of three years.

(4 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q8

CH 1 • TOPIC 7**3 marks**

- 8 Hamish buys a new car for \$20 000
The car depreciates in value by 19% each year.

Work out the value of the car at the end of 3 years.
Give your answer to the nearest \$.

\$.....

(Total for Question 8 is 3 marks)**Q20****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q8

CH 1 • TOPIC 7**3 marks**

- 8 Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.

Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

Q21 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q6

CH 1 • TOPIC 7 6 marks

6 Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	Service charge band
$x \geq 700\,000$	A
$600\,000 \leq x < 700\,000$	B
$500\,000 \leq x < 600\,000$	C
$400\,000 \leq x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

- (a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %

(Total for Question 6 is 6 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q9

CH 1 • TOPIC 7 3 marks

- 9 Kazi buys a car for 700 000 taka.
The value of the car depreciates by 12% each year.

Work out the value of the car at the end of 3 years.
Give your answer correct to the nearest taka.

..... taka

(Total for Question 9 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q7

CH 1 • TOPIC 7 3 marks

- 7 Hermione buys a boat for \$26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

\$.....

(Total for Question 7 is 3 marks)

Q24**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q5

CH 1 • TOPIC 7 6 marks

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs

(Total for Question 5 is 6 marks)

03 Compare and Changing Rates

Different rates in different years: one multiplier for each year, multiply them all. To compare offers, work out each final amount (or interest), then decide in a sentence.

WHEN YOU SEE → YOU DO

“4% for the first year, then 1.5% each year”

→ $\times 1.04 \times 1.015 \times 1.015 \dots$

“which bank should she choose?”

→ final amount of each, then choose

“receive 1150 dollars interest each year”

→ simple: 3×1150

“the average of 1% and 5% is 3% ...”

→ show $1.01 \times 1.05 \neq 1.03^2$

WORKED MODEL • NOVEMBER 2021 P2H

\$25 000: Cyclone Bank 4.5% compound for 3 years; Tornado Bank \$1150 interest each year. How much more interest from Cyclone?

1 $25\,000 \times 1.045^3 - 25\,000 = 3529.15$

compound interest

2 $3 \times 1150 = 3450$

simple interest

ANSWER \$79 more

IDEA 03

Practise: Compare and Changing Rates

My questions first, then selected practice, then real past papers.

Q25 **DR ESLAM** Changing rate

CH 1 • TOPIC 7 3 marks

£1000 earns 5% in year 1 and 2% in year 2 (compound). Find the total after 2 years.

SELECTED PRACTICE

Q26**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 4 marks

3 Peter has £20 000 to invest in a savings account for 2 years.

He finds information about two savings accounts.

Bonus Saver	Fixed Rate
Compound interest	Compound interest
4% for the first year then 1.5% each year	2.5% each year

Peter wants to have as much money as possible in his savings account at the end of 2 years.

Which of these savings accounts should he choose?

(4 marks)

Q27**PRACTICE**

Practice · Hard

CH 1 • TOPIC 7 4 marks

- 1 Viv wants to invest £2000 for 2 years in the same bank.

The International Bank

Compound Interest

4% for the first year
1% for each extra year

The Friendly Bank

Compound Interest

5% for the first year
0.5% for each extra year

At the end of 2 years, Viv wants to have as much money as possible.

Which bank should she invest her £2000 in?

(4 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 6 marks

- 3 (a)** Katie invests £200 in a savings account for 2 years.

The account pays compound interest at an annual rate of

3.3% for the first year 1.5% for the second year

Work out the total amount of money in Katie's account at the end of 2 years.

(3 marks)

- (b)** Katie travels to work by train. The cost of her weekly train ticket increases by 12.5% to £225

Katie's weekly pay increases by 5% to £535.50

Compare the increase in the amount of money Katie has to pay for her weekly train ticket with the increase in her weekly pay.

(3 marks)

Q29**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 6 marks**10 (a)** Here are the interest rates for two accounts.

Account A	Account B
Interest: 3% per year compound interest.	Interest: 4% for the first year, 3% for the second year and 2% for the third year.
No withdrawals until the end of three years.	Withdrawals allowed at any time.

Derrick has £10000 he wants to invest.

Calculate which account would give him most money if he invests his money for 3 years.

Give the difference in the interest to the nearest penny.

Account by p

(5 marks)**(b)** Explain why he might **not** want to use Account A.**(1 mark)**

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q7

CH 1 • TOPIC 7 4 marks

- 7 Ali and Badia each have 25 000 dollars to invest.

Cyclone Bank	Tornado Bank
Invest 25 000 dollars 4.5% compound interest per year for 3 years	Invest 25 000 dollars Receive 1150 dollars interest each year for 3 years

Ali invests in the Cyclone Bank for 3 years.

Badia invests in the Tornado Bank for 3 years.

By the end of the 3 years, Ali will have received more interest than Badia.

How much more?

Show your working clearly.

Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 7 is 4 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q11

CH 1 • TOPIC 7 5 marks

- 11 Zhou invests some money for 2 years.
He invests \$2500 with Bank A and \$3000 with Bank B.

Bank A	
Invests \$2500	
amount of money invested	: total amount of interest after 2 years = 20 : 3
Bank B	
Invests \$3000	
4% per year compound interest for 2 years	

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

\$

(Total for Question 11 is 5 marks)

04 Work Backwards: Rate, Start or Time

Three things in the formula; you are given two. Divide to find the start, take a root to find the rate, try years to find the time.

WHEN YOU SEE → YOU DO

“find the value of P ” (the start)

→ amount $\div$ multiplier ^{n}

“find x ” (the rate)

→ multiplier = $\sqrt[n]{\text{amount} \div \text{start}}$, then -1 , $\times 100$

“ $x\%$ in the second and third year”

→ divide out the known year first, then $\sqrt{\quad}$

“the least time in whole years”

→ try $n = 1, 2, 3, \dots$ until it passes

Amount $\div$ start

Root (years)

Minus 1

$\times 100$

WORKED MODEL • JANUARY 2022 P1H

200 000 yen, 1.8% in year 1 and $x\%$ in years 2 and 3. Total 209 754 yen. Find x .

$$1 \quad \frac{209\,754}{200\,000 \times 1.018} = 1.030225$$

divide out year 1

$$2 \quad \sqrt{1.030225} = 1.015$$

two years → square root

ANSWER $x = 1.5$

× **TRAP** Dividing the percentage by the years ($6.1\% \div 3$). Rates multiply — you need the root.

IDEA 04

Practise: Work Backwards: Rate, Start or Time

My questions first, then selected practice, then real past papers.

Q32

DR ESLAM

Find the rate

CH 1 • TOPIC 7

3 marks

An investment grows from £4000 to £4840 in 2 years at $r\%$ compound interest. Find r .

SELECTED PRACTICE

Q33**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 6 marks

- 2 (a)** The population of a city increased by 5.2% for the year 2014

At the beginning of 2015 the population of the city was 1 560 000

Lin assumes that the population will continue to increase at a constant rate of 5.2% each year.

Use Lin's assumption to estimate the population of the city at the beginning of 2017
Give your answer correct to 3 significant figures.

(3 marks)

- (b)** i) Use Lin's assumption to work out the year in which the population of the city will reach 2 000 000
ii) If Lin's assumption about the rate of increase of the population is too low, how might this affect your answer to (b)(i)?

(3 marks)

Q34**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 **3 marks**

9 Mirek invests £6000 at a compound interest rate of 1.5% per year.

He wants to earn more than £1000 interest.

Work out the **least** time, in whole years, that this will take.

.....years

(3 marks)

Q35**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 7 8 marks

4 (a) Katy invests £2000 in a savings account for 3 years.

The account pays compound interest at an annual rate of

2.5% for the first year

$x\%$ for the second year

$x\%$ for the third year

There is a total amount of £2124.46 in the savings account at the end of 3 years.

Work out the rate of interest in the second year.

(4 marks)

(b) Katy goes to work by train.

The cost of her weekly train ticket increases by 12.5% to £225

Work out the cost of her weekly train ticket before this increase.

(4 marks)

Q36**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 7 3 marks

- 8 Hans invests \$550 at a rate of $x\%$ per year compound interest.
At the end of 10 years, the value of the investment is \$638.30, correct to the nearest cent.

Find the value of x .

$x = \dots\dots\dots$ [3]

(3 marks)

PAST PAPERS

Q37**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q11

CH 1 • TOPIC 7 3 marks

- 11 Max invests \$6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.

The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.

Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

$\dots\dots\dots\%$

(Total for Question 11 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q13

CH 1 • TOPIC 7 3 marks

- 13 Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of $x\%$ per year.
At the end of 6 years, there is a total of \$8877.62 in the account.
Work out the value of x .
Give your answer correct to 2 decimal places.

 $x = \dots\dots\dots$ **(Total for Question 13 is 3 marks)**

Q39**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q14

CH 1 • TOPIC 7 4 marks

14 Simon bought a house at the beginning of 2018

The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020

The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

(Total for Question 14 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q11

CH 1 • TOPIC 7 3 marks

- 11 Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and $x\%$ for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of x

Give your answer correct to one decimal place.

 $x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q13

CH 1 • TOPIC 7 3 marks

- 13 Feruzi invests 80 000 Kenyan shillings (KES)

He invests the money for 3 years at $x\%$ compound interest each year.

At the end of 3 years, the total interest he receives is 6151.25 KES

Work out the value of x

 $x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q13

CH 1 • TOPIC 7 3 marks

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q12

CH 1 • TOPIC 7 3 marks

12 Osman buys a car for \$16 000

The car depreciates at a rate of 12% each year for the first 2 years.

In the third year, the car depreciates at a rate of $x\%$

At the end of 3 years, the value of the car is \$11 461.12

Work out the value of x

$\dots\dots\dots$

(Total for Question 12 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE GROWTH

$$\times(1 + r)^n$$

ZONE DEPRECIATION

$$\times(1 - r)^n$$

- | | | |
|-----------|-------------------------|---|
| 01 | Growth | Start $\times$ multiplier ⁿ ; interest = amount – start. |
| 02 | Depreciation | Multiplier below 1; the % is never multiplied by the years. |
| 03 | Compare / change | One multiplier per year; decide with the final amounts. |
| 04 | Backwards | Start: divide. Rate: root. Time: try $n = 1, 2, 3 \dots$ |

BOTTOM LINE amount = start $\times$ multiplierⁿ $\rightarrow$ fill two boxes, find the third.

Answers

Compound Interest and Depreciation · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • COMPOUND GROWTH

Q1	Dr Eslam • Growth	$5000 \times 1.03^2 = £5304.50$
Q2	Practice • Medium	£8112
Q3	Practice • Medium	(a) 3% (b) £167 107.20
Q4	Practice • Medium	£12 272.71
Q5	Practice • Medium	2248 rupees
Q6	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q8	£1512
Q7	Edexcel 4MA1 • January 2021 • Paper 1HR • Q7	53 687 yen
Q8	Edexcel 4MA1 • January 2021 • Paper 1H • Q7	(a) 44 349 yuan (b) $P = 32\,600$
Q9	Edexcel 4MA1 • May 2021 • Paper 2H • Q11	(a) 145 000 yuan (b) 127 000 yuan
Q10	Edexcel 4MA1 • January 2022 • Paper 2HR • Q5	\$7754
Q11	Edexcel 4MA1 • June 2022 • Paper 2H • Q9	\$52 651
Q12	Edexcel 4MA1 • June 2023 • Paper 2H • Q8	\$765
Q13	Edexcel 4MA1 • June 2024 • Paper 2H • Q9	4948 koruna
Q14	Edexcel 4MA1 • May 2024 • Paper 1HR • Q5	5740 euros
Q15	Edexcel 4MA1 • November 2024 • Paper 2H • Q9	22 174 euros
Q16	Edexcel 4MA1 • June 2025 • Paper 2H • Q7	4900 Swiss francs

IDEA 02 • DEPRECIATION

- Q17** Dr Eslam • Depreciation $8000 \times 0.9^3 = £5832$
- Q18** Practice • Medium $15\,000 \times 0.77 \times 0.82^2 = £7766 > £7500$: it lost less than half
- Q19** Edexcel 4MA1 • January 2020 • Paper 2HR • Q8 \$10 629
- Q20** Edexcel 4MA1 • June 2022 • Paper 1HR • Q8 \$9396
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q6 (a) 2020 value 610 000: band B both years — No (b) 27.75%
- Q22** Edexcel 4MA1 • May 2023 • Paper 1HR • Q9 477 030 taka
- Q23** Edexcel 4MA1 • November 2023 • Paper 2H • Q7 \$20 869
- Q24** Edexcel 4MA1 • November 2025 • Paper 2H • Q5 (a) 16% (b) 491.52 Swiss francs

IDEA 03 • COMPARE AND CHANGING RATES

- Q25** Dr Eslam • Changing rate $1000 \times 1.05 \times 1.02 = \text{£}1071$
- Q26** Practice • Medium Bonus £21 112, Fixed £21 012.50: Bonus Saver
- Q27** Practice • Hard International £2100.80, Friendly £2110.50: The Friendly Bank
- Q28** Practice • Hard (a) £209.70 (b) ticket up £25.00, pay up £25.50
- Q29** Practice • Hard (a) A by 103p (£10 927.27 vs £10 926.24) (b) he cannot withdraw for 3 years
- Q30** Edexcel 4MA1 • November 2021 • Paper 2H • Q7 3529.15 – 3450: \$79
- Q31** Edexcel 4MA1 • June 2025 • Paper 1HR • Q11 A: \$375, B: \$244.80: \$130.20 more

IDEA 04 • WORK BACKWARDS: RATE, START OR TIME

- Q32** Dr Eslam • Find the rate $\sqrt{1.21} = 1.1: r = 10$
- Q33** Practice • Hard (a) 1730 000 (b)(i) 2020 (ii) it would reach 2 000 000 sooner
- Q34** Practice • Hard 11 years
- Q35** Practice • Very Hard (a) 1.8% (b) £200
- Q36** Practice • Very Hard $x = 1.5$
- Q37** Edexcel 4MA1 • January 2020 • Paper 1H • Q11 2.1%
- Q38** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q13 $x = 1.75$
- Q39** Edexcel 4MA1 • January 2021 • Paper 2HR • Q14 10%
- Q40** Edexcel 4MA1 • January 2022 • Paper 1H • Q11 $x = 1.5$
- Q41** Edexcel 4MA1 • June 2023 • Paper 2HR • Q13 $x = 2.5$
- Q42** Edexcel 4MA1 • June 2025 • Paper 1H • Q13 $x = 3.75$
- Q43** Edexcel 4MA1 • June 2025 • Paper 2HR • Q12 $x = 7.5$

Fractions, Decimals and Percentages

Three ways to write the same number. Convert to compare; recurring decimals become fractions with one algebra trick: multiply, subtract, divide.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Fractions, Decimals and Percentages

01 Convert and compare

62 out of 80 as a %

65% or $\frac{5}{8}$?

order $0.2\dot{4}\dot{6}$, $0.24\dot{6}$...

02 Recurring decimals: pure

$$0.\dot{3}\dot{6} = \frac{4}{11}$$

$$0.\dot{7}0\dot{2} = \frac{26}{37}$$

$$4.\dot{5}\dot{7}$$

03 Recurring decimals: delayed start

$$0.5\dot{7}\dot{2} = \frac{63}{110}$$

$$0.28\dot{1}\dot{3}$$

$0.4\dot{x}$ in terms of x

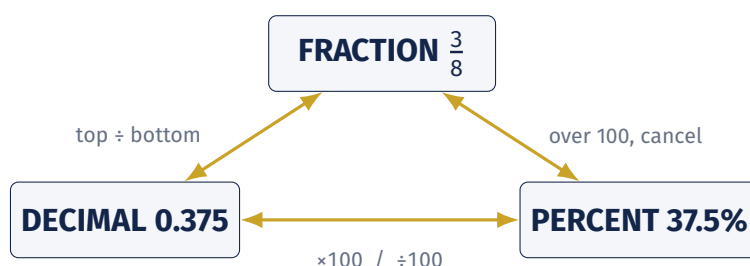
04 Mixed problems

$\frac{3}{4}$ stay, 15% college

Family Railcard

Venn: house and museum

THE TRIANGLE Any one form $\rightarrow$ the other two:



01 Convert and Compare

To compare or order, put everything in the same form — usually decimals. For a mark “out of”, divide and multiply by 100.

WHEN YOU SEE → YOU DO

“62 out of 80”

→ $62 \div 80 \times 100 = 77.5\%$

“did she get a higher percentage in ...”

→ convert both, then compare

“65% of 80 minutes or $\frac{5}{8}$ of 80 minutes”

→ work out both amounts

“write in order”, recurring dots

→ write each one out to 6 digits

“the lowest mark needed for 60%”

→ 60% of the total, minus what she already has

WORKED MODEL • ORDER RECURRING DECIMALS

Order $0.2\dot{4}\dot{6}$, $0.24\dot{6}$, $0.\dot{2}4\dot{6}$, 0.246 .

1 $0.246464 \dots$, $0.246666 \dots$, $0.246246 \dots$, 0.246

write them out

ANSWER 0.246 , $0.\dot{2}4\dot{6}$, $0.2\dot{4}\dot{6}$, $0.24\dot{6}$

IDEA 01

Practise: Convert and Compare

My questions first, then selected practice.

Q1 DR ESLAM Compare

CH 1 • TOPIC 8 2 marks

Which is bigger: $\frac{7}{20}$ or 36%? Show your working.

SELECTED PRACTICE

Q2**PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 2 marks

2 Sasha takes a music exam.

The table shows the result that Sasha can get for different percentages in her music exam.

Percentage	Result
50% - 69%	Pass
70% - 84%	Merit
85% - 100%	Distinction

Sasha gets 62 out of 80 in her music exam.

What result does Sasha get?

You must show your working.

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 8**2 marks**

- 3 Karen got 32 out of 80 in a maths test.
She got 38% in an English test.

Karen wants to know if she got a higher percentage in maths or in English.
Did Karen get a higher percentage in maths or in English?

(2 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 4 marks

4 Celina and Zoe both sing in a band.

One evening the band plays for 80 minutes.

Celina sings for 65% of the 80 minutes.

Zoe sings for $\frac{5}{8}$ of the 80 minutes.

Celina sings for more minutes than Zoe sings.

Work out for how many more minutes.

You must show all your working.

(4 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 8 1 mark

12 Circle the largest number.

0. $\dot{5}$	0.55	0.545	0.5 $\dot{4}\dot{5}$
--------------	------	-------	----------------------

(1 mark)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

- 2 Write these numbers in order of size.
Start with the smallest number.

$0.2\dot{4}\dot{6}$	$0.24\dot{6}$	$0.\dot{2}4\dot{6}$	0.246
---------------------	---------------	---------------------	---------

(2 marks)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 4 marks

- 11 James is taking three examination papers in Spanish. Here are his first two results.

Paper 1: $\frac{43}{80}$ Paper 2: $\frac{38}{65}$

Paper 3 is out of 95.

The marks in each of the three papers are added together.

Find the lowest mark that James needs in Paper 3 to achieve 60% of the total marks.

(4 marks)

02 Recurring Decimals: Pure

The repeat starts straight after the point. Multiply by $10^{(\text{length of the repeat})}$, subtract, divide.

WHEN YOU SEE → YOU DO

“use algebra to show that”

→ $x = \dots$, then $10x$, $100x$ or $1000x$

2 digits repeat: $0.\dot{3}\dot{6}$

→ $100x$

3 digits repeat: $0.\dot{3}0\dot{6}$

→ $1000x$

a whole number in front: $4.\dot{5}\dot{7}$

→ do the decimal part, add the 4 back

$x = \text{the decimal}$

Multiply by 10^n

Subtract

Divide and cancel

WORKED MODEL • JUNE 2025 P2H

Show that $0.\dot{7}0\dot{2} = \frac{26}{37}$.

1 $x = 0.702702 \dots$

name it

2 $1000x = 702.702702 \dots$

3 digits repeat

3 $999x = 702$

subtract: the tails cancel

ANSWER $x = \frac{702}{999} = \frac{26}{37}$

IDEA 02

Practise: Recurring Decimals: Pure

My questions first, then selected practice, then real past papers.

Q8 DR ESLAM Pure repeat

CH 1 • TOPIC 8 2 marks

Use algebra to show that $0.\dot{4}\dot{5} = \frac{5}{11}$.

SELECTED PRACTICE**Q9** PRACTICE Practice • Medium

CH 1 • TOPIC 8 3 marks

3 Prove that the recurring decimal $0.\dot{3}\dot{6} = \frac{4}{11}$

(3 marks)

Q10**PRACTICE**

Practice · Medium

CH 1 · TOPIC 8 2 marks

4 Show that the recurring decimal $0.\dot{3}9\dot{6} = \frac{44}{111}$

(2 marks)

Q11**PRACTICE**

Practice · Medium

CH 1 · TOPIC 8 2 marks

5 Use algebra to show that the recurring decimal $0.\dot{4}1\dot{7} = \frac{139}{333}$

(2 marks)

Q12**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 5 marks

12 (a) **Without using a calculator**, show that $0.\dot{1}\dot{9}$ can be written as $\frac{19}{99}$.

(3 marks)

(b) Explain how $\frac{19}{99} = 0.\dot{1}\dot{9}$ can be used to find $\frac{19}{990}$ as a decimal and write down its value.

$$\frac{19}{990} = \dots\dots\dots$$

(2 marks)

PAST PAPERS

Q13**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q15

CH 1 • TOPIC 8 5 marks

15 (a) Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(b) Show that $\frac{2}{6 - 3\sqrt{2}}$ can be written in the form $\frac{a + \sqrt{a}}{b}$

where a and b are integers.
Show your working clearly.

(Total for Question 15 is 5 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q18

CH 1 • TOPIC 8 2 marks

18 Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$

(Total for Question 18 is 2 marks)

Q15**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q18

CH 1 · TOPIC 8 2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

03 Recurring Decimals: Delayed Start

Some digits come before the repeat. Use **two** multiples of x whose decimal tails are identical, then subtract them.

WHEN YOU SEE → YOU DO

$0.5\dot{7}\dot{2}$ (one digit, then a 2-digit repeat)

→ $1000x$ and $10x \rightarrow 990x$

$0.2\dot{5}$ (one digit, then a 1-digit repeat)

→ $100x$ and $10x \rightarrow 90x$

$0.281\dot{3}$

→ $10\,000x$ and $100x \rightarrow 9900x$

“find $0.4\dot{x}$ in terms of x ”

→ $100n - 10n = 90n: \frac{36 + x}{90}$

ZONE RULE

line up the tails

write both multiples out; the digits after the point must match exactly

ZONE CHECK

calculator

type the fraction: the display must show your decimal

WORKED MODEL • JANUARY 2020 P1HR

Show that $0.5\dot{7}\dot{2} = \frac{63}{110}$.

1 $1000x = 572.7272 \dots$, $10x = 5.7272 \dots$

same tails

2 $990x = 567$

subtract

ANSWER $x = \frac{567}{990} = \frac{63}{110}$

× **TRAP** Subtracting x from $100x$ when the repeat is delayed: the tails do not match, and nothing cancels.

IDEA 03

Practise: Recurring Decimals: Delayed Start

My questions first, then selected practice, then real past papers.

Q16 **DR ESLAM** Delayed repeat

CH 1 • TOPIC 8 2 marks

Use algebra to show that $0.1\dot{6} = \frac{1}{6}$.

SELECTED PRACTICE

Q17 **PRACTICE** Practice • Easy

CH 1 • TOPIC 8 2 marks

5 Prove algebraically that the recurring decimal $0.2\dot{5}$ has the value $\frac{23}{90}$

(2 marks)

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 2 marks

8 Use algebra to show that the recurring decimal $0.2\dot{5}\dot{4} = \frac{14}{55}$

(2 marks)

Q19**PRACTICE**

Practice • Hard

CH 1 • TOPIC 8 2 marks

5 Prove algebraically that the recurring decimal $0.3\dot{1}\dot{8}$ can be written as $\frac{7}{22}$

(2 marks)

Q20**PRACTICE**

Practice · Hard

CH 1 · TOPIC 8 3 marks

- 12 Write $0.4\dot{1}\dot{6}$ as a fraction in its simplest form.
You must **show full working** in support of your answer.

(3 marks)

PAST PAPERS**Q21****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 1HR · Q13

CH 1 · TOPIC 8 4 marks

- 13 (a) Use algebra to show that $0.5\dot{7}\dot{2} = \frac{63}{110}$

Given that y is a prime number,

- (b) express $\frac{3}{2 - \sqrt{y}}$ in the form $\frac{a + b\sqrt{y}}{c - y}$ where a , b and c are integers.

(Total for Question 13 is 4 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q16

CH 1 • TOPIC 8 2 marks

16 Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \frac{557}{1980}$

(Total for Question 16 is 2 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q18

CH 1 • TOPIC 8 3 marks

18 $0.4\dot{x}$ is a recurring decimal.

x is a whole number such that $1 \leq x \leq 9$

Find, in terms of x , the recurring decimal $0.4\dot{x}$ as a fraction.

Give your fraction in its simplest form.

Show clear algebraic working.

(Total for Question 18 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • June 2022 • Paper 2HR • Q17**CH 1 • TOPIC 8** **2 marks**

17 Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

(Total for Question 17 is 2 marks)

Q25 **PAST PAPER** Edexcel 4MA1 • May 2023 • Paper 1HR • Q16**CH 1 • TOPIC 8** **2 marks**

16 Use algebra to show that $0.1\dot{7}\dot{6} = \frac{35}{198}$

(Total for Question 16 is 2 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q15

CH 1 • TOPIC 8 5 marks

15 (a) Use algebra to show that $0.3\dot{7}\dot{2} = \frac{41}{110}$

(b) Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

(Total for Question 15 is 5 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q15

CH 1 • TOPIC 8 2 marks

15 Use algebra to show that $0.7\dot{6}\dot{3} = \frac{42}{55}$

(Total for Question 15 is 2 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q16

CH 1 • TOPIC 8 2 marks

16 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

(Total for Question 16 is 2 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q16

CH 1 • TOPIC 8 2 marks

16 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

(Total for Question 16 is 2 marks)

04 Mixed Problems

Fractions, decimals and percentages of one whole in one question: change each part to a number of people (or pounds), then finish the story.

WHEN YOU SEE → YOU DO

“15% ... $\frac{3}{4}$... the rest”

→ work out each part of the total, subtract from the total

“ $\frac{1}{3}$ off adult tickets, 60% off child tickets”

→ price after each discount

“percentage profit”

→ total income vs total cost

“you may use the Venn diagram”

→ put each group in its region, then multiply by its price

WORKED MODEL • STUDENTS

240 students: 15% college, $\frac{3}{4}$ stay at school, the rest do not know. How many do not know?

1 $0.15 \times 240 = 36$, $\frac{3}{4} \times 240 = 180$

each part

ANSWER $240 - 36 - 180 = 24$

IDEA 04

Practise: Mixed Problems

My questions first, then selected practice.

Q30 **DR ESLAM** Parts of a whole

CH 1 • TOPIC 8 3 marks

200 people: 30% walk, $\frac{2}{5}$ take the bus, the rest drive. How many drive?

SELECTED PRACTICE**Q31** **PRACTICE** Practice · Easy

CH 1 • TOPIC 8 4 marks

- 1 Mr Mason asks 240 Year 11 students what they want to do next year. 15% of the students want to go to college.

$\frac{3}{4}$ of the students want to stay at school.

The rest of the students do not know.

Work out the number of students who do not know.

(4 marks)

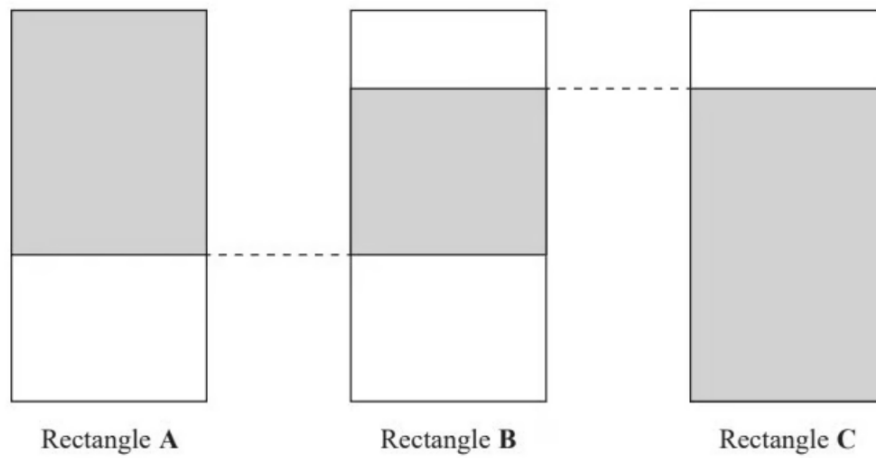
Q32**PRACTICE**

Practice • Easy

CH 1 • TOPIC 8

3 marks

11 The diagram shows three identical rectangles.



$\frac{5}{8}$ of rectangle **A** is shaded.

80% of rectangle **C** is shaded.

What fraction of rectangle **B** is shaded?

(3 marks)

Q33**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8**4 marks**

- 1 Mr Brown and his 2 children are going to London by train.

An adult ticket costs £24

A child ticket costs £12

Mr Brown has a Family Railcard.

Family Railcard gives

$\frac{1}{3}$ off adult tickets

60% off child tickets

Work out the total cost of the tickets when Mr Brown uses his Family Railcard.

(4 marks)

Q34**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8 **3 marks**

6 The people working for a company work in Team **A** or in Team **B**.

number of people in Team **A** : number of people in Team **B** = 3 : 4

$\frac{4}{5}$ of Team **A** work full time.

24% of Team **B** work full time.

Work out what fraction of the people working for the company work full time.

Give your fraction in its simplest form.

(3 marks)

Q35**PRACTICE**

Practice • Medium

CH 1 • TOPIC 8

5 marks

9 Josh buys and sells books for a living.

He buys 120 books for £4 each.

He sells $\frac{1}{2}$ of the books for £5 each.

He sells 40% of the books for £7 each.

He sells the rest of the books for £8 each.

Calculate Josh's percentage profit.

(5 marks)

Q36

PRACTICE Practice · Hard

CH 1 · TOPIC 8 5 marks

10 At a country park there is a house, a museum and a garden.

The table shows the prices per person to visit the park.

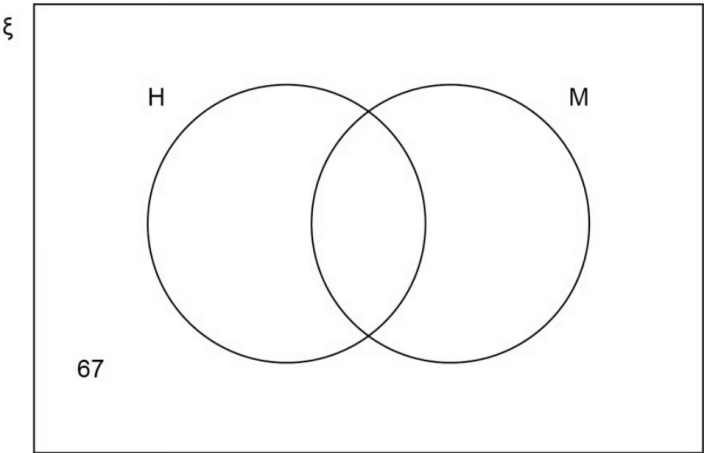
	Price per person
Garden only	Free
House and museum	£12.50
House only	£8
Museum only	£7

One day, 480 people visit the park.

- 67 visit the garden **only**.
- 40% visit the house **and** the museum.
- $\frac{3}{8}$ visit the house **only**.

The rest visit the museum **only**.

In total, how much do the 480 people pay to visit the park?
You may use the Venn diagram to help you.



(5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|--|
| 01 | Convert | Same form to compare; write recurring decimals out. |
| 02 | Pure recurring | $10^n x$ where n = length of the repeat; subtract; divide. |
| 03 | Delayed | Two multiples with the same tail; subtract; cancel. |
| 04 | Problems | Each part as an amount, then the story. |

BOTTOM LINE Recurring decimal → name it x → two multiples with the same tail → subtract → divide.

Answers

Fractions, Decimals and Percentages • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CONVERT AND COMPARE

- | | | |
|-----------|--------------------|--|
| Q1 | Dr Eslam • Compare | $\frac{7}{20} = 35\% < 36\%: 36\%$ |
| Q2 | Practice • Easy | 77.5%: Merit |
| Q3 | Practice • Easy | maths 40% > 38%: maths |
| Q4 | Practice • Easy | 52 – 50 = 2 minutes |
| Q5 | Practice • Easy | 0.5 |
| Q6 | Practice • Medium | 0.246, $0.\dot{2}4\dot{6}$, $0.2\dot{4}\dot{6}$, $0.24\dot{6}$ |
| Q7 | Practice • Medium | 144 – 81 = 63 |

IDEA 02 • RECURRING DECIMALS: PURE

- Q8** Dr Eslam • Pure repeat $100x - x = 45, x = \frac{45}{99} = \frac{5}{11}$
- Q9** Practice • Medium $100x - x = 36: \frac{36}{99} = \frac{4}{11}$
- Q10** Practice • Medium $999x = 396: \frac{44}{111}$
- Q11** Practice • Medium $999x = 417: \frac{139}{333}$
- Q12** Practice • Medium (a) $99x = 19$ (b) divide by 10: $0.0\dot{1}\dot{9}$
- Q13** Edexcel 4MA1 • January 2021 • Paper 1HR • Q15 (a) $99x = 57, 4 + \frac{57}{99}$ (b) $\frac{2 + \sqrt{2}}{3}$
- Q14** Edexcel 4MA1 • June 2024 • Paper 2HR • Q18 $999x = 306: \frac{34}{111}$
- Q15** Edexcel 4MA1 • June 2025 • Paper 2H • Q18 $999x = 702: \frac{26}{37}$

IDEA 03 • RECURRING DECIMALS: DELAYED START

Q16	Dr Eslam • Delayed repeat	$100x - 10x = 15, x = \frac{15}{90} = \frac{1}{6}$
Q17	Practice • Easy	$100x - 10x = 23: \frac{23}{90}$
Q18	Practice • Medium	$990x = 252: \frac{14}{55}$
Q19	Practice • Hard	$990x = 315: \frac{7}{22}$
Q20	Practice • Hard	$\frac{5}{12}$
Q21	Edexcel 4MA1 • January 2020 • Paper 1HR • Q13	(a) $990x = 567$ (b) $\frac{6 + 3\sqrt{y}}{4 - y}$
Q22	Edexcel 4MA1 • May 2021 • Paper 1H • Q16	$9900x = 2785: \frac{557}{1980}$
Q23	Edexcel 4MA1 • November 2021 • Paper 1H • Q18	$\frac{36 + x}{90}$
Q24	Edexcel 4MA1 • June 2022 • Paper 2HR • Q17	$990x = 342: \frac{19}{55}$
Q25	Edexcel 4MA1 • May 2023 • Paper 1HR • Q16	$990x = 175: \frac{35}{198}$
Q26	Edexcel 4MA1 • November 2023 • Paper 2H • Q15	(a) $990x = 369$ (b) $\sqrt{135}$
Q27	Edexcel 4MA1 • November 2024 • Paper 2H • Q15	$990x = 756: \frac{42}{55}$
Q28	Edexcel 4MA1 • June 2025 • Paper 1HR • Q16	$990x = 606: \frac{101}{165}$
Q29	Edexcel 4MA1 • November 2025 • Paper 2H • Q16	$990x = 375: \frac{25}{66}$

IDEA 04 • MIXED PROBLEMS

- Q30** Dr Eslam • Parts of a whole $200 - 60 - 80 = 60$
- Q31** Practice • Easy 24
- Q32** Practice • Easy $\frac{17}{40}$
- Q33** Practice • Medium $16 + 9.60 = \text{£}25.60$
- Q34** Practice • Medium $\frac{12}{25}$
- Q35** Practice • Medium 52.5%
- Q36** Practice • Hard $\text{£}4127$

Rounding, Estimation and Bounds



DR ESLAM AHMED
eliteigcse.com

A rounded number hides a range: half the unit below, half the unit above. For the biggest answer, choose each bound to push the answer up.

THE TOPIC ON ONE MAP

Rounding, Estimation and Bounds

01 Rounding and error intervals

4.3 kg to 2 s.f.

$\dots \leq L < \dots$

18 000 to the nearest 1000

truncated: $8.3 \leq y < 8.4$

02 Estimation

each number to 1 s.f.

is his answer sensible?

over- or underestimate?

03 Bounds in a formula

upper bound of $\frac{2a - c}{d}$

lower bound of $\frac{t}{a - h}$

sine rule with bounds

04 Bounds in context: decide

could the speed be over 80?

enough soup?

"to a suitable degree of accuracy"

HALF THE UNIT The rounded value sits in the middle of its interval:



01 Rounding and Error Intervals

Find the unit of the rounding, halve it, then go that far down and up. The lower bound is included ($\leq$), the upper bound is not ($<$).

WHEN YOU SEE → YOU DO

“correct to 1 d.p.” / “to 2 s.f.” / “to the nearest 5”

→ unit 0.1 / last significant place / 5 → halve it

“write down the error interval”

→ lower $\leq x <$ upper

“minimum / maximum number of people”

→ whole numbers: 17 500 and 18 499

“truncated” / “the display began 8.3”

→ $8.3 \leq y < 8.4$ (not half the unit)

a perimeter from rounded sides

→ multiply both bounds

Find the unit

Halve it

– half : lower

+ half : upper

WORKED MODEL • JUNE 2025 P1HR

A ship is 142.8 m, correct to 1 decimal place. Write down the bounds.

1 unit 0.1, half 0.05

1 d.p.

ANSWER $142.75 \leq L < 142.85$

× **TRAP** 142.84 is not the upper bound. The upper bound is 142.85 — the value it cannot reach.

IDEA 01

Practise: Rounding and Error Intervals

My questions first, then selected practice, then real past papers.

Q1

DR ESLAM

Half the unit

CH 1 • TOPIC 9 2 marks

$w = 6.4$ kg correct to 1 decimal place. Write the error interval for w .

SELECTED PRACTICE

Q2

PRACTICE

Practice • Easy

CH 1 • TOPIC 9 2 marks

- 1 A number, n , is rounded to 2 decimal places.
The result is 4.76
Using inequalities, write down the error interval for n .

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 9**2 marks**

7 The length of a book is 33.8 cm, correct to one decimal place.

i) Write down the lower bound of the length of the book.

..... cm [1]

ii) Write down the upper bound of the length of the book.

..... cm [1]

(2 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 9**3 marks**

12 (a) The length of each side of a regular pentagon is 8.4 cm to 1 decimal place.

Complete the error interval for the length of one side.

.....cm $\leq$ length <cm

(2 marks)

(b) Complete the error interval for the perimeter.

.....cm $\leq$ perimeter <cm

(1 mark)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 4 marks

13 (a) The length of a pipe is 6 metres to the nearest metre.

Complete the error interval for the length of the pipe.

.....m $\leq$ length <m

(2 marks)

(b) The length of a different pipe is 4 metres to the nearest metre.

Olly says,

“The total length of the two pipes is 11 metres to the nearest metre.”

Give an example to show that he could be correct.

(2 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1HR • Q11

CH 1 • TOPIC 9 4 marks

11 The weight of a cat is 4.3 kg correct to 2 significant figures.

(a) Write down the upper bound of the weight of the cat.

..... kg

(b) Write down the lower bound of the weight of the cat.

..... kg

$$G = e - f$$

$e = 17$ correct to the nearest integer

$f = 9.4$ correct to one decimal place

(c) Work out the upper bound for the value of G .

.....

(Total for Question 11 is 4 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q4

CH 1 • TOPIC 9 2 marks

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

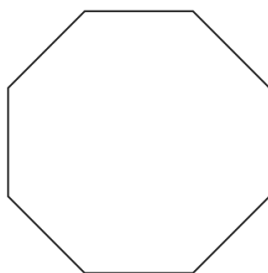


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm

- (b) Write down the upper bound of the length of each side of the octagon.

..... mm

(Total for Question 4 is 2 marks)

Q8**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q2

CH 1 • TOPIC 9 2 marks

- 2 The length of a table is measured as 1.4 metres correct to one decimal place.

- (a) Write down the upper bound of the length of the table.

..... metres

- (b) Write down the lower bound of the length of the table.

..... metres

(Total for Question 2 is 2 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q2

CH 1 • TOPIC 9 2 marks

2 The length of a ship is 142.8 m, correct to 1 decimal place.

(i) Write down the lower bound of the length of the ship.

..... m

(ii) Write down the upper bound of the length of the ship.

..... m

(Total for Question 2 is 2 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 1H • Q7

CH 1 • TOPIC 9 2 marks

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm

(Total for Question 7 is 2 marks)

02 Estimation

Round every number to 1 significant figure, then calculate without a calculator. For a square root, round to a nearby square number.

WHEN YOU SEE → YOU DO

“work out an estimate”

→ each number to 1 s.f., show the rounded line

“check his answer is sensible”

→ estimate, then compare the size

$\sqrt{63.5 \times 101.7}$

→ $\sqrt{64 \times 100} = 80$ (nice squares)

“overestimate or underestimate?”

→ look at which numbers you rounded up / down

WORKED MODEL • NOVEMBER 2025 P1H

Estimate $\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$.

1 $\frac{2^2 \times 60}{\sqrt{9}}$

1 s.f. (and a square)

2 $\frac{240}{3}$

calculate

ANSWER 80

OVER OR UNDER? Top rounded up or bottom rounded down → the estimate is too big (overestimate).

IDEA 02

Practise: Estimation

My questions first, then selected practice, then real past papers.

Q11 DR ESLAM 1 s.f.

CH 1 • TOPIC 9 2 marks

Estimate $\frac{49.6 \times 3.82}{0.198}$.

SELECTED PRACTICE**Q12** PRACTICE Practice • Easy

CH 1 • TOPIC 9 3 marks

14 Asha worked out $\frac{326.8 \times (6.94 - 3.4)}{59.4}$

She got an answer of 19.5, correct to 3 significant figures.

Write each number correct to 1 significant figure to decide if Asha's answer is reasonable.

(3 marks)

Q13**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 2 marks

- 10 Nav has worked out $\frac{68.3 \times 42.8}{0.021}$ on his calculator.

His answer is 139 201.9048

Without using a calculator and using suitable approximations, check that his answer is sensible.

Show your working clearly.

(2 marks)

Q14**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9 4 marks

10

Competition
a prize every 2014 seconds

In a competition, a prize is won every 2014 seconds.

Work out an estimate for the number of prizes won in 24 hours. You must show your working.

(4 marks)

Q15**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 9 4 marks

- 6 (a) One uranium atom has a mass of 3.95×10^{-22} grams.

Work out an estimate for the number of uranium atoms in 1 kg of uranium.

(3 marks)

- (b) Is your answer to (a) an underestimate or an overestimate?

Give a reason for your answer.

(1 mark)**PAST PAPERS****Q16****PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q2

CH 1 · TOPIC 9 2 marks

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

03 Bounds in a Formula

To make an answer as **big** as possible: add the big ones, subtract the small one, multiply by the big one, divide by the small one.

WHEN YOU SEE → YOU DO

“upper bound of $a + b / a \times b$ ”

→ upper + upper, upper × upper

“upper bound of $a - b$ ”

→ upper - **lower**

“upper bound of $\frac{a}{b}$ ”

→ upper ÷ **lower**

a formula such as $\frac{2a - c}{d}$

→ decide each letter one at a time

“show your working clearly”

→ write the bound you chose for every letter

ZONE BIGGEST

Upper bound of the answer

$$\frac{\text{UP} + \text{UP}}{\text{LOW}}, \quad \text{UP} - \text{LOW}$$

push up everything on top, down everything underneath

ZONE SMALLEST

Lower bound of the answer

$$\frac{\text{LOW} + \text{LOW}}{\text{UP}}, \quad \text{LOW} - \text{UP}$$

the opposite of every choice

WORKED MODEL • JANUARY 2020 P1H

$P = \frac{2a - c}{d}$, $a = 58.4$ (3 s.f.), $c = 20$ (2 s.f.), $d = 3.6$ (2 s.f.). Upper bound of P ?

1 $a = 58.45, c = 19.5, d = 3.55$

up · low · low

2
$$\frac{2(58.45) - 19.5}{3.55}$$

substitute

ANSWER 27.44

× **TRAP** Using upper - upper for the upper bound of a subtraction. Subtract the **smallest** possible amount.

IDEA 03

Practise: Bounds in a Formula

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Biggest

CH 1 • TOPIC 9 3 marks

$x = 12$ (nearest whole), $y = 3.4$ (1 d.p.). Find the upper bound of $\frac{x}{y}$.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 9 4 marks

- 1 (a) The value of p is 4.3
The value of q is 0.4

Both p and q are given correct to the nearest 0.1

Write down the lower bound for p .

(1 mark)

(b) $r = p + \frac{1}{q}$

Work out the upper bound for r .
You must show all your working.

(3 marks)

Q19**PRACTICE**

Practice · Medium

CH 1 · TOPIC 9 3 marks

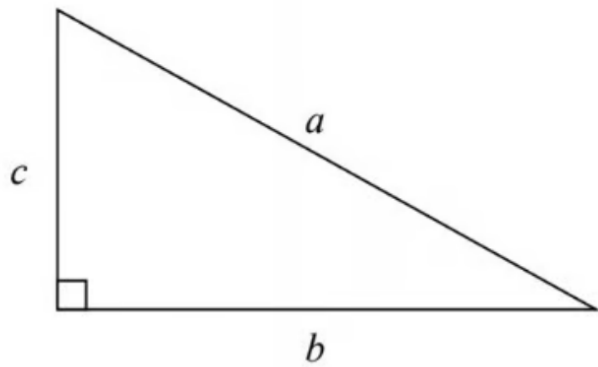
$$4 \quad D = \frac{x}{y}$$

 $x = 99.7$ correct to 1 decimal place. $y = 67$ correct to 2 significant figures.Work out an upper bound for D .

(3 marks)

Q20**PRACTICE**

Practice · Hard

CH 1 · TOPIC 9 4 marks**4**

a is 8.3 cm correct to the nearest mm

b is 6.1 cm correct to the nearest mm

Calculate the upper bound for c .

You must show your working.

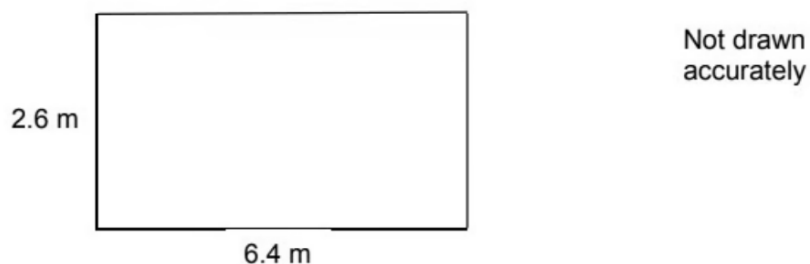
(4 marks)

Q21**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 9 5 marks

- 7 The dimensions of a rectangular floor are to the nearest 0.1 metres.



A force of 345 Newtons is applied to the floor.
The force is to the nearest 5 Newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the upper bound of the pressure.
Give your answer to 4 significant figures.
You **must** show your working.

.....N/m² [5]

(5 marks)

PAST PAPERS

Q22 **PAST PAPER** Edexcel 4MA1 • January 2020 • Paper 1H • Q16**CH 1 • TOPIC 9** **3 marks**

16 $P = \frac{2a - c}{d}$

$a = 58.4$ correct to 3 significant figures.

$c = 20$ correct to 2 significant figures.

$d = 3.6$ correct to 2 significant figures.

Work out the upper bound for the value of P .

Show your working clearly.

Give your answer correct to 2 decimal places.

.....

(Total for Question 16 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q19

CH 1 • TOPIC 9 3 marks

19 The diagram shows rectangle $ABCD$ with rectangle $EFGH$ cut out to form the shaded region.

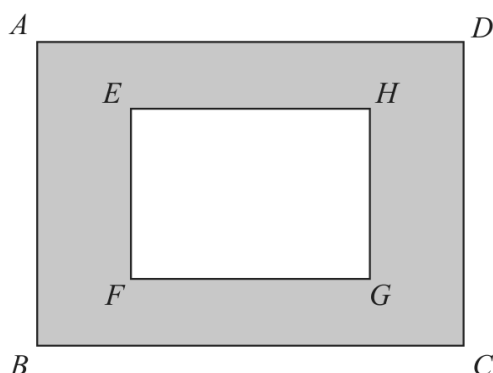


Diagram **NOT**
accurately drawn

$AD = 8.3$ cm correct to one decimal place

$DC = 7.2$ cm correct to one decimal place

$EH = 6.2$ cm correct to one decimal place

$HG = 5.3$ cm correct to one decimal place

Work out the upper bound of the area of the shaded region.

Show your working clearly.

..... cm^2

(Total for Question 19 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1H • Q15**CH 1 • TOPIC 9** **3 marks****15**

$$a = \frac{v - u}{t}$$

 $v = 9.6$ correct to 1 decimal place $u = 3.8$ correct to 1 decimal place $t = 1.84$ correct to 2 decimal placesCalculate the upper bound for the value of a .

Give your answer as a decimal correct to 2 decimal places.

Show your working clearly.

.....

(Total for Question 15 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q19

CH 1 • TOPIC 9 3 marks

19 $k = \frac{t}{a - h}$

 $t = 14$ correct to 2 significant figures $a = 7.8$ correct to 2 significant figures $h = 3.4$ correct to 2 significant figuresWork out the lower bound for the value of k .

Show your working clearly.

.....

(Total for Question 19 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q20

CH 1 • TOPIC 9 3 marks

20 $P = \frac{t - w}{y}$

$t = 9.7$ correct to 1 decimal place

$w = 5.9$ correct to 1 decimal place

$y = 3$ correct to 1 significant figure

Calculate the upper bound for the value of P .
Show your working clearly.

.....
(Total for Question 20 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 1H • Q13

CH 1 • TOPIC 9 3 marks

13 Here is a triangle XYZ .

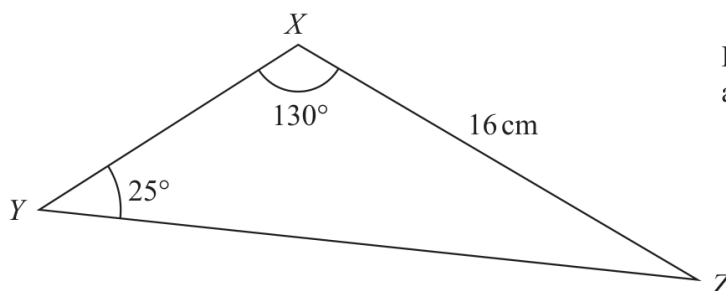


Diagram **NOT**
accurately drawn

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

(Total for Question 13 is 3 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q18

CH 1 • TOPIC 9 3 marks

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

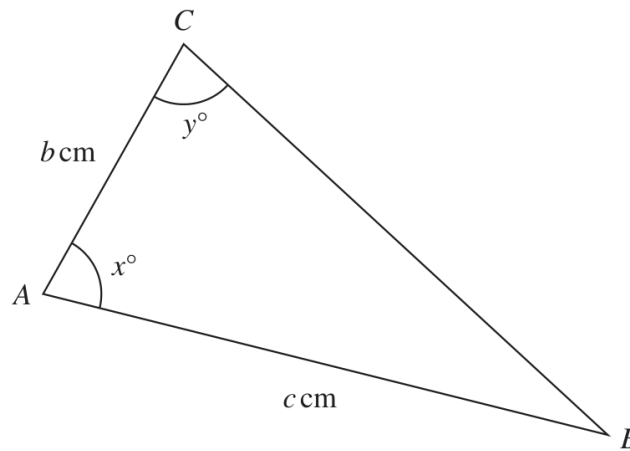
Work out the lower bound of the value of A

Show your working clearly.

(Total for Question 18 is 3 marks)

Q29 PAST PAPER Edexcel 4MA1 • January 2022 • Paper 1HR • Q22

CH 1 • TOPIC 9 4 marks

22 The diagram shows triangle ABC Diagram **NOT**
accurately drawn

$c = 11.5$ correct to one decimal place
 $x = 80$ correct to the nearest whole number
 $y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b

Show your working clearly.

Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q18

CH 1 • TOPIC 9 3 marks

$$18 \quad X = \frac{2a - b}{f}$$

 $a = 7.5$ correct to 1 decimal place. $b = 3.42$ correct to 2 decimal places. $f = 2$ correct to the nearest whole number.Work out the upper bound of the value of X

Show your working clearly.

(Total for Question 18 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q17

CH 1 • TOPIC 9 3 marks

$$17 \quad P = a(c + y)$$

 $a = 8.3$ correct to 2 significant figures $c = 2$ correct to 1 significant figure $y = 15$ correct to the nearest 5Work out the upper bound for the value of P

Show your working clearly.

(Total for Question 17 is 3 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q19

CH 1 • TOPIC 9 3 marks

19 $G = \frac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number

$f = 6.62$ correct to 2 decimal places

$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G

Give your answer correct to 3 decimal places.

Show your working clearly.

.....
(Total for Question 19 is 3 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q21

CH 1 • TOPIC 9 3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

(Total for Question 21 is 3 marks)

04 Bounds in Context: Decide

“Could it ...?” and “definitely ...?” are decided by the extreme value. “Suitable degree of accuracy” — round both bounds until they agree.

WHEN YOU SEE → YOU DO

“could the speed have been greater than 80?”

→ work out the **upper** bound, compare

“does he definitely have enough?”

→ worst case: least supply, most needed

“by considering bounds ... suitable degree of accuracy”

→ find both bounds, round both until they match

“greatest number of spheres / panels”

→ whole numbers: round down (or up) the division

Upper bound

Lower bound

Round both

Same? That is the answer

WORKED MODEL • NOVEMBER 2025 P2H

$D = \frac{n}{p - q}$, $n = 10.3$ (1 d.p.), $p = 7.24$, $q = 4.39$ (2 d.p.). Find D to a suitable degree of accuracy.

$$1 \quad \text{UB} = \frac{10.35}{7.235 - 4.395} = 3.644$$

biggest

$$2 \quad \text{LB} = \frac{10.25}{7.245 - 4.385} = 3.584$$

smallest

ANSWER $D = 3.6$ (both bounds round to 3.6 to 2 s.f.)

IDEA 04

Practise: Bounds in Context: Decide

My questions first, then selected practice, then real past papers.

Q34 **DR ESLAM** Could it?

CH 1 • TOPIC 9 3 marks

A car travels 120 km (nearest km) in 1.5 hours (1 d.p.). Could its average speed have been more than 82 km/h?

SELECTED PRACTICE**Q35** **PRACTICE** Practice • Hard

CH 1 • TOPIC 9 4 marks

- 1 Dan does an experiment to find the value of π .
He measures the circumference and the diameter of a circle.

He measures the circumference, C , as 170 mm to the nearest millimetre.
He measures the diameter, d , as 54 mm to the nearest millimetre.

Dan uses $\pi = \frac{C}{d}$ to find the value of π .

Calculate the upper bound and the lower bound for Dan's value of π .

(4 marks)

Q36**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9**4 marks**

13 Edith's van can safely carry a maximum load of 920 kilograms.

She wants to use her van to carry

30 sacks of potatoes, each of mass 25 kilograms to the nearest kilogram

and

20 sacks of carrots, each of mass 7.5 kilograms to 1 decimal place.

Can she definitely use her van safely in one journey?

You **must** show your working.

(4 marks)

Q37**PRACTICE**

Practice • Hard

CH 1 • TOPIC 9 4 marks

17 A solid metal cube has sides of length 125 mm, correct to 3 significant figures.

The cube is melted down and the metal used to make solid spheres.

The volume of each sphere is to be 140 cm^3 , correct to the nearest 10 cm^3

Work out the greatest number of spheres that could be made from the metal.

Show your working clearly

(4 marks)

Q38**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 9 3 marks

- 8 A £1 coin weighs 8.75 g, correct to the nearest 0.01 g. Mitul weighs the contents of a large bag of £1 coins. The coins weigh 2.63 kg, correct to the nearest 10 g.

Mitul says

I am sure that the bag contains exactly £300 because, using bounds,
 $2625 \div 8.755 = 299.8$ to 1 decimal place.

Show that Mitul may not be correct.

[3]

(3 marks)

PAST PAPERS

Q39**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q17

CH 1 • TOPIC 9 4 marks

- 17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2HR • Q18

CH 1 • TOPIC 9 4 marks

18 Kaidan and Sonja went on two different car journeys.

For Kaidan's journey

distance = 80 km correct to the nearest 5 km

time = 2.7 hours correct to 1 decimal place

For Sonja's journey

distance = 33 km correct to 2 significant figures

time = 1 hour correct to the nearest 0.1 hour

Kaidan says,

“My average speed could have been greater than Sonja's average speed.”

By considering bounds, show that Kaidan is correct.

Show your working clearly.

(Total for Question 18 is 4 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q25

CH 1 • TOPIC 9 4 marks

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.
Give your answer in g/cm^3 correct to 2 decimal places.
Show your working clearly.

..... g/cm^3

(Total for Question 25 is 4 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q18

CH 1 • TOPIC 9 4 marks

18 Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.

The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.

Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.

Show your working clearly.

£.....

(Total for Question 18 is 4 marks)

Q43**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2HR • Q19

CH 1 • TOPIC 9 3 marks

19 Aviv goes on a cycle journey.

For the cycle journey

average speed = 19 km/h correct to the nearest whole number

time = 1.5 hours correct to one decimal place

Work out the upper bound for the distance Aviv travels.

Give your answer correct to 3 significant figures.

..... km

(Total for Question 19 is 3 marks)

Q44 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2H • Q22

CH 1 • TOPIC 9 4 marks

22 The diagram shows a square inside rectangle $ABCD$

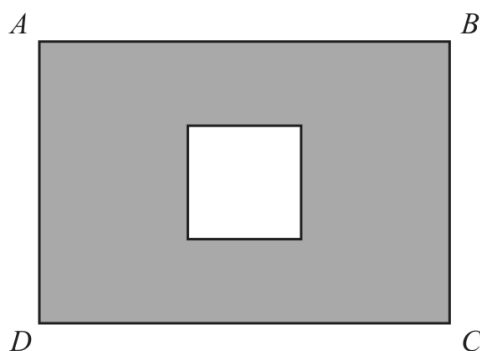


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square $= 4.1 \text{ cm}$ correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.

Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Q45**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q24

CH 1 • TOPIC 9 5 marks

24 $D = \frac{n}{p - q}$

 $n = 10.3$ correct to 1 decimal place $p = 7.24$ correct to 2 decimal places $q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy.
Show your working clearly.

.....
(Total for Question 24 is 5 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE BIGGEST**UP on top, LOW below**

UP – LOW

ZONE SMALLEST**LOW on top, UP below**

LOW – UP

- | | | |
|-----------|------------------------|--|
| 01 | Error intervals | Half the unit each way: $LB \leq x < UB$. |
| 02 | Estimate | Everything to 1 s.f.; nice squares under roots. |
| 03 | Formula bounds | Choose each letter to push the answer the way you want. |
| 04 | Decide | Extreme case for “could / definitely”; round both bounds to agree. |

BOTTOM LINE Unit → half → bounds → for each letter ask: does this make my answer bigger or smaller?

Answers

Rounding, Estimation and Bounds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • ROUNDING AND ERROR INTERVALS

- | | | |
|------------|---|---|
| Q1 | Dr Eslam • Half the unit | $6.35 \leq w < 6.45$ |
| Q2 | Practice • Easy | $4.755 \leq n < 4.765$ |
| Q3 | Practice • Easy | 33.75 cm, 33.85 cm |
| Q4 | Practice • Easy | (a) $8.35 \leq l < 8.45$ (b) $41.75 \leq P < 42.25$ |
| Q5 | Practice • Medium | (a) $5.5 \leq l < 6.5$ (b) e.g. $6.4 + 4.6 = 11$ |
| Q6 | Edexcel 4MA1 • January 2020 • Paper 1HR • Q11 | (a) 4.35 kg (b) 4.25 kg (c) $17.5 - 9.35 = 8.15$ |
| Q7 | Edexcel 4MA1 • January 2022 • Paper 2H • Q4 | (a) 17.75 mm (b) 18.25 mm |
| Q8 | Edexcel 4MA1 • November 2024 • Paper 2H • Q2 | (a) 1.45 m (b) 1.35 m |
| Q9 | Edexcel 4MA1 • June 2025 • Paper 1HR • Q2 | (i) 142.75 m (ii) 142.85 m |
| Q10 | Edexcel 4MA1 • November 2025 • Paper 1H • Q7 | (a) 474.5 g (b) 125 cm |

IDEA 02 • ESTIMATION

Q11 Dr Eslam • 1 s.f.

$$\frac{50 \times 4}{0.2} = 1000$$

Q12 Practice • Easy

$$\frac{300 \times 4}{60} = 20: \text{reasonable}$$

Q13 Practice • Medium

$$\frac{70 \times 40}{0.02} = 140\,000: \text{sensible}$$

Q14 Practice • Hard

$$\frac{90\,000}{2000} \approx 45$$

Q15 Practice • Very Hard

(a) $\frac{1000}{4 \times 10^{-22}} = 2.5 \times 10^{24}$ (b) underestimate:
the mass was rounded up

Q16 Edexcel 4MA1 • November 2025
• Paper 1H • Q2

$$\frac{2^2 \times 60}{3} = 80$$

IDEA 03 • BOUNDS IN A FORMULA

- Q17** Dr Eslam • Biggest $\frac{12.5}{3.35} = 3.73$
- Q18** Practice • Medium (a) 4.25 (b) $4.35 + \frac{1}{0.35} = 7.21$
- Q19** Practice • Medium $\frac{99.75}{66.5} = 1.5$
- Q20** Practice • Hard $\sqrt{8.35^2 - 6.05^2} = 5.75 \text{ cm}$
- Q21** Practice • Very Hard $\frac{347.5}{2.55 \times 6.35} = 21.46$
- Q22** Edexcel 4MA1 • January 2020 • Paper 1H • Q16 $\frac{2(58.45) - 19.5}{3.55} = 27.44$
- Q23** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q19 $8.35 \times 7.25 - 6.15 \times 5.25 = 28.25 \text{ cm}^2$
- Q24** Edexcel 4MA1 • January 2021 • Paper 1H • Q15 $\frac{9.65 - 3.75}{1.835} = 3.22$
- Q25** Edexcel 4MA1 • January 2021 • Paper 1HR • Q19 $\frac{13.5}{7.85 - 3.35} = 3$
- Q26** Edexcel 4MA1 • May 2021 • Paper 1H • Q20 $\frac{9.75 - 5.85}{2.5} = 1.56$
- Q27** Edexcel 4MA1 • November 2021 • Paper 1H • Q13 $\frac{16.5 \sin 125^\circ}{\sin 24.5^\circ} = 32.6 \text{ cm}$
- Q28** Edexcel 4MA1 • January 2022 • Paper 1H • Q18 $3.445 - \frac{1.95^2}{4.5} = 2.6$
- Q29** Edexcel 4MA1 • January 2022 • Paper 1HR • Q22 $\frac{11.55 \sin 26^\circ}{\sin 74.5^\circ} = 5.25 \text{ cm}$
- Q30** Edexcel 4MA1 • June 2022 • Paper 2H • Q18 $\frac{2(7.55) - 3.415}{1.5} = 7.79$
- Q31** Edexcel 4MA1 • November 2023 • Paper 2H • Q17 $8.35(2.5 + 17.5) = 167$

- Dr Eslam Ahmed • eliteigcse.com • WhatsApp 011 2000 9622
- Q32** Edexcel 4MA1 • June 2024 • Paper 2H • Q19 $\frac{7.5}{2(6.625) - 3(1.15)} = 0.765$

IDEA 04 • BOUNDS IN CONTEXT: DECIDE

- Q34** Dr Eslam • Could it? $UB = \frac{120.5}{1.45} = 83.1 > 82$: yes
- Q35** Practice • Hard $UB \frac{170.5}{53.5} = 3.19$, $LB \frac{169.5}{54.5} = 3.11$
- Q36** Practice • Hard $\max 30(25.5) + 20(7.55) = 916 \leq 920$: Yes
- Q37** Practice • Hard $\frac{125.5^3 \div 1000}{135} = 14.6$: 14 spheres
- Q38** Practice • Very Hard bounds $\frac{2625}{8.755} = 299.8$ to $\frac{2635}{8.745} = 301.3$:
could be £301
- Q39** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q17 $\frac{5.025}{1.835 \times 10^{-3}} = 2738.4 \text{ kg/m}^3$
- Q40** Edexcel 4MA1 • June 2022 • Paper 2HR • Q18 Kaidan $UB \frac{82.5}{2.65} = 31.1 > 31.0$ Sonja $LB \frac{32.5}{1.05} =$
- Q41** Edexcel 4MA1 • June 2023 • Paper 2H • Q25 $\frac{265}{\frac{4}{3}(2.75)^3} = 9.56 \text{ g/cm}^3$
- Q42** Edexcel 4MA1 • May 2023 • Paper 1HR • Q18 $\max 42 \text{ panels, min } 36: 6 \times 8.65 = \text{£}51.90$
- Q43** Edexcel 4MA1 • June 2024 • Paper 2HR • Q19 $19.5 \times 1.55 = 30.2 \text{ km}$
- Q44** Edexcel 4MA1 • June 2025 • Paper 2H • Q22 $UB 92.3, LB 85.7: X = 90 (1 \text{ s.f.})$
- Q45** Edexcel 4MA1 • November 2025 • Paper 2H • Q24 $UB 3.64, LB 3.58: D = 3.6$

Surds

Keep the root exact. Split out square factors, expand like brackets, and clear a surd from the bottom with its conjugate.



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THE TOPIC ON ONE MAP

Surds

01 Simplify a surd

$$\sqrt{20} = 2\sqrt{5} \quad \sqrt{45} + \sqrt{20} \quad \sqrt{675} = n\sqrt{27} \quad (\sqrt{3})^5$$

02 Expand brackets with surds

$$(\sqrt{12} - \sqrt{3})^2 \quad (8 - \sqrt{x})(5 + \sqrt{x}) \quad (a - b)(a + b)$$

03 Rationalise: one term

$$10/\sqrt{5} \quad (\sqrt{20} + \sqrt{80})/\sqrt{3} \quad \text{spot the mistake}$$

04 Rationalise: two terms

$$6/(3 - \sqrt{7}) \quad \text{"in the form } a + \sqrt{b}\text{"} \quad (\sqrt{2} - 1)^2 \text{ on the bottom}$$

05 Surds in problems

$$\text{rectangle width} \quad \text{cuboid volume} \quad \text{maximum area}$$

THREE RULES Everything in this topic comes from these:

ZONE 1

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

split out a square factor:

$$\sqrt{12} = \sqrt{4}\sqrt{3} = 2\sqrt{3}$$

ZONE 2

$$\sqrt{a} \times \sqrt{a} = a$$

a surd times itself is a whole number

ZONE 3

$$(a + \sqrt{b})(a - \sqrt{b})$$

$= a^2 - b$: the conjugate kills the surd

01 Simplify a Surd

Find the biggest square number that divides the number under the root. Take its root outside. Only **like** surds add.

WHEN YOU SEE → YOU DO

“without a calculator, show that $\sqrt{20} = 2\sqrt{5}$ ”

→ $\sqrt{4 \times 5} = \sqrt{4}\sqrt{5}$

$\sqrt{45} + \sqrt{20}$

→ simplify each first: $3\sqrt{5} + 2\sqrt{5} = 5\sqrt{5}$

“in the form $n\sqrt{27}$ ”

→ divide: $\sqrt{675} \div \sqrt{27} = \sqrt{25} = 5$

$(\sqrt{3})^5$

→ pairs: $\sqrt{3}\sqrt{3}\sqrt{3}\sqrt{3}\sqrt{3} = 9\sqrt{3}$

Biggest square factor

Split the root

Root the square

square numbers	4	9	16	25	36	49	64	100
example	$\sqrt{12}$	$\sqrt{45}$	$\sqrt{32}$	$\sqrt{75}$	$\sqrt{72}$	$\sqrt{98}$	$\sqrt{128}$	$\sqrt{300}$
simplified	$2\sqrt{3}$	$3\sqrt{5}$	$4\sqrt{2}$	$5\sqrt{3}$	$6\sqrt{2}$	$7\sqrt{2}$	$8\sqrt{2}$	$10\sqrt{3}$

× **TRAP** $\sqrt{9} + \sqrt{16} \neq \sqrt{25}$. Roots do not add like that: $3 + 4 = 7$, not 5.

IDEA 01

Practise: Simplify a Surd

My questions first, then selected practice.

Q1 **DR ESLAM** Square factor

CH 1 • TOPIC 10 2 marks

Simplify (a) $\sqrt{50}$ (b) $\sqrt{12} + \sqrt{27}$ **SELECTED PRACTICE****Q2** **PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

3 Show that $\frac{10}{\sqrt{5}}$ can be rewritten as $2\sqrt{5}$

(2 marks)

Q3**PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

- 7 Show that $\sqrt{45} + \sqrt{20} = 5\sqrt{5}$
Show your working clearly.

(2 marks)

Q4**PRACTICE** Practice • Medium

CH 1 • TOPIC 10 2 marks

- 8 Without using a calculator, show that $\sqrt{20} = 2\sqrt{5}$.

(2 marks)

Q5**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 3 marks

- 5 Show that $\sqrt{5}(\sqrt{8} + \sqrt{18})$ can be written in the form $a\sqrt{10}$ and state the value of a . You must show your full working.

(3 marks)

Q6**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 5 marks

- 7 (a) Show that $\sqrt{3} + \sqrt{12}$ can be rewritten as $3\sqrt{3}$

(2 marks)

(b) Show that $\left(\frac{1}{\sqrt{3}}\right)^7 = \frac{\sqrt{3}}{81}$

(3 marks)

02 Expand Brackets with Surds

Expand exactly like algebra (grid or FOIL). Every $\sqrt{a} \times \sqrt{a}$ becomes a ; collect whole numbers and surds separately.

WHEN YOU SEE → YOU DO

$$(\sqrt{12} - \sqrt{3})^2$$

→ write it twice: $(\)(\)$ — not $12 - 3$

$$(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$$

→ expand, then match the whole numbers and the surds

$$(a - b)(a + b) \text{ with } a = \sqrt{8} + 4$$

→ use $a^2 - b^2$, or subtract then add first

“simplifies to an integer”

→ show the surd terms cancel

WORKED MODEL • JANUARY 2021 P1H

$(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$, x prime, y integer. Find x and y .

1 $40 + 8\sqrt{x} - 5\sqrt{x} - x$

expand

2 $(40 - x) + 3\sqrt{x}$

collect

3 $40 - x = 21, 3 = y$

match the parts

ANSWER $x = 19, y = 3$

IDEA 02

Practise: Expand Brackets with Surds

My questions first, then selected practice, then real past papers.

Q7 DR ESLAM Expand

CH 1 • TOPIC 10 2 marks

Expand and simplify $(2 + \sqrt{3})^2$.

SELECTED PRACTICE**Q8** PRACTICE Practice • Medium

CH 1 • TOPIC 10 4 marks

1 (a) Show that $\frac{12}{\sqrt{3}}$ can be rewritten as $4\sqrt{3}$

(2 marks)

(b) Show that $(\sqrt{2} + \sqrt{8})^2 = 18$

(2 marks)

Q9**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 4 marks

1 (a) Show that $\frac{5}{\sqrt{2}}$ can be written as $\frac{5\sqrt{2}}{2}$

(2 marks)

(b) Show that $(2 + \sqrt{3})^2 - (2 - \sqrt{3})^2 = 8\sqrt{3}$

(2 marks)

Q10**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10 3 marks

4 Simplify fully $\frac{(6 - \sqrt{5})(6 + \sqrt{5})}{\sqrt{31}}$

You must show your working.

(3 marks)

Q11**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10**4 marks**

$$1 \quad a = \sqrt{8} + 2$$

$$b = \sqrt{8} - 2$$

$$T = a^2 - b^2$$

Show that $T = c\sqrt{2}$ stating the value of c

(4 marks)**PAST PAPERS****Q12****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1H • Q18

CH 1 • TOPIC 10**3 marks**

18 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer,

find the value of x and the value of y .

Show each stage of your working clearly.

 $x = \dots\dots\dots$
 $y = \dots\dots\dots$
(Total for Question 18 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q13

CH 1 • TOPIC 10**3 marks**

13 $a = \sqrt{8} + 4$

$b = \sqrt{8} - 4$

 $(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$ Find the value of y

Show your working clearly.

$y = \dots\dots\dots$

(Total for Question 13 is 3 marks)

03 Rationalise: One Term

A single surd on the bottom? Multiply top and bottom by that surd. The bottom becomes a whole number.

WHEN YOU SEE → YOU DO

$$\frac{10}{\sqrt{5}}$$

$$\rightarrow \times \frac{\sqrt{5}}{\sqrt{5}}: \frac{10\sqrt{5}}{5} = 2\sqrt{5}$$

$$\frac{5}{\sqrt{12}}$$

$$\rightarrow \text{simplify first: } \sqrt{12} = 2\sqrt{3}, \text{ then } \times \sqrt{3}$$

“find her mistake”

$$\rightarrow \text{check each line: } \sqrt{12} \times \sqrt{12} = 12, \text{ and } \sqrt{12} = 2\sqrt{3}$$

$$\frac{\sqrt{20} + \sqrt{80}}{\sqrt{3}} \quad \text{“in the form } \sqrt{a}\text{”}$$

$$\rightarrow \text{simplify the top, then square it all}$$

WORKED MODEL • ONE SURD

Rationalise $\frac{5}{\sqrt{12}}$.

$$1 \quad \frac{5}{2\sqrt{3}}$$

simplify first

$$2 \quad \frac{5 \times \sqrt{3}}{2\sqrt{3} \times \sqrt{3}} = \frac{5\sqrt{3}}{6}$$

$\times \sqrt{3}$ top and bottom

ANSWER $\frac{5\sqrt{3}}{6}$

IDEA 03

Practise: Rationalise: One Term

My questions first, then selected practice.

Q14 DR ESLAM One term

CH 1 • TOPIC 10 2 marks

Rationalise the denominator of $\frac{6}{\sqrt{2}}$.

SELECTED PRACTICE**Q15** PRACTICE Practice • Medium

CH 1 • TOPIC 10 3 marks

6 Show that $\frac{\sqrt{20} + \sqrt{80}}{\sqrt{3}}$ can be expressed in the form $\sqrt{a}$ where a is an integer.

Show your working clearly.

(3 marks)

Q16

PRACTICE

Practice • Hard

CH 1 • TOPIC 10 2 marks

3 (a) Martin did this question.

Rationalise the denominator of $\frac{14}{2 + \sqrt{3}}$

Here is how he answered the question.

$$\begin{aligned}\frac{14}{2 + \sqrt{3}} &= \frac{14 \times (2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} \\ &= \frac{28 - 14\sqrt{3}}{4 + 2\sqrt{3} - 2\sqrt{3} + 3} \\ &= \frac{28 - 14\sqrt{3}}{7} \\ &= 4 - 2\sqrt{3}\end{aligned}$$

Martin's answer is wrong.

Find Martin's mistake.

(1 mark)

(b) Sian did this question.

Rationalise the denominator of $\frac{5}{\sqrt{12}}$

Here is how she answered the question.

$$\begin{aligned}\frac{5}{\sqrt{12}} &= \frac{5\sqrt{12}}{\sqrt{12} \times \sqrt{12}} \\ &= \frac{5 \times 3\sqrt{2}}{12} \\ &= \frac{5\sqrt{2}}{4}\end{aligned}$$

Sian's answer is wrong.

Find Sian's mistake.

(1 mark)

Q17**PRACTICE**

Practice • Hard

CH 1 • TOPIC 10

3 marks

- 15 Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

(3 marks)**Q18****PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10

3 marks

- 3 Show that $\frac{1}{1 + \frac{1}{\sqrt{2}}}$ can be written as $2 - \sqrt{2}$

(3 marks)

04 Rationalise: Two Terms — the Conjugate

Bottom is $a + \sqrt{b}$? Multiply top and bottom by $a - \sqrt{b}$ (change the sign). The bottom becomes $a^2 - b$.

WHEN YOU SEE → YOU DO

$$\frac{6}{3 - \sqrt{7}}$$

→ $\times \frac{3 + \sqrt{7}}{3 + \sqrt{7}}$, bottom = $9 - 7 = 2$

“in the form $a + \sqrt{b}$ / $a - \sqrt{b}$ ”

→ rationalise, then $2\sqrt{3} = \sqrt{12}$

$(\sqrt{2} - 1)^2$ on the bottom

→ expand it first: $3 - 2\sqrt{2}$

“show each stage of your working”

→ conjugate line, expanded top, bottom $a^2 - b$

Change the sign

× top and bottom

Bottom = $a^2 - b$

Simplify

WORKED MODEL • JUNE 2025 P2HR

Show that $\frac{4}{3\sqrt{5} + 7}$ can be written as $a - \sqrt{b}$.

1
$$\frac{4(3\sqrt{5} - 7)}{(3\sqrt{5} + 7)(3\sqrt{5} - 7)}$$

conjugate

2
$$\frac{12\sqrt{5} - 28}{45 - 49} = \frac{12\sqrt{5} - 28}{-4}$$

bottom $a^2 - b$

3
$$7 - 3\sqrt{5}$$

divide

ANSWER $7 - \sqrt{45}$

× **TRAP** $(3\sqrt{5})^2 = 45$, not 15. Square the 3 as well.

IDEA 04

Practise: Rationalise: Two Terms

My questions first, then selected practice, then real past papers.

Q19 DR ESLAM Conjugate

CH 1 • TOPIC 10 3 marks

Rationalise $\frac{4}{3 - \sqrt{5}}$. Give your answer as $a + \sqrt{b}$.

SELECTED PRACTICE**Q20** PRACTICE Practice • Hard

CH 1 • TOPIC 10 3 marks

8 Rationalise the denominator of $\frac{4}{7 - \sqrt{5}}$

Show each stage of your working.

Give your answer in the form $a + b\sqrt{5}$ where a and b are fractions in their simplest forms.

(3 marks)

Q21 PRACTICE Practice • Hard

CH 1 • TOPIC 10 2 marks

12 Express $\frac{2}{\sqrt{3}-1}$ in the form $p + \sqrt{q}$

where p and q are integers.
Show your working clearly.

(2 marks)

Q22 PRACTICE Practice • Very Hard

CH 1 • TOPIC 10 3 marks

5 Rationalise the denominator of $\frac{a + \sqrt{4b}}{a - \sqrt{4b}}$ where a is an integer and b is a prime number.

Simplify your answer.

(3 marks)

Q23**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 10

4 marks

10 Show that $\frac{(4 + 2\sqrt{5})}{\sqrt{5} - 1}$ can be simplified to $\frac{3\sqrt{5} + 7}{2}$

(4 marks)**PAST PAPERS****Q24****PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q19

CH 1 • TOPIC 10

3 marks

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q17

CH 1 • TOPIC 10**3 marks**

17 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$ where a and b are integers.

Show each stage of your working clearly.

.....

(Total for Question 17 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q17

CH 1 • TOPIC 10**3 marks**

17 Show that $\frac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

(Total for Question 17 is 3 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q21

CH 1 • TOPIC 10**4 marks**

- 21 Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

(Total for Question 21 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1H • Q16

CH 1 • TOPIC 10**3 marks**

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2} - 1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$

Show your working clearly.

(Total for Question 16 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q16

CH 1 • TOPIC 10**3 marks**

- 16 Show that $\frac{2\sqrt{3}}{\sqrt{3}-1}$ can be written in the form $a + \sqrt{a}$ where a is an integer.

Show your working clearly.

(Total for Question 16 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q17

CH 1 • TOPIC 10**3 marks**

- 17 Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

(Total for Question 17 is 3 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q17

CH 1 • TOPIC 10**4 marks**

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

(b) Show that $\frac{5-\sqrt{2}}{\sqrt{2}-1}$ can be written in the form $a+b\sqrt{2}$ where a and b are integers.

(Total for Question 17 is 4 marks)

Q32 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 1H • Q15

CH 1 • TOPIC 10 4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k = \dots\dots\dots$

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(Total for Question 15 is 4 marks)

Q33 PAST PAPER Edexcel 4MA1 • June 2025 • Paper 2HR • Q16

CH 1 • TOPIC 10 3 marks

16 Show that $\frac{4}{3\sqrt{5} + 7}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show each stage of your working.

(Total for Question 16 is 3 marks)

Q34 **PAST PAPER** Edexcel 4MA1 • November 2025 • Paper 1H • Q17**CH 1 • TOPIC 10** **3 marks**

- 17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

05 Surds in Problems

Lengths are given exactly. Set up the equation as usual, then rationalise to find the unknown.

WHEN YOU SEE → YOU DO

“area 18, length $\sqrt{7} + 1$, find the width” → width = $\frac{18}{\sqrt{7} + 1}$, rationalise

volume of a cuboid with $x, x, 2\sqrt{5} - 3$ → $x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$

“ x for which R is a maximum” → vertex of the quadratic, then rationalise

WORKED MODEL • JANUARY 2020 P2H

A rectangle has area 18 cm^2 and length $(\sqrt{7} + 1) \text{ cm}$. Find the width as $a\sqrt{b} + c$.

$$1 \quad \frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1}$$

width = area ÷ length

$$2 \quad \frac{18(\sqrt{7} - 1)}{6}$$

bottom $7 - 1$

ANSWER $3\sqrt{7} - 3$

IDEA 05

Practise: Surds in Problems

My questions first, then real past papers.

PAST PAPERS**Q35****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2H • Q20

CH 1 • TOPIC 10**3 marks**

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1)\text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

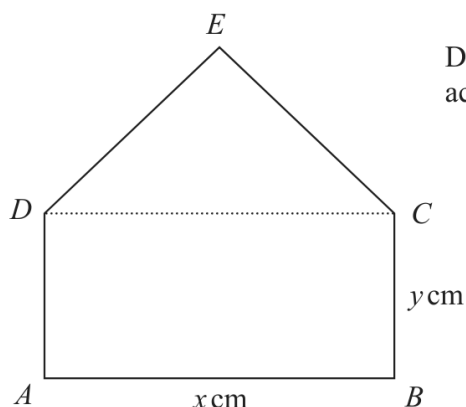
Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

(Total for Question 20 is 3 marks)

Q36 PAST PAPER Edexcel 4MA1 • November 2021 • Paper 1H • Q19

CH 1 • TOPIC 10 6 marks

19 $ABCED$ is a five-sided shape.Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.
 CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x = \dots\dots\dots$

(ii) Explain why the maximum value of R is given by this value of x .

.....

(Total for Question 19 is 6 marks)

Q37 PAST PAPER Edexcel 4MA1 • June 2023 • Paper 2H • Q23

CH 1 • TOPIC 10 4 marks

23 The diagram shows a cuboid with a square cross section.

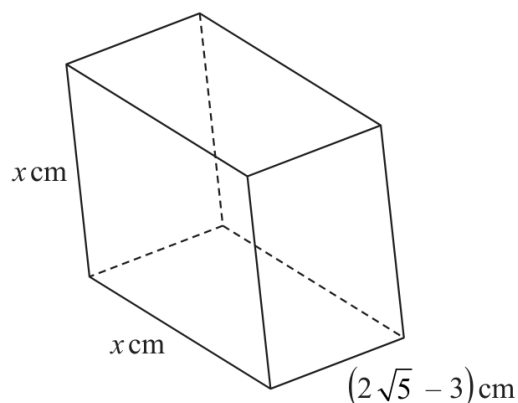


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

(Total for Question 23 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 1

$$\sqrt{ab} = \sqrt{a}\sqrt{b}$$

square factor out

ZONE 2

$$\sqrt{a}\sqrt{a} = a$$

ZONE 3

$$(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$$

- | | | |
|-----------|------------------|---|
| 01 | Simplify | Biggest square factor out; only like surds add. |
| 02 | Expand | Like algebra; $\sqrt{a}\sqrt{a} = a$; match parts. |
| 03 | One term | Multiply top and bottom by the surd. |
| 04 | Two terms | Conjugate; bottom $a^2 - b$. |
| 05 | Problems | Set up, then rationalise. |

BOTTOM LINE Never a surd on the bottom → one term: times itself → two terms: change the sign.

Answers

Surds • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • SIMPLIFY A SURD

- Q1** Dr Eslam • Square factor (a) $5\sqrt{2}$ (b) $2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$
- Q2** Practice • Medium $\frac{10\sqrt{5}}{5} = 2\sqrt{5}$
- Q3** Practice • Medium $3\sqrt{5} + 2\sqrt{5}$
- Q4** Practice • Medium $\sqrt{4}\sqrt{5}$
- Q5** Practice • Hard $\sqrt{40} + \sqrt{90} = 2\sqrt{10} + 3\sqrt{10}: a = 5$
- Q6** Practice • Hard (a) $\sqrt{3} + 2\sqrt{3}$ (b) $\frac{1}{27\sqrt{3}} = \frac{\sqrt{3}}{81}$

IDEA 02 • EXPAND BRACKETS WITH SURDS

- Q7** Dr Eslam • Expand $4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$
- Q8** Practice • Medium (a) $\frac{12\sqrt{3}}{3}$ (b) $2 + 2\sqrt{16} + 8 = 18$
- Q9** Practice • Hard (a) $\frac{5\sqrt{2}}{2}$ (b) $(7 + 4\sqrt{3}) - (7 - 4\sqrt{3})$
- Q10** Practice • Hard $\frac{31}{\sqrt{31}} = \sqrt{31}$
- Q11** Practice • Very Hard $T = 16\sqrt{2}: c = 16$
- Q12** Edexcel 4MA1 • January 2021 • Paper 1H • Q18 $x = 19, y = 3$
- Q13** Edexcel 4MA1 • January 2022 • Paper 2HR • Q13 $a^2 - b^2 = 32\sqrt{2} = 8\sqrt{32}: y = 8$

IDEA 03 • RATIONALISE: ONE TERM

Q14 Dr Eslam • One term

$$\frac{6\sqrt{2}}{2} = 3\sqrt{2}$$

Q15 Practice • Medium

$$\frac{6\sqrt{5}}{\sqrt{3}} = \sqrt{60}$$

Q16 Practice • Hard

(a) $4 - 3 = 1$ on the bottom, not 7 (b) $\sqrt{12} \times 3$
 wrong: $\sqrt{12} = 2\sqrt{3}$, answer $\frac{5\sqrt{3}}{6}$
 $\sqrt{135}$

Q17 Practice • Hard

Q18 Practice • Very Hard

$$\frac{\sqrt{2}}{\sqrt{2} + 1} = \sqrt{2}(\sqrt{2} - 1) = 2 - \sqrt{2}$$

IDEA 04 • RATIONALISE: TWO TERMS

Q19	Dr Eslam • Conjugate	$\frac{4(3 + \sqrt{5})}{4} = 3 + \sqrt{5}$
Q20	Practice • Hard	$\frac{7}{11} + \frac{1}{11}\sqrt{5}$
Q21	Practice • Hard	$1 + \sqrt{3}$
Q22	Practice • Very Hard	$\frac{a^2 + 4a\sqrt{b} + 4b}{a^2 - 4b}$
Q23	Practice • Very Hard	$\frac{(4 + 2\sqrt{5})(\sqrt{5} + 1)}{4} = \frac{14 + 6\sqrt{5}}{4}$
Q24	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q19	$9 + 3\sqrt{7}$
Q25	Edexcel 4MA1 • November 2021 • Paper 2H • Q17	$\sqrt{20} + 2$
Q26	Edexcel 4MA1 • January 2022 • Paper 2H • Q17	$-6 + \sqrt{48}$
Q27	Edexcel 4MA1 • June 2022 • Paper 1HR • Q21	$17 + \sqrt{288}$
Q28	Edexcel 4MA1 • June 2022 • Paper 1H • Q16	$12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12 = 2\sqrt{32} + 12$
Q29	Edexcel 4MA1 • June 2023 • Paper 2HR • Q16	$3 + \sqrt{3}$
Q30	Edexcel 4MA1 • May 2024 • Paper 1H • Q17	$2 + \sqrt{5}$
Q31	Edexcel 4MA1 • November 2024 • Paper 1H • Q17	(a) $n = 5$ (b) $3 + 4\sqrt{2}$
Q32	Edexcel 4MA1 • June 2025 • Paper 1H • Q15	(a) $k = 9$ (b) $9 + \sqrt{18}$
Q33	Edexcel 4MA1 • June 2025 • Paper 2HR • Q16	$7 - \sqrt{45}$
Q34	Edexcel 4MA1 • November 2025 • Paper 1H • Q17	$-10 - \sqrt{112}$

IDEA 05 • SURDS IN PROBLEMS

Q35Edexcel 4MA1 • January 2020 •
Paper 2H • Q20

$$3\sqrt{7} - 3$$

Q36Edexcel 4MA1 • November 2021
• Paper 1H • Q19

(b)(i) $x = \frac{100}{6 - \sqrt{3}}$ (ii) R is a quadratic in x with
negative x^2 : the vertex is the maximum
 $x^2 = 9 + 4\sqrt{5}$: $x = 2 + \sqrt{5}$

Q37Edexcel 4MA1 • June 2023 • Pa-
per 2H • Q23

Using a Calculator

It does exactly what you type: brackets round every top and bottom, the right key, the full display, then round.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Using a Calculator

01 Fractions, brackets and the full display

$$\frac{38.5 \times 14.2}{18.4 - 5.9}$$

“all the figures on your display”

then to 1 s.f.

02 Powers, roots and standard form

$$\sqrt[3]{340}$$

$$256^{0.25}$$

$$\sqrt[5]{3.2 \times 10^{11}}$$

$\times 10^x$ key

03 Trigonometry on the calculator

$$\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$$

degree mode

04 Check it is sensible

approximations

within 0.1 of 3.14?

KEY MAP The keys that cost marks when they are missing:

ZONE ()

Brackets

round every top,
every bottom
and everything
under a root

ZONE ■ / □

Fraction key

types the
fraction line for
you

ZONE $x^{\square}$

Power key

$1.3^{2.5}$, $32^{0.4}$

ZONE $\times 10^x$

Standard form

2.8×10^{-3} in one
go

01 Fractions, Brackets and the Full Display

Use the fraction key, or put brackets around the whole top and the whole bottom. Write **every** figure first — that is the method mark — then round.

WHEN YOU SEE → YOU DO

“write down all the figures on your calculator display”

→ copy the whole display, no rounding

“give your answer as a decimal”

→ press $S \leftrightarrow D$

“to 1 s.f. / 4 d.p.” in part (b)

→ round your part (a) answer

a long fraction bar

→ $(\text{top}) \div (\text{bottom})$

Fraction key

Type top then bottom

Write full display

Then round

WORKED MODEL • FULL DISPLAY

Work out $\frac{38.5 \times 14.2}{18.4 - 5.9}$. (b) Give it to 1 s.f.

1 $(38.5 \times 14.2) \div (18.4 - 5.9)$

brackets

2 43.736

full display

ANSWER (b) 40

× **TRAP** $38.5 \times 14.2 \div 18.4 - 5.9$ without brackets gives 23.8 — the calculator subtracts 5.9 only at the end.

IDEA 01

Practise: Fractions, Brackets and the Full Display

My questions first, then selected practice.

Q1 **DR ESLAM** Full display

CH 1 • TOPIC 11 2 marks

Work out $\frac{12.6 + 3.9}{2.5 \times 0.8}$. Write down all the figures on your display.

SELECTED PRACTICE**Q2** **PRACTICE** Practice • Easy

CH 1 • TOPIC 11 3 marks

1 (a) Use your calculator to work out $\frac{38.5 \times 14.2}{18.4 - 5.9}$

Write down all the figures on your calculator display.
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 1 significant figure.

(1 mark)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 3 marks

3 (a) Use your calculator to work out $\frac{\sqrt{7056}}{0.35 \times 12.8}$

Write down all the figures on your calculator display.
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 1 significant figure.

(1 mark)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 2 marks

7 Work out $\frac{\sqrt{17 + 4^2}}{7.3^2}$

Write down all the figures on your calculator display.

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 11 1 mark

11 Calculate.

$$\frac{5.39 - 0.98}{0.743 - 0.0743}$$

(1 mark)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11 3 marks

2 (a) Use your calculator to work out $\frac{\sqrt{70.25}}{4.2 - 2.37}$

Write down all the figures on your calculator display
You must give your answer as a decimal.

(2 marks)

(b) Write your answer to part (a) correct to 4 decimal places.

(1 mark)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11

2 marks**10** Calculate.

$$\frac{16.379 - 0.879}{4.2} \times 1.241$$

Give your answer correct to 2 significant figures.

(2 marks)

02 Powers, Roots and Standard Form

Use the power key for any power, the $\sqrt{\quad}$ key for any root, and the $\times 10^x$ key for standard form. Everything under a root goes inside the root.

WHEN YOU SEE → YOU DO

$$\sqrt[3]{340}, \sqrt[5]{3.2 \times 10^{11}}$$

→ $\sqrt{\quad}$ key (SHIFT)

$$1.3^{2.5}, 256^{0.25}$$

→ power key $x^{\square}$

$$10^{\frac{3}{4}}$$

→ power key with a fraction: $10^{(3\div 4)}$

“give your answer in standard form”

→ convert the display: $4730 = 4.73 \times 10^3$

WORKED MODEL • STANDARD FORM

Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$.

1 top = 273 000

use $\times 10^x$

2 273 000 $\div$ 0.06 = 4 550 000

divide

ANSWER 4.55×10^6

IDEA 02

Practise: Powers, Roots and Standard Form

My questions first, then selected practice.

Q8

DR ESLAM

Root key

CH 1 • TOPIC 11

1 mark

Work out $\sqrt[3]{2744}$.**SELECTED PRACTICE****Q9**

PRACTICE

Practice • Easy

CH 1 • TOPIC 11

2 marks

4 Work out $(5.6 \times 10^6) + (2.3 \times 10^5)$

(2 marks)

Q10**PRACTICE**

Practice · Easy

CH 1 · TOPIC 11 1 mark

10 Calculate.

$$\frac{5}{8} + \sqrt[3]{340}$$

(1 mark)

Q11**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 2 marks

6 Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

(2 marks)

Q12**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 1 mark

11 Calculate $\sqrt{256^{0.25} + 4 \times 8}$.

(1 mark)

Q13**PRACTICE**

Practice • Medium

CH 1 • TOPIC 11 1 mark

$$13 \quad \sqrt[3]{2^3 + 2}$$

(1 mark)

Q14**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 3 marks

5 (a) Use your calculator to work out $\frac{29^2 - 4.6}{\sqrt{35 - 1.9^3}}$

Write down all the figures on your calculator display.

(2 marks)

(b) Write your answer to part (a) correct to 4 significant figures.

(1 mark)

03 Trigonometry on the Calculator

Check the calculator is in **degrees** (a small D on the screen). Close every bracket after sin, cos, tan.

WHEN YOU SEE → YOU DO

sin, cos, tan **in the question**

→ Degree mode first

$\tan 60^\circ + 1$

→ $\tan(60) + 1$ — close the bracket before +1

$(\sin 33^\circ)^2$

→ $(\sin(33))^2$

WORKED MODEL • TRIG INSIDE A ROOT

Work out $\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$.

1 $\sqrt{}$ then fraction: $(\tan(60) + 1) / (\tan(60) - 1)$

brackets

ANSWER 1.931851653

× **TRAP** In radian mode $\tan 60$ is 0.32 — every answer is wrong. Look for the D.

IDEA 03

Practise: Trigonometry on the Calculator

My questions first, then selected practice.

Q15 DR ESAM Degrees

CH 1 • TOPIC 11 2 marks

Work out $\frac{5 \sin 30^\circ}{\cos 60^\circ}$.

SELECTED PRACTICE**Q16** PRACTICE Practice • Medium

CH 1 • TOPIC 11 2 marks

- 1 Calculate the value of $\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$

Write down all the figures on your calculator display.

You must give your answer as a decimal.

(2 marks)

Q17**PRACTICE**

Practice · Medium

CH 1 · TOPIC 11 3 marks

4 (a) Use your calculator to work out $\sqrt{\frac{\sin 25^\circ + \sin 40^\circ}{\cos 25^\circ - \cos 40^\circ}}$

Write down all the figures on your calculator display.

(2 marks)

(b) Write your answer to part (a) correct to 2 decimal places.

(1 mark)

Q18**PRACTICE**

Practice · Hard

CH 1 · TOPIC 11 3 marks

1 (a) Find the reciprocal of 2.5

(1 mark)

(b) Work out $\sqrt[3]{\frac{4.3 \times \tan 39^\circ}{23.4 - 6.06}}$

Give your answer correct to 3 significant figures.

(2 marks)

Q19**PRACTICE**

Practice · Hard

CH 1 · TOPIC 11 2 marks

3 Work out $\sqrt{\frac{2.5 \times \sin 43^\circ}{8.2^2 - 50.5}}$

Give your answer correct to 3 significant figures.

(2 marks)

Q20**PRACTICE**

Practice · Hard

CH 1 · TOPIC 11 1 mark

6 Use your calculator to work out $\sqrt{1 - (\sin 33^\circ)^2}$.

(1 mark)

04 Check It Is Sensible

Round each number to 1 significant figure (or a nice square), work it out in your head, then compare with the calculator answer.

WHEN YOU SEE → YOU DO

“use approximations to check”

→ each number to 1 s.f., show the working

“is within 0.1 of 3.14”

→ calculate, then $3.04 < \text{value} < 3.24$

WORKED MODEL • APPROXIMATION

$19.42^2 - \sqrt[3]{1006} \div 4.95 = 375.11 \dots$ Check it is sensible.

1 $20^2 - \sqrt[3]{1000} \div 5 = 400 - 2$

approximate

ANSWER $398 \approx 375$: sensible

IDEA 04

Practise: Check It Is Sensible

My questions first, then selected practice.

Q21 DR ESLAM Estimate

CH 1 • TOPIC 11 2 marks

Estimate $\frac{41.2 \times 9.8}{0.49}$.

SELECTED PRACTICE**Q22** PRACTICE Practice • Medium

CH 1 • TOPIC 11 2 marks

7 An approximation for the value of π is given by

$$4 \left(1 - \frac{22}{57} + \frac{22}{85} - \frac{22}{105} + \frac{22}{117} - \frac{22}{242} \right)$$

Use your calculator to show that this approximation is within 0.1 of 3.14

(2 marks)

Q23**PRACTICE**

Practice • Hard

CH 1 • TOPIC 11 3 marks

- 4 (a) Use your calculator to work out $19.42^2 - \sqrt[3]{1006} \div 4.95$

Write down your full calculator display.

(1 mark)

- (b) Use approximations to check that your answer to part (a) is sensible.

You **must** show your working.

(2 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Full display | Fraction key or brackets; write every figure, then round. |
| 02 | Powers / roots | $x^{\square}$, $\sqrt[\square]{}$, $\times 10^x$; convert to standard form. |
| 03 | Trig | Degree mode; close every bracket. |
| 04 | Check | 1 s.f. estimate, compare. |

BOTTOM LINE Brackets → right key → full display → then round → estimate to check.

Answers

Using a Calculator • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • FRACTIONS, BRACKETS AND THE FULL DISPLAY

Q1	Dr Eslam • Full display	8.25
Q2	Practice • Easy	(a) 43.736 (b) 40
Q3	Practice • Easy	(a) 18.75 (b) 20
Q4	Practice • Easy	0.1077981356
Q5	Practice • Easy	6.594885599
Q6	Practice • Medium	(a) 4.580069567 (b) 4.5801
Q7	Practice • Medium	4.6

IDEA 02 • POWERS, ROOTS AND STANDARD FORM

Q8	Dr Eslam • Root key	14
Q9	Practice • Easy	5.83×10^6
Q10	Practice • Easy	7.604532047
Q11	Practice • Medium	4.55×10^6
Q12	Practice • Medium	6
Q13	Practice • Medium	2.15443469
Q14	Practice • Hard	(a) 157.668255 (b) 157.7

IDEA 03 • TRIGONOMETRY ON THE CALCULATOR

- Q15** Dr Eslam • Degrees $\frac{5 \times 0.5}{0.5} = 5$
- Q16** Practice • Medium 1.931851653
- Q17** Practice • Medium (a) 2.75603957 (b) 2.76
- Q18** Practice • Hard (a) 0.4 (b) 0.586
- Q19** Practice • Hard 0.319
- Q20** Practice • Hard 0.8386705679

IDEA 04 • CHECK IT IS SENSIBLE

- Q21** Dr Eslam • Estimate $\frac{40 \times 10}{0.5} = 800$
- Q22** Practice • Medium 3.0418..., and $3.04 < 3.0418 < 3.24$: within 0.1
- Q23** Practice • Hard (a) 375.1121656 (b) $20^2 - 10 \div 5 = 398$: sensible

Ratio Toolkit

Two methods solve every ratio question: the Multiplier for recipes and scaling, and the Value of 1 Part for sharing.



DR ESLAM AHMED
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THE TOPIC ON ONE MAP

Ratio Toolkit

01 Recipes and scaling: the Multiplier

10 pancakes → 30

enough flour?

greatest number of biscuits

02 Simplify and $1 : n$

2 cm : 5 m

720 : 540

map scale $1 : n$

03 Sharing: the Value of 1 Part

£630 in $5 : 13$

12 more goals

36 pigs

04 Linking two ratios

$7 : 2$ and $5 : 9$

flour : butter : sugar

05 Ratio problems

coins worth 85p

counters added

best buy of paint

VALUE OF 1 PART Three ways to find one part — the question tells you which:

ZONE A

Given the TOTAL

$$1 \text{ part} = \frac{\text{total}}{\text{sum of parts}}$$

ZONE B

Given a DIFFERENCE

$$1 \text{ part} = \frac{\text{difference}}{\text{diff. in parts}}$$

ZONE C

Given ONE SHARE

$$1 \text{ part} = \frac{\text{that share}}{\text{its parts}}$$

01 Recipes and Scaling: the Multiplier

Find the Multiplier = $\frac{\text{new}}{\text{old}}$, then multiply every ingredient. For “greatest number”, find each ingredient’s multiplier — the **smallest** one wins.

WHEN YOU SEE → YOU DO

“ingredients for 16 ... he makes 24”

→ multiplier $\frac{24}{16} = 1.5$

“does she have enough flour?”

→ amount needed vs amount she has

“greatest number she can make”

→ each ingredient: have ÷ need — take the smallest

“160 cm weighs 17.8 g ... 210 cm”

→ $17.8 \times \frac{210}{160}$

Multiplier = new ÷ old

× every amount

Units and rounding

WORKED MODEL • GREATEST NUMBER

8 shortbread use 120 g butter, 60 g sugar, 180 g flour. Tariq has 330 g, 200 g, 450 g.

1 $330 \div 120 = 2.75$, $200 \div 60 = 3.3$, $450 \div 180 = 2.5$

each multiplier

2 smallest = 2.5 (flour runs out first)

the limit

ANSWER $2.5 \times 8 = 20$ biscuits

IDEA 01

Practise: Recipes and Scaling: the Multiplier

My questions first, then selected practice.

Q1 DR ESLAM Multiplier

CH 1 • TOPIC 12 2 marks

A recipe for 6 people uses 450 g of rice. How much rice for 10 people?

SELECTED PRACTICE

Q2**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12

4 marks

- 1 (a) Here are the ingredients needed to make 10 pancakes.

PancakesIngredients to make **10** pancakes300 *ml* of milk

120 g of flour

2 eggs

Matthew makes 30 pancakes.

Work out how much flour he uses.

(2 marks)

- (b) Tara makes some pancakes.

She uses 750 *ml* of milk.

Work out how many pancakes she makes.

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12

3 marks

- 3 This is a list of ingredients for making chicken soup for 4 people.

Ingredients for **4** people

60 g butter
300 g chicken
150 ml cream
1 onion
640 ml chicken stock

Bill is going to make chicken soup for 6 people.

Work out the amount of each ingredient he needs.

(3 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12**3 marks**

12 18 rice cakes weigh a total of 130 g. There are 329 calories in 100 g of rice cakes.

How many calories are there in one rice cake?

..... calories

(3 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12

4 marks

2 Here is a list of ingredients for making 18 mince pies.

Ingredients for mince pies

225 g of butter
350 g of flour
100 g of sugar
280 g of mincemeat
1 egg

Elaine wants to make 45 mince pies.

Elaine has

1 kg of butter
1 kg of flour
500 g of sugar
600 g of mincemeat
6 eggs

Does Elaine have enough of each ingredient to make 45 mince pies?
You must show clearly how you got your answer.

(4 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 3 marks

5 There are 1200 students at a school.

Kate is helping to organise a party.
She is going to order pizza.

Kate takes a sample of 60 of the students at the school.
She asks each student to tell her **one** type of pizza they want.

The table shows information about her results.

Pizza	Number of students
ham	20
salami	15
vegetarian	8
margarita	17

Work out how much ham pizza Kate should order.

Write down any assumption you make **and** explain how this could affect your answer.

(3 marks)

Q7**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12

5 marks

- 1 Colin works on 5 days each week.
Each day he drives from his home to work and from work to his home.

Colin pays £3.50 each day to use the car park at work.

The distance from Colin's home to work is 18 miles.
Colin's car uses one gallon of petrol every 45.2 miles.

1 litre of petrol costs 136.9p

1 gallon = 4.546 litres

Work out the total cost for Colin to use his car for work each week.
You must show all your working.

(5 marks)

02 Simplify and $1 : n$

Same units first, then divide both sides by the same number. For $1 : n$, divide both sides by the first number.

WHEN YOU SEE → YOU DO

“2 cm to 5 m in the form $1 : n$ ”

→ same units: $2 : 500 \rightarrow 1 : 250$

“in its simplest form”

→ divide by the HCF: $720 : 540 = 4 : 3$

“A is half of B”

→ $A : B = 1 : 2$

“scale $1 : 200$... length of the model”

→ real $\div 200$

WORKED MODEL • $1 : n$

Write 50 grams : 1 kilogram in the form $1 : n$.

1 50 : 1000

same units

ANSWER 1 : 20

× **TRAP** $2 \text{ cm} : 5 \text{ m} = 2 : 5$ is wrong — change metres to centimetres first.

IDEA 02

Practise: Simplify and $1 : n$

My questions first, then selected practice.

Q8 DR ESLAM $1 : n$

CH 1 • TOPIC 12 2 marks

Write $40 \text{ cm} : 2 \text{ m}$ in the form $1 : n$.**SELECTED PRACTICE****Q9** PRACTICE Practice • Easy

CH 1 • TOPIC 12 1 mark

9 A is half of B .Work out the ratio $A : B$

Circle your answer.

1 : 2	2 : 1	1 : 3	3 : 1
-------	-------	-------	-------

(1 mark)

Q10**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12 2 marks

- 11 The ratio 50 grams to 1 kilogram can be written in the form $1 : n$.

Find the value of n .

$n = \dots\dots\dots$

(2 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 4 marks

11 (a) Emma has a digital photo.

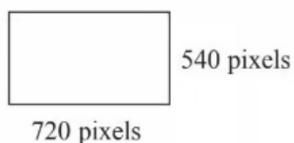


Diagram **NOT**
accurately drawn

The photo has a width of 720 pixels.

The photo has a height of 540 pixels.

Write down the ratio of the width of the photo to the height of the photo.

Give your ratio in its simplest form.

(2 marks)

(b) Emma wants the ratio of the width of the photo to the height of the photo to be 3 : 2

She reduces the number of pixels in the height of the photo.

The width of the photo is still 720 pixels.

The ratio of the width of the photo to the new height of the photo is 3 : 2

Work out the new height of the photo.

(2 marks)

Q12 **PRACTICE** Practice • Medium

CH 1 • TOPIC 12 3 marks

15 A plane has a length of 73 metres.

A scale model is made of the plane.
The scale of the model is 1 : 200

Work out the length of the scale model.
Give your answer in centimetres.

(3 marks)

Q13 **PRACTICE** Practice • Hard

CH 1 • TOPIC 12 3 marks

11 Work out

cube root of 512 : reciprocal of 0.4

Give your answer in the form $n : 1$

(3 marks)

Q14 **PRACTICE** Practice • Very HardCH 1 • TOPIC 12 **3 marks**

- 3 The surface gravity of a planet can be worked out using the formula

$$g = \frac{6.67 \times 10^{-11} m}{r^2}$$

where

m kilograms is the mass of the planet

r metres is the radius of the planet

For the Earth and Jupiter here are the values of m and r .

Earth	Jupiter
$m = 5.98 \times 10^{24}$	$m = 1.90 \times 10^{27}$
$r = 6.378 \times 10^6$	$r = 7.149 \times 10^7$

Work out the ratio of the surface gravity of Earth to the surface gravity of Jupiter.

Write your answer in the form $1 : n$

(3 marks)

03 Sharing: the Value of 1 Part

Step 1: find the value of 1 part (Zone A, B or C). Step 2: multiply by the number of parts you need.

WHEN YOU SEE → YOU DO

“share £630 in the ratio 5 : 13”

→ Zone A: $630 \div 18$

“Simon scored 12 more goals than Arjun”

→ Zone B: $12 \div (5 - 2)$

“95 girls, boys : girls = 4 : 5”

→ Zone C: $95 \div 5$

“the total that Asha and Julie receive is \$36”

→ Zone A on their parts only

Which zone?

Value of 1 part

× parts you need

WORKED MODEL • MAY 2023 P1H

Goals Arjun : Simon : Kath = 2 : 5 : 8. Simon scored 12 more than Arjun. How many did Kath score?

1 $12 \div (5 - 2) = 4$

Zone B: 1 part

ANSWER $8 \times 4 = 32$ goals

IDEA 03

Practise: Sharing: the Value of 1 Part

My questions first, then selected practice, then real past papers.

Q15 DR ESLAM Zone A

CH 1 • TOPIC 12 2 marks

Share £360 in the ratio 2 : 3 : 4.

Q16 DR ESLAM Zone B

CH 1 • TOPIC 12 2 marks

Boys : girls = 5 : 3. There are 18 more boys than girls. How many students?

SELECTED PRACTICE

Q17**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12 5 marks

6 (a) Jon shares £700 equally between his two children, Ellie and Maddie.

Ellie gives £125 of her share of the money to Maddie.

Write down the ratio of the amount of money Ellie now has to the amount of money Maddie now has.

(2 marks)

(b) Jenny shares £630 between her two children, Daniel and Rose, in the ratio 5 : 13

Work out how much money Jenny gives to each child.

(3 marks)

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12

6 marks

9 (a) Babajan makes breakfast cereal.

She mixes nuts, raisins and oats in the ratio 3 : 2 : 5 by weight.

On Monday, Babajan uses 60 grams of nuts.

Work out the weight of raisins and the weight of oats she uses to make the breakfast cereal.

(3 marks)

(b) On Tuesday, Babajan makes 300 grams of the breakfast cereal.

500 grams of nuts cost £8

Work out the cost of the nuts used to make 300 grams of the breakfast cereal.

(3 marks)

Q19**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12**3 marks**

- 4 Frank, Mary and Seth shared some sweets in the ratio 4 : 5 : 7 Seth got 18 more sweets than Frank.

Work out the total number of sweets they shared.

(3 marks)**Q20****PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12**3 marks**

- 4 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(3 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1H • Q1

CH 1 • TOPIC 12 3 marks

You must write down all the stages in your working.

- 1 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2:5:8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

.....

(Total for Question 1 is 3 marks)

04 Linking Two Ratios

Two ratios share one item. Make that item the **same number** in both (multiply each ratio), then join them into one three-part ratio.

WHEN YOU SEE → YOU DO

“chickens : ducks = 7 : 2, ducks : pigs = 5 : 9”

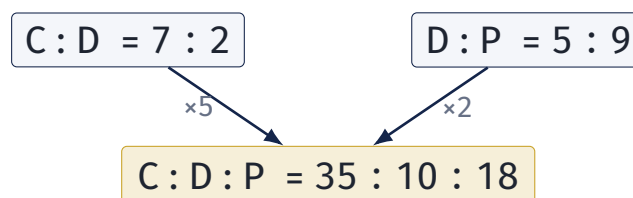
→ ducks → LCM 10: 35 : 10 : 18

“flour : butter = 6 : 5, butter : sugar = 3 : 2”

→ butter → 15: 18 : 15 : 10

“there are 36 pigs”

→ Zone C on the joined ratio



WORKED MODEL • MAY 2021 P1H

Chickens : ducks = 7 : 2, ducks : pigs = 5 : 9, 36 pigs. How many chickens?

1 35 : 10 : 18

ducks made equal (10)

2 $36 \div 18 = 2$

Zone C: 1 part

ANSWER $35 \times 2 = 70$ chickens

IDEA 04

Practise: Linking Two Ratios

My questions first, then real past papers.

Q22 DR ESLAM Link

CH 1 • TOPIC 12 3 marks

$a : b = 3 : 4$ and $b : c = 6 : 5$. Write $a : b : c$ in its simplest form.

PAST PAPERS**Q23** PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q4

CH 1 • TOPIC 12 3 marks

- 4 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is 7:2

The ratio of the number of ducks to the number of pigs is 5:9

There are 36 pigs on the farm.

Work out the number of chickens on the farm.

(Total for Question 4 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 2H • Q5**CH 1 • TOPIC 12** **3 marks**

5 Avril bakes a cake.

She uses flour, butter and sugar such that

$$\text{weight of flour : weight of butter} = 6 : 5$$

$$\text{weight of butter : weight of sugar} = 3 : 2$$

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

..... grams

(Total for Question 5 is 3 marks)

05 Ratio Problems

The ratio counts things, but the question may be about value, weight or cost. Use the parts, then turn them into what the question asks.

WHEN YOU SEE → YOU DO

“2p and 5p coins in ratio 1 : 3, total 85p”

→ value of one group: $1 \times 2 + 3 \times 5 = 17p$

“she puts more red counters in ... now 1 : 2”

→ yellow stays the same — find new red, subtract

“ratio now 10 : 11 — more than 90 people?”

→ list the possible totals

containers of paint, “make 40% profit”

→ cost per litre of the mix, then $\times 1.4$

WORKED MODEL • JUNE 2023 P2H

Nancy has 2p and 5p coins in ratio 1 : 3, total 85p. How many more 5p coins than 2p?

1 one group: $1 \times 2p + 3 \times 5p = 17p$

value of one group

2 $85 \div 17 = 5$ groups: 5 two-pence, 15 five-pence

number of groups

ANSWER $15 - 5 = 10$ more

IDEA 05

Practise: Ratio Problems

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q25****PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 4 marks

8 5 schools sent some students to a conference.

One of the schools sent both boys and girls.

This school sent 16 boys.

The ratio of the number of boys it sent to the number of girls it sent was 1:2

The other 4 schools sent only girls.

Each of the 5 schools sent the same number of students.

Work out the total number of students sent to the conference by these 5 schools.

(4 marks)

Q26**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12 3 marks

- 12 Purple paint is made by mixing red paint and blue paint in the ratio 5 : 2
Yan has 30 litres of red paint and 9 litres of blue paint.

What is the **maximum** amount of purple paint he can make?

.....litres

(3 marks)

Q27**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12 3 marks

- 5 The ratio $(y + x) : (y - x)$ is equivalent to $k : 1$

Show that $y = \frac{x(k+1)}{k-1}$

(3 marks)

Q28**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12 4 marks

Volume of a sphere = $\frac{4}{3} \pi r^3$ where r is the radius

Volume of a cone = $\frac{1}{3} \pi r^2 h$ where r is the radius and h is the perpendicular height

11

A sphere has radius $2x$ cm

A cone has

radius $3x$ cm

perpendicular height h cm

The sphere and the cone have the same volume.

Work out radius of cone : perpendicular height of cone

Give your answer in the form $a : b$ where a and b are integers.

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q5

CH 1 • TOPIC 12**6 marks**

- 5 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5
- On Sunday, all the children either go sailing or go climbing.
- $\frac{16}{25}$ of the boys go climbing.
- Twice as many girls go sailing as go climbing.
- Work out how many children go sailing on Sunday.

.....

(Total for Question 5 is 6 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q5

CH 1 • TOPIC 12**4 marks**

- 5 Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

.....

(Total for Question 5 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE A total

$\div$ sum of parts

ZONE B difference

$\div$ difference in parts

ZONE C one share

$\div$ its parts

- | | |
|----------------------|---|
| 01 Multiplier | new $\div$ old, times everything; greatest number: smallest multiplier. |
| 02 Simplify | Same units; $1 : n$ divide by the first number. |
| 03 Sharing | Value of 1 part (A, B or C), then $\times$ parts. |
| 04 Linking | Make the shared item equal, join the ratios. |
| 05 Problems | Parts $\rightarrow$ value / weight / cost; watch what stays the same. |

BOTTOM LINE Scaling $\rightarrow$ Multiplier • Sharing $\rightarrow$ Value of 1 Part • two ratios $\rightarrow$ make the middle equal.

Answers

Ratio Toolkit · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RECIPES AND SCALING: THE MULTIPLIER

- Q1** Dr Eslam · Multiplier $450 \times \frac{10}{6} = 750 \text{ g}$
- Q2** Practice · Easy (a) 360 g (b) 25 pancakes
- Q3** Practice · Easy 90 g, 450 g, 225 ml, 1.5 onions, 960 ml
- Q4** Practice · Easy 23.8 calories
- Q5** Practice · Medium needs 562.5 g butter, 875 g flour, 250 g sugar, 700 g mincemeat, 3 eggs: not enough mincemeat — No
- Q6** Practice · Medium 400 ham pizza portions; assumes the sample is representative
- Q7** Practice · Hard £42.28

IDEA 02 • SIMPLIFY AND 1 : n

- Q8** Dr Eslam · 1 : n $40 : 200 = 1 : 5$
- Q9** Practice · Easy 1 : 2
- Q10** Practice · Easy $n = 20$
- Q11** Practice · Medium (a) 4 : 3 (b) 480 pixels
- Q12** Practice · Medium 36.5 cm
- Q13** Practice · Hard $8 : 2.5 = 3.2 : 1$
- Q14** Practice · Very Hard 1 : 2.53

IDEA 03 • SHARING: THE VALUE OF 1 PART

- Q15** Dr Eslam • Zone A 1 part = £40: £80, £120, £160
- Q16** Dr Eslam • Zone B 1 part = 9: $8 \times 9 = 72$
- Q17** Practice • Easy (a) $225 : 475 = 9 : 19$ (b) Daniel £175, Rose £455
- Q18** Practice • Medium (a) raisins 40 g, oats 100 g (b) £1.44
- Q19** Practice • Hard 96
- Q20** Practice • Very Hard 147 euros
- Q21** Edexcel 4MA1 • May 2023 • Paper 1H • Q1 32 goals

IDEA 04 • LINKING TWO RATIOS

- Q22** Dr Eslam • Link 9 : 12 : 10
- Q23** Edexcel 4MA1 • May 2021 • Paper 1H • Q4 35 : 10 : 18: 70 chickens
- Q24** Edexcel 4MA1 • November 2023 • Paper 2H • Q5 18 : 15 : 10: 216 g

IDEA 05 • RATIO PROBLEMS

- Q25** Practice · Medium 240
- Q26** Practice · Hard 31.5 litres
- Q27** Practice · Very Hard $y + x = k(y - x) \Rightarrow y(k - 1) = x(k + 1)$
- Q28** Practice · Very Hard $h = \frac{32}{9}x: 27 : 32$
- Q29** Edexcel 4MA1 · January 2020 · Paper 1H · Q5 57
- Q30** Edexcel 4MA1 · June 2023 · Paper 2H · Q5 10

Ratio Problem Solving

Ratio mixed with fractions, percentages and changes: draw the parts, fill in what you know.



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THE TOPIC ON ONE MAP

Ratio Problem Solving

01 Ratio with fractions and percentages

$\frac{3}{10}$ into euros, rest 9 : 5 : 2

30% female

Joe gets 35%

02 Combining two or three ratios

almonds : cashews : peanuts

$e : g$ from $e : f, f : g$

$AB : BC : CD$

03 Before and after

each gives £10 to Pau

rent 2018 vs 2019

counters added

04 Ratio with algebra

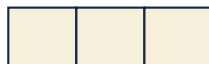
ages: 7 older, twice as old

$5a = 4c$

$(y + x) : (y - x) = k : 1$

BAR MODEL Draw the parts — every ratio problem becomes counting boxes:

Danil 3



Gabriel 5



Hadley 9



4 parts = 196 → 1 part = 49

01 Ratio with Fractions and Percentages

Change every fraction or percentage into parts of the same whole. What is left after a fraction is shared in a ratio — do the fraction first.

WHEN YOU SEE → YOU DO

“ $\frac{3}{10}$ into euros, the rest in 9 : 5 : 2”

→ rest = $\frac{7}{10}$, then split into 16 parts

“30% are female, 1295 are male”

→ male = 70%: 1% = 18.5

“Joe gets 35% in the ratio 7 : n”

→ 7 parts = 35%

“ $\frac{2}{3}$... $\frac{2}{5}$ of those ...”

→ multiply the fractions

WORKED MODEL • JANUARY 2021 P1H

20 160 rand: $\frac{3}{10}$ to euros, the rest in dollars : rupees : francs = 9 : 5 : 2.
How many more into dollars than francs?

1 rest = $\frac{7}{10} \times 20\,160 = 14\,112$

fraction first

2 $14\,112 \div 16 = 882$ per part

Zone A

ANSWER $(9 - 2) \times 882 = 6174$ rand

IDEA 01

Practise: Ratio with Fractions and Percentages

My questions first, then selected practice, then real past papers.

Q1 DR ESLAM Fraction first

CH 1 • TOPIC 13 3 marks

£600: $\frac{1}{4}$ is saved, the rest is shared by Ali and Ben in the ratio 2 : 7. How much does Ben get?

SELECTED PRACTICE

Q2 PRACTICE Practice • Medium

CH 1 • TOPIC 13 3 marks

- 2 30% of the people at a concert are female.
1295 of the people at the concert are male.

Work out the number of people at the concert who are female.

(3 marks)

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 13 3 marks

8 Anna and Lionel share \$675 in the ratio 4 : 5

Lionel gives $\frac{3}{5}$ of his share of the money to his mother.

How much money does Lionel give to his mother?

\$.....

(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 1 · TOPIC 13

4 marks

11 A bag contains some counters.

- There are 300 counters in the bag.
- There are only red, white and blue counters in the bag.
- The probability of picking a blue counter is $\frac{23}{50}$
- The ratio of red counters to white counters is 2 : 1.

Calculate the number of red counters in the bag.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 13 **5 marks**

10 The fares for a train journey are shown in the table below.

From London to Marseille	Standard fare	Premier fare
Adult	\$84	\$140
Child	\$60	\$96

For one journey from London to Marseille, the ratio

number of adults : number of children = 11 : 2.

There were 220 adults in total on this journey.

All of the children and 70% of the adults paid the standard fare.

The remaining adults paid the premier fare.

Calculate the total of the fares paid by the adults and the children.

\$

(5 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q1

CH 1 • TOPIC 13

3 marks

You must write down all the stages in your working.

- 1** Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6
The **total** amount of money that Asha and Julie receive is \$36

Work out the amount of money that Brendon receives.

\$.....

(Total for Question 1 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q1

CH 1 · TOPIC 13 4 marks

You must write down all the stages in your working.

- 1** Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

..... rand

(Total for Question 1 is 4 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q2

CH 1 · TOPIC 13 3 marks

- 2 Danil, Gabriel and Hadley share some money in the ratios 3:5:9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(Total for Question 2 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q6

CH 1 · TOPIC 13 4 marks

- 6 Eli and Peta share \$275 in the ratio 2:3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

02 Combining Two or Three Ratios

Make the shared item the same number in both ratios (use the LCM), then read any pair from the combined ratio.

WHEN YOU SEE → YOU DO

“ $e : f = 2 : 3$ and $f : g = 5 : 4$... find $e : g$ ”

→ $f \rightarrow 15: 10 : 15 : 12, e : g = 5 : 6$

“what fraction are sherberts?”

→ its part over the total parts

“there are less than 100 pens”

→ the total is a multiple of the sum

“ $AB : BD = 1 : 5, AC : CD = 7 : 11$ ”

→ same total length: 6 and 18 → use 18

WORKED MODEL • CASHEWS

1025 nuts. Almonds : cashews = 1 : 3, cashews : peanuts = 5 : 7. How many cashews?

1 cashews → 15: 5 : 15 : 21

combine

2 $1025 \div 41 = 25$

1 part

ANSWER $15 \times 25 = 375$ cashews

IDEA 02

Practise: Combining Two or Three Ratios

My questions first, then selected practice.

Q10 DR ESLAM Combine

CH 1 • TOPIC 13 3 marks

$x : y = 3 : 4$ and $y : z = 6 : 7$. Find $x : z$ in its simplest form.

SELECTED PRACTICE**Q11** PRACTICE Practice • Medium

CH 1 • TOPIC 13 2 marks

1 In a box of pens, there are

three times as many red pens as green pens

and two times as many green pens as blue pens.

For the pens in the box, write down

the ratio of the number of red pens to the number of green pens to the number of blue pens.

(2 marks)

Q12**PRACTICE**

Practice · Medium

CH 1 · TOPIC 13

4 marks

12 There is a total of 250 men, women and children on a train.

The ratio of men to women is 4 : 5.

The ratio of women to children is 10 : 7.

How many men are on the train?

(4 marks)**Q13****PRACTICE**

Practice · Hard

CH 1 · TOPIC 13

3 marks

4 There are only blue pens, green pens and red pens in a box.

The ratio of the number of blue pens to the number of green pens is 2 : 5

The ratio of the number of green pens to the number of red pens is 4 : 1

There are less than 100 pens in the box.

What is the greatest possible number of red pens in the box?

(3 marks)

Q14 **PRACTICE** Practice · Hard

CH 1 · TOPIC 13 3 marks

10 $e : f = 2 : 3$ and $f : g = 5 : 4$

Work out $e : g$

Give your answer in its simplest form.

(3 marks)

Q15 **PRACTICE** Practice · Very Hard

CH 1 · TOPIC 13 3 marks

4 The points A , B , C and D lie in order on a straight line.

$$AB : BD = 1 : 5$$

$$AC : CD = 7 : 11$$

Work out $AB : BC : CD$

(3 marks)

Q16

PRACTICE

Practice · Very Hard

CH 1 · TOPIC 136 marks

11 Some people who work for a company are in the following age groups.

Group A	Group B	Group C
Under 30 years	30 to 50 years	Over 50 years

The ratio of the number in group A to the number in group B is 7 : 10. The ratio of the number in group B to the number in group C is 4 : 3.

i) Find the ratio of the number in group A to the number in group C. Give your answer in its simplest form.

..... : [3]

ii) There are 45 people in group C.

Find the total number of people who work for the company.

[3]
(6 marks)

03 Before and After

Something changes — money given away, counters added. Work out the actual amounts before, apply the change, then write the new ratio. Look for what stays the same.

WHEN YOU SEE → YOU DO

“Sam and Tia each give £10 to Pau”

→ shares first, then +20, -10, -10

“Arjun paid the same rent both years”

→ make Arjun’s parts equal in both ratios

“she adds red counters, now 1 : 2”

→ yellow does not change

“the ratio is now 10 : 11 — more than 90?”

→ totals are multiples of 21

Amounts before

Apply the change

New ratio

Simplify

WORKED MODEL • JUNE 2025 P2H

Pau, Sam, Tia share £240 as 3 : 4 : 5. Sam and Tia each give £10 to Pau. New ratio?

1 $240 \div 12 = 20$: 60, 80, 100

amounts before

2 80, 70, 90

apply the change

ANSWER 8 : 7 : 9

IDEA 03

Practise: Before and After

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Before and after

CH 1 • TOPIC 13 3 marks

Red : blue counters = 2 : 5, 35 blue. 5 red are added. Find the new ratio of red : blue.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 13 4 marks

8 Arjun and Gretal each pay rent.

In 2018, the ratio of the amount each paid in rent was Arjun : Gretal = 5 : 7.

In 2019, the ratio of the amount each paid in rent was Arjun : Gretal = 9 : 13.

Arjun paid the same amount of rent in both 2018 and 2019.

Gretal paid \$290 more rent in 2019 than she did in 2018.

Work out the amount Arjun paid in rent in 2019.

\$

(4 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q6

CH 1 • TOPIC 13**4 marks**

- 6 Pau, Sam and Tia share £240 in the ratios 3:4:5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

04 Ratio with Algebra

When the information is a sentence (“7 years older”, “twice as old”) or an equation ($5a = 4c$), write each quantity in terms of one letter, then form the ratio.

WHEN YOU SEE → YOU DO

“Kiaria is 7 years older than Jay; Martha is twice Kiaria”

→ $J, J + 7, 2(J + 7)$

“ b is two thirds of c ; $5a = 4c$ ”

→ choose c so all are whole: $c = 15$

“ $(y + x) : (y - x) = k : 1$ — show that ...”

→ $y + x = k(y - x)$, rearrange

WORKED MODEL • AGES

Kiaria is 7 years older than Jay, Martha is twice as old as Kiaria, sum 77. Find Jay : Kiaria : Martha.

1 $J + (J + 7) + 2(J + 7) = 77 \Rightarrow 4J = 56$

one letter

2 $14 : 21 : 42$

ages

ANSWER $2 : 3 : 6$

IDEA 04

Practise: Ratio with Algebra

My questions first, then selected practice.

Q20 **DR ESLAM** One letterCH 1 • TOPIC 13 **3 marks**

Amy is 4 years older than Bea. Cara is 3 times Bea's age. Total 49. Find Amy : Bea : Cara.

SELECTED PRACTICE**Q21** **PRACTICE** Practice • HardCH 1 • TOPIC 13 **4 marks**

- 3 Kiaria is 7 years older than Jay.
Martha is twice as old as Kiaria.
The sum of their three ages is 77

Find the ratio of Jay's age to Kiaria's age to Martha's age.

(4 marks)

Q22**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 13 **3 marks**

6 b is two thirds of c .

$$5a = 4c$$

Work out the ratio $a : b : c$

Give your answer in its simplest form where a , b and c are integers.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Fractions / % | Everything as parts of one whole; fraction first. |
| 02 | Combine | Shared item → LCM; read any pair. |
| 03 | Before / after | Real amounts, change them, new ratio — find what stays fixed. |
| 04 | Algebra | One letter for everything, then the ratio. |

BOTTOM LINE Draw the parts → value of 1 part → amounts → then the change or the fraction.

Answers

Ratio Problem Solving • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RATIO WITH FRACTIONS AND PERCENTAGES

Q1	Dr Eslam • Fraction first	rest £450: $450 \div 9 \times 7 = \text{£}350$
Q2	Practice • Medium	555
Q3	Practice • Medium	\$225
Q4	Practice • Hard	108
Q5	Practice • Very Hard	\$24 576
Q6	Edexcel 4MA1 • January 2020 • Paper 1HR • Q1	\$13.50
Q7	Edexcel 4MA1 • January 2021 • Paper 1H • Q1	6174 rand
Q8	Edexcel 4MA1 • January 2022 • Paper 1H • Q2	147 euros
Q9	Edexcel 4MA1 • November 2025 • Paper 1H • Q6	$30 + 52.80 = \$82.80$

IDEA 02 • COMBINING TWO OR THREE RATIOS

Q10	Dr Eslam • Combine	$9 : 12 : 14$; $x : z = 9 : 14$
Q11	Practice • Medium	$6 : 2 : 1$
Q12	Practice • Medium	80
Q13	Practice • Hard	15
Q14	Practice • Hard	$5 : 6$
Q15	Practice • Very Hard	$3 : 4 : 11$
Q16	Practice • Very Hard	(i) $14 : 15$ (ii) 147

IDEA 03 • BEFORE AND AFTER

Q17	Dr Eslam • Before and after	$14 \text{ red} \rightarrow 19$; $19 : 35$
Q18	Practice • Very Hard	\$6525
Q19	Edexcel 4MA1 • June 2025 • Paper 2H • Q6	$8 : 7 : 9$

IDEA 04 • RATIO WITH ALGEBRA

Q20	Dr Eslam • One letter	$5B + 4 = 49$, $B = 9$; $13 : 9 : 27$
Q21	Practice • Hard	$2 : 3 : 6$
Q22	Practice • Very Hard	$12 : 10 : 15$

Exchange Rates and Best Buys



DR ESLAM AHMED
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Exchange rates: multiply one way, divide the other. Best buys: same unit, same currency, then compare and say which.

THE TOPIC ON ONE MAP

Exchange Rates and Best Buys

01 — Converting currencies

\$306 into euros at $\text{€}1 = \$1.125$

100-koruna notes

conversion graph

02 — Compare prices in two currencies

camera in Hong Kong

boots in London, Geneva, Paris

petrol UK vs USA

03 — Best buys

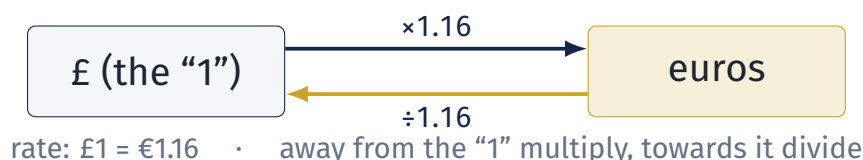
trays of plants

crisps packs

discount offers

water meter or not

MULTIPLY OR DIVIDE? Read the rate as “1 of these = so many of those”:



01 Converting Currencies

Find the side of the rate that says “1”. Going **away** from it: multiply. Going **towards** it: divide. Check the size makes sense.

WHEN YOU SEE → YOU DO

“£1 = 25.82 koruna ... change £200”

→ away from the 1: $\times 25.82$

“€1 = \$1.125 ... change \$306 into euros”

→ towards the 1: $\div 1.125$

“only 100-koruna notes, as many as possible”

→ round **down** to whole notes

“complete: \$1 = ... Namibian dollars”

→ divide: $3891 \div 300$

a conversion graph

→ read across and down; for big amounts scale up

WORKED MODEL • TOWARDS THE 1

Adele changes \$306 into euros. €1 = \$1.125.

1 dollars → euros is towards the “1”

decide

2 $306 \div 1.125$

divide

ANSWER €272

× **TRAP** Multiplying the wrong way gives €344 — more euros than dollars, though a euro is worth more. Always check the size.

IDEA 01

Practise: Converting Currencies

My questions first, then selected practice.

Q1 **DR ESLAM** Away from the 1

CH 1 • TOPIC 14 2 marks

£1 = 5.2 dirhams. Change £350 into dirhams.

Q2 **DR ESLAM** Towards the 1

CH 1 • TOPIC 14 2 marks

£1 = \$1.25. Change \$450 into pounds.

SELECTED PRACTICE

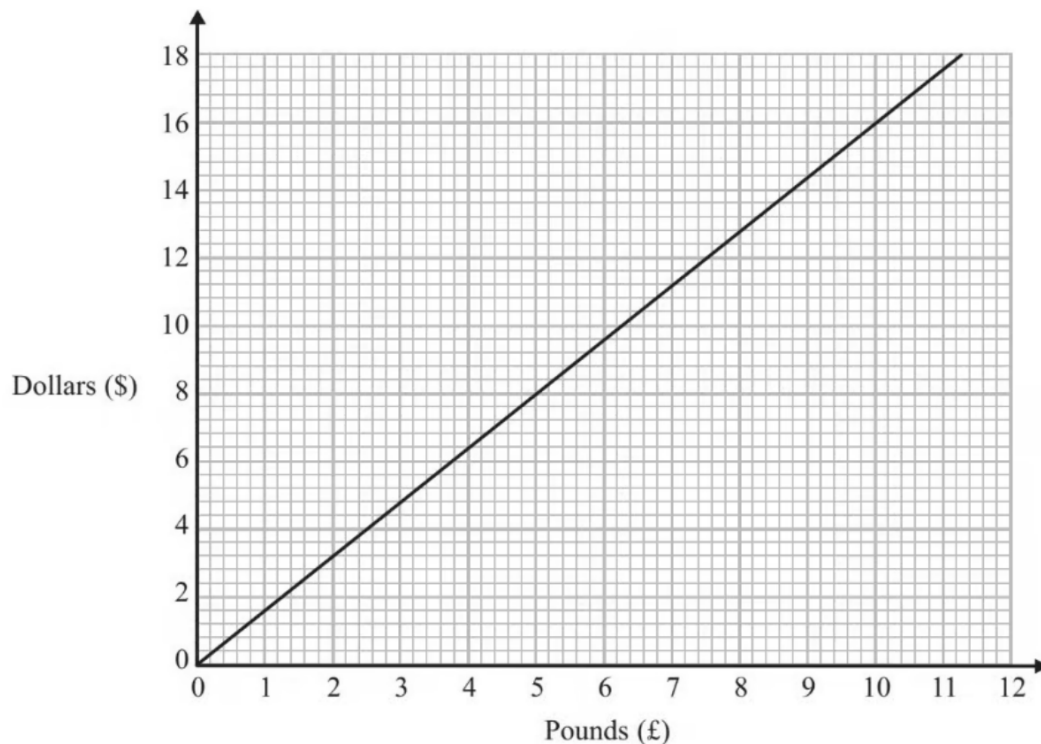
Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 14

5 marks

6 (a) You can use this conversion graph to change between pounds (£) and dollars (\$).



Use the conversion graph to change £5 to dollars.

(1 mark)

(b) Ella has \$200 and £800

Her hotel bill is \$600

Ella pays the bill with the \$200 and some of the pounds.

Use the conversion graph to work out how many pounds she has left.

(4 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 14 2 marks

- 8 Rashid changes 30 000 rupees to dollars when the exchange rate is \$1 = 68.14 rupees.

How many dollars does he receive?

\$

(2 marks)

Q5**PRACTICE**

Practice • Easy

CH 1 • TOPIC 14 2 marks

- 10 Adele changes \$306 into euros (€) when the exchange rate is €1 = \$1.125.

Calculate the number of euros she receives.

€

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 5 marks

7 (a) Asif is going on holiday to Turkey.

The exchange rate is $\text{£}1 = 3.5601$ lira.

Asif changes $\text{£}550$ to lira.

Work out how many lira he should get.

Give your answer to the nearest lira.

(2 marks)

(b) Asif sees a pair of shoes in Turkey. The shoes cost 210 lira.

Asif does not have a calculator.

He uses $\text{£}2 = 7$ lira to work out the approximate cost of the shoes in pounds.

Use $\text{£}2 = 7$ lira to show that the approximate cost of the shoes is $\text{£}60$

(2 marks)

(c) Is using $\text{£}2 = 7$ lira instead of using $\text{£}1 = 3.5601$ lira a sensible start to Asif's method to work out the cost of the shoes in pounds?

You must give a reason for your answer.

(1 mark)

Q7**PRACTICE**

Practice · Medium

CH 1 · TOPIC 14 2 marks

- 9 A new model locomotive costs \$64.

Calculate the cost of the locomotive in rupees when the exchange rate is 1 rupee = \$0.0154.
Give your answer correct to the nearest 10 rupees.

..... rupees

(2 marks)

Q8**PRACTICE**

Practice · Medium

CH 1 · TOPIC 14 2 marks

- 11 The Muller family are on holiday in New Zealand.

They change some euros (€) and receive \$1962 (New Zealand dollars).
The exchange rate is €1 = \$1.635 .

Calculate the number of euros they change.

€

(2 marks)

02 Compare Prices in Two Currencies

Convert everything into **one** currency (usually pounds), and into the **same unit** (per litre, per kg). Then compare and give the difference.

WHEN YOU SEE → YOU DO

“how much cheaper in Hong Kong?”

→ convert, subtract

“London, Geneva or Paris?”

→ all three into pounds

“per US gallon ... 1 gallon = 3.79 litres”

→ price per litre in the same currency

“give your answer in dollars, to the nearest cent”

→ convert the other price into dollars

Same currency

Same unit

Compare

Say which and by how much

WORKED MODEL • PETROL

UK: £1.24 per litre. USA: \$3.15 per US gallon. 1 gallon = 3.79 L, £1 = \$1.47.
Which is cheaper?

1 $3.15 \div 3.79 = \$0.831$ per litre

same unit

2 $0.831 \div 1.47 = £0.565$ per litre

same currency

ANSWER USA is cheaper ($£0.57 < £1.24$)

IDEA 02

Practise: Compare Prices in Two Currencies

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Compare

CH 1 • TOPIC 14 3 marks

A watch costs £84 in London and €99 in Rome. £1 = €1.15. Where is it cheaper, and by how much?

SELECTED PRACTICE**Q10** PRACTICE Practice • Medium

CH 1 • TOPIC 14 3 marks

4 Ben goes on holiday to Hong Kong.

In Hong Kong, Ben sees a camera costing HK\$3179.55

In London, an identical camera costs £285

The exchange rate is £1 = HK\$12.30

Ben buys the camera in Hong Kong.

How much cheaper is the camera in Hong Kong than in London?

(3 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 **2 marks**

12 A dressmaker charges \$35 or 2300 rupees to make a dress.

Calculate the difference in price when the exchange rate is 1 rupee = \$0.0153 .

Give your answer in rupees.

..... rupees

(2 marks)

Q12**PRACTICE**

Practice • Hard

CH 1 • TOPIC 14

3 marks

4 Identical pairs of boots are sold in London, in Geneva and in Paris.

These boots have a price of

£115 in London

189 Swiss francs in Geneva

174 euros in Paris

The exchange rates are

£1 = 1.39 Swiss francs

£1 = 1.27 euros

Are the boots the best value for money in London or in Geneva or in Paris?

You must show how you get your answer.

(3 marks)

Q13**PRACTICE**

Practice · Hard

CH 1 · TOPIC 14 3 marks

- 10** On holiday Maria paid 2.25 euros for a newspaper when the exchange rate was \$1 = 0.9416 euros.
At home Maria paid \$1.13 for the newspaper.

Calculate the difference in price.

Give your answer in dollars, correct to the nearest cent.

\$

(3 marks)

PAST PAPERS

Q14 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 2H · Q12

CH 1 · TOPIC 14 4 marks

- 12 Astrid wants to buy some oil.
She can buy the oil from either Dane Oil or Arctic Oil.

Here is information about the price that each company will charge Astrid.

Dane Oil	Arctic Oil
(4.2×10^5) litres for 2 500 000 Krone	(8.6×10^5) litres for 770 000 Dollars

Astrid wants to get the better value for money for the oil.

$$1 \text{ Dollar} = 6.57 \text{ Krone}$$

From which company should she buy her oil, Dane Oil or Arctic Oil?
You must show your working.

(Total for Question 12 is 4 marks)

03 Best Buys

Work out the cost of **one unit** (per plant, per ml, per bag) for every option. The smallest cost per unit is the best value.

WHEN YOU SEE → YOU DO

“which size is the best value for money?”

→ price ÷ quantity for each

“60% off, then 15% off” vs “ $\frac{2}{3}$ off”

→ final price of each

“buy 2, get 3 more at half price”

→ price of exactly the number needed

“should he have a water meter?”

→ yearly cost of each option

ZONE COST PER UNIT

price ÷ amount

smallest wins

ZONE AMOUNT PER £

amount ÷ price

biggest wins

WORKED MODEL • COFFEE

12 sachets £5.65, 20 sachets £9.20, 35 sachets £15.75. Best value?

1 47.1p, 46p, 45p per sachet

price ÷ sachets

ANSWER the large box (45p per sachet)

× **TRAP** Comparing the prices themselves. The large box costs most but is cheapest per sachet.

IDEA 03

Practise: Best Buys

My questions first, then selected practice.

Q15 **DR ESLAM** Best buy

CH 1 • TOPIC 14 3 marks

500 ml for £1.40 or 750 ml for £2.05. Which is the better value?

SELECTED PRACTICE**Q16** **PRACTICE** Practice • Easy

CH 1 • TOPIC 14 4 marks

- 1 Plants are sold in three different sizes of tray.

A small tray of 30 plants costs £6.50

A medium tray of 40 plants costs £8.95

A large tray of 50 plants costs £10.99

Kaz wants to buy the tray of plants that is the best value for money.

Which size tray of plants should she buy?

You must show all your working.

(4 marks)

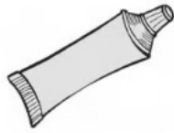
Q17**PRACTICE**

Practice · Easy

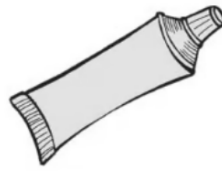
CH 1 · TOPIC 14

4 marks

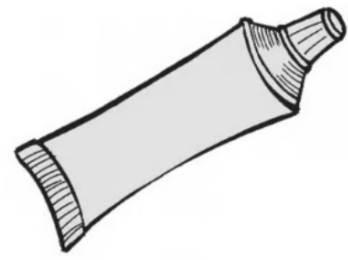
3 Toothpaste is sold in three different sizes of tube.



50ml



75ml



125ml

A 50 ml tube costs £ 1.09

A 75 ml tube costs £ 1.68

A 125 ml tube costs £ 2.69

Which tube of toothpaste is the best value for money?

You must show all your working.

(4 marks)

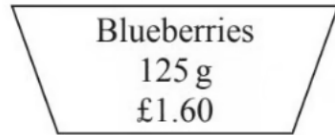
Q18**PRACTICE**

Practice · Easy

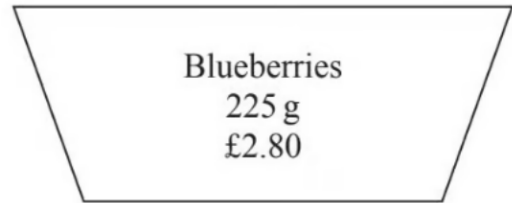
CH 1 · TOPIC 14

3 marks

5 Blueberries are sold in small cartons and in large cartons.



small carton



large carton

There are 125 g of blueberries in a small carton.

Each small carton costs £1.60

There are 225 g of blueberries in a large carton.

Each large carton costs £2.80

Which size of carton is the better value for money? You must show your working.

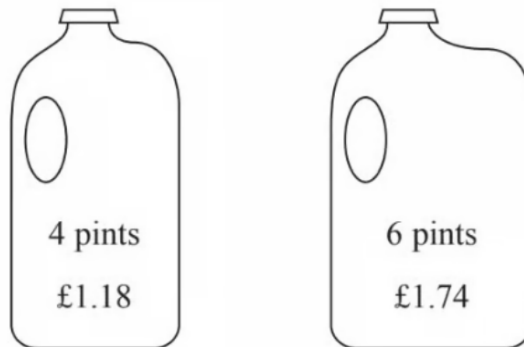
(3 marks)

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 14 3 marks

2 Milk is sold in two sizes of bottle.



A 4 pint bottle of milk costs £1 .18

A 6 pint bottle of milk costs £1 .74

Which bottle of milk is the best value for money?

You must show all your working.

(3 marks)

Q20**PRACTICE**

Practice • Hard

CH 1 • TOPIC 14 5 marks

- 2 Henry is thinking of having a water meter.

These are the two ways he can pay for the water he uses.

Water Meter

A charge of £28.20 per year

plus

91.22p for every cubic metre of water used

1 cubic metre = 1000 litres

No Water Meter

A charge of £107 per year

Henry uses an average of 180 litres of water each day.

Use this information to determine whether or not Henry should have a water meter.

(5 marks)

Q21**PRACTICE**

Practice · Hard

CH 1 · TOPIC 14 3 marks

9 Bottles of drink are for sale at three shops.

The normal price of a bottle is the same at each shop.

Shop A Buy 1 bottle Get 2 more bottles at half price	Shop B Buy 2 bottles Get 3 more bottles at half price
Shop C 30% off a bottle	

What is the cheapest way to buy **exactly** 8 bottles?

You can buy from more than one shop.

You **must** show your working.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------|--|
| 01 | Convert | Away from the “1”: multiply. Towards it: divide. Check the size. |
| 02 | Compare | Same currency, same unit, then the difference. |
| 03 | Best buy | Cost per unit — smallest wins; say which. |

BOTTOM LINE Find the “1” → multiply away, divide towards → compare like with like.

Answers

Exchange Rates and Best Buys • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • CONVERTING CURRENCIES

Q1	Dr Eslam • Away from the 1	$350 \times 5.2 = 1820$ dirhams
Q2	Dr Eslam • Towards the 1	$450 \div 1.25 = £360$
Q3	Practice • Easy	(a) \$7.50 (b) \$400 = £267: about £533 left
Q4	Practice • Easy	\$440.27
Q5	Practice • Easy	€272
Q6	Practice • Medium	(a) 1958 lira (b) $210 \div 7 \times 2 = 60$ (c) yes: $3.5 \approx 3.56$
Q7	Practice • Medium	4160 rupees
Q8	Practice • Medium	€1200

IDEA 02 • COMPARE PRICES IN TWO CURRENCIES

Q9	Dr Eslam • Compare	Rome: $99 \div 1.15 = £86.09$: London cheaper by £2.09
Q10	Practice • Medium	£258.50: £26.50 cheaper
Q11	Practice • Medium	12 rupees
Q12	Practice • Hard	Geneva £135.97, Paris £137.01: London (£115)
Q13	Practice • Hard	\$2.39 abroad: \$1.26 more
Q14	Edexcel 4MA1 • January 2020 • Paper 2H • Q12	Dane \$0.906 per litre, Arctic \$0.895: Arctic Oil

IDEA 03 • BEST BUYS

Q15	Dr Eslam • Best buy	0.28p vs 0.273p per ml: the 750 ml bottle
Q16	Practice • Easy	21.7p, 22.4p, 22.0p per plant: small tray
Q17	Practice • Easy	2.18p, 2.24p, 2.15p per ml: 125 ml
Q18	Practice • Easy	1.28p vs 1.24p per g: large carton
Q19	Practice • Medium	29.5p vs 29p per pint: 6 pints
Q20	Practice • Hard	meter £88.13 < £107: yes, have a meter
Q21	Practice • Hard	Shop A twice (6 bottles for 4) + 2 from Shop C: 5.4 bottle-prices

Direct & Inverse Proportion

Write the $\propto$, find k , write the link, then substitute. Every proportion question is the same four moves: L-I-N-K.



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THE TOPIC ON ONE MAP

Direct & Inverse Proportion

01 Proportion in words

cost of n items

workers and days

02 The LINK method

$$y = kx$$

$$y = \frac{k}{x}$$

03 Squares, cubes and roots

$$M = kh^3$$

$$y = \frac{k}{\sqrt{x}}$$

04 Graphs and tables

line through O

read a grid point

05 Two links

$$y \propto \frac{1}{\sqrt{x}}, x \propto T^3$$

06 Percentage change

$$1.15^2$$

$$2^3$$

01 Proportion in Words

No letters, no k ? Then it is a words question. Ask one thing first: **more of one — more or less of the other?**

WHEN YOU SEE → YOU DO

“in direct proportion”, cost of n items, recipe

→ find **one** unit, then multiply

“workers / machines / people ... days / hours”

→ inverse: find the **total work** first

“assume they work at the same rate”

→ comment: if slower, it takes longer

“paid for each hour or part of an hour”

→ round the hours **up**

Direct

more items → more cost divide, then multiply

Inverse

more workers → fewer days multiply (total work), then divide

1

Direct: find one

5 pens cost £3.40 → 1 pen = $3.40 \div 5 = £0.68$ → 8 pens = £5.44.

2

Inverse: find the total work

workers $\times$ days = total work (worker-days). Then divide by the new number of workers.

WORKED MODEL • MACHINES

6 machines make all the boxes in 9 days. Days 1, 2, 3 only 3, 4, 5 machines work; then all 6. Total days?

1 $6 \times 9 = 54$ machine-days

total work

2 $3 + 4 + 5 = 12$ done in 3 days

first days

3 $(54 - 12) \div 6 = 7$ more days

the rest

ANSWER $3 + 7 = 10$ days

× **TRAP** Inverse is **not** “divide the days”. More workers → fewer days: always go through the total work.

IDEA 01

Practise: Proportion in Words

My questions first, then selected practice.

Q1 DR ESLAM Find one

CH 1 • TOPIC 15 2 marks

5 pens cost £3.40. Work out the cost of 8 of these pens.

Q2 DR ESLAM Total work

CH 1 • TOPIC 15 2 marks

6 workers build a wall in 10 days. How many days would 4 workers take, working at the same rate?

SELECTED PRACTICE

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 3 marks

- 1 Yesterday it took 5 cleaners $4\frac{1}{2}$ hours to clean all the rooms in a hotel.

There are only 3 cleaners to clean all the rooms in the hotel today.

Each cleaner is paid £8.20 for each hour or part of an hour they work.

How much will each cleaner be paid today?

(3 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15

4 marks

9 (a) To complete a task in 15 days a company needs

4 people each working for 8 hours per day.

The company decides to have

5 people each working for 6 hours per day.

Assume that each person works at the same rate.

How many days will the task take to complete?

You **must** show your working.

(3 marks)

(b) Comment on how the assumption affects your answer to part (a).

(1 mark)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 8 marks

- 14 (a)** Sam and two friends put letters in envelopes on Monday. The three of them take two hours to put 600 letters in envelopes.

On Tuesday Sam has three friends helping.

Working at the same rate, how many letters should the **four** of them be able to put in envelopes in two hours?

(2 marks)

- (b)** Working at the same rate, how much longer would it take **four** people to put 1000 letters in envelopes than it would take **five** people?

(4 marks)

- (c)** Sam says

It took two hours for three people to put 600 letters in envelopes.

If I assume they work all day, then in one day three people will put 7200 letters in envelopes because $600 \times 12 = 7200$.

Why is Sam's assumption not reasonable? What effect has Sam's assumption had on her answer?

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15

6 marks

15 (a) Donald swims 3 lengths of a swimming pool in 93 seconds.

i) Use this information to show that he could swim 100 lengths in under 55 minutes.

[4]

ii) What assumption did you make in part (i)?

[1]

(5 marks)

(b) Donald tries to swim the 100 lengths in under 55 minutes.

Suggest one reason why he might not achieve this.

(1 mark)

Q7**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 3 marks

15 15 machines work at the same rate. Together, the 15 machines can complete an order in 8 hours.

3 of the machines break down after working for 6 hours. The other machines carry on working until the order is complete.

In total, how many hours does **each** of the other machines work?

.....hours

(3 marks)

Q8**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 4 marks

- 11** Beth and Mia translate documents from Spanish into English.
A set of documents that would take Beth 8 days would take Mia 10 days.

Beth starts to translate the documents.

After 2 days Beth and Mia both work on translating the documents.

How many **more** days will it take to complete the work?

You **must** show your working.

.....days

(4 marks)

02 The LINK Method: Direct and Inverse

Every algebra proportion question is solved with the same four steps. Learn the name: **L-I-N-K**.

WHEN YOU SEE → YOU DO

“y is (directly) proportional to x”,
“varies as”

$$\rightarrow y = kx$$

“y is inversely proportional to x”

$$\rightarrow y = \frac{k}{x}$$

“find a formula / equation linking”

→ stop after step 3

“find the value of ...”

→ do all 4 steps

1

L — Layout

Write $\propto$, then the equation with k : $y \propto \frac{1}{x} \Rightarrow y = \frac{k}{x}$.

2

I — Insert

Put the **first pair** of numbers in to find k . Never round k .

3

N — New equation

Rewrite with the value of k . This is the linking formula.

4

K — Keep solving

Substitute the new value and solve.

WORKED MODEL • INVERSE

y is inversely proportional to x . $y = 36$ when $x = 1.5$. Find y when $x = 6$.

1 $y = \frac{k}{x}$ L

2 $36 = \frac{k}{1.5} \Rightarrow k = 54$ I

3 $y = \frac{54}{x}$ N

4 $y = \frac{54}{6}$ K

ANSWER $y = 9$

CHECK Direct: x up $\rightarrow y$ up. Inverse: x up $\rightarrow y$ down. If not, look again at step 1.

IDEA 02

Practise: The LINK Method: Direct and Inverse

My questions first, then selected practice.

Q9 DR ESLAM Direct

CH 1 • TOPIC 15 3 marks

y is directly proportional to x . $y = 20$ when $x = 4$. Find y when $x = 7$.

Q10 DR ESLAM Inverse

CH 1 • TOPIC 15 3 marks

p is inversely proportional to q . $p = 6$ when $q = 4$. Find p when $q = 3$.

SELECTED PRACTICE

Q11 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

2 p is inversely proportional to t .

When $t = 4$, $p = 12$

Find the value of p when $t = 6$

(3 marks)

Q12 **PRACTICE** Practice · Medium

CH 1 • TOPIC 15 3 marks

3 d is inversely proportional to c

When $c = 280$, $d = 25$

Find the value of d when $c = 350$

(3 marks)

Q13**PRACTICE**

Practice · Medium

CH 1 • TOPIC 15

3 marks

4 y is inversely proportional to x

When $x = 1.5$, $y = 36$

Find the value of y when $x = 6$

(3 marks)**Q14****PRACTICE**

Practice · Medium

CH 1 • TOPIC 15

3 marks

12 y is inversely proportional to x .
 $y = 0.04$ when $x = 80$.

Find the value of y when $x = 32$.

 $y = \dots\dots\dots$ **(3 marks)**

Q15**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 1 mark

- 16 $V = \frac{k}{H}$ where k is a constant. Which **two** statements are correct?

Tick **two** boxes.

☐	V is directly proportional to H
☐	V is inversely proportional to H
☐	V is directly proportional to $\frac{1}{H}$
☐	V is inversely proportional to $\frac{1}{H}$

(1 mark)

Q16**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 1 mark

17 $xy = c$ where c is a constant. Circle the correct statement.

y is directly proportional to x	y is directly proportional to $\frac{1}{x}$
y is inversely proportional to $\frac{1}{x}$	x is directly proportional to y

(1 mark)

03 Squares, Cubes and Roots

Same four steps. The only new thing: the power sits on x in step 1 — and you undo it at the end.

WHEN YOU SEE → YOU DO

“the square of d ”, “ d^2 ”

$$\longrightarrow y = kd^2 \text{ or } y = \frac{k}{d^2}$$

“varies as the cube of h ”

$$\longrightarrow M = kh^3 \rightarrow \text{at the end take } \sqrt[3]{}$$

“the square root of q ”, “ $\sqrt{q}$ ”

$$\longrightarrow y = k\sqrt{q} \text{ or } y = \frac{k}{\sqrt{q}}$$

“the positive value of x ”

$$\longrightarrow \text{square root} \rightarrow \text{take the } + \text{ root}$$

directly proportional to the square of x

$$y = kx^2$$

inversely proportional to the cube of x

$$y = \frac{k}{x^3}$$

directly proportional to the square root of x

$$y = k\sqrt{x}$$

inversely proportional to $\sqrt{x}$

$$y = \frac{k}{\sqrt{x}}$$

WORKED MODEL • JUNE 2022 P2H

M varies directly as the cube of h . $M = 4$ when $h = 0.5$. Find h when $M = 500$.

1 $M = kh^3$

L

2 $4 = k(0.5)^3 = 0.125k \Rightarrow k = 32$

I

3 $M = 32h^3$

N

4 $500 = 32h^3 \Rightarrow h^3 = 15.625$

K

ANSWER $h = \sqrt[3]{15.625} = 2.5$

× **TRAP** Put the power on x **before** you find k : $4 = k \times 0.5$ is wrong — it is $4 = k \times 0.5^3$.

IDEA 03

Practise: Squares, Cubes and Roots

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Square

CH 1 • TOPIC 15 3 marks

y is directly proportional to the square of x . $y = 50$ when $x = 5$. Find y when $x = 6$.

Q18 DR ESLAM Inverse square

CH 1 • TOPIC 15 3 marks

y is inversely proportional to x^2 . $y = 3$ when $x = 2$. Find y when $x = 6$.

SELECTED PRACTICE

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 15 4 marks

6 (a) T is directly proportional to the cube of r

$$T = 21.76 \text{ when } r = 4$$

Find a formula for T in terms of r

(3 marks)

(b) Work out the value of T when $r = 6$

(1 mark)

Q20**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 3 marks

4 The intensity of the sound, I watts/m², received from a loudspeaker is inversely proportional to the square of the distance, d metres, from the loudspeaker.

When $d = 2$, $I = 30$ Work out the value of I when $d = 10$

(3 marks)

Q21**PRACTICE**

Practice · Hard

CH 1 • TOPIC 15 5 marks

14 (a) y is inversely proportional to $\sqrt{x}$

$$y = 4 \quad \text{when} \quad x = 9$$

Work out an equation connecting y and x .

(3 marks)

(b) Work out the value of y when $x = 25$

(2 marks)

Q22**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 5 marks

- 9 (a) y is directly proportional to the cube of x

$$y = 20h \text{ when } x = h \quad (h \neq 0)$$

Find a formula for y in terms of x and h

$$y = \dots\dots\dots$$

(3 marks)

- (b) Find x in terms of h when $y = 67.5h$

Give your answer in its simplest form.

$$x = \dots\dots\dots$$

(2 marks)

PAST PAPERS

Q23**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q14

CH 1 • TOPIC 15**3 marks**

14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

(Total for Question 14 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • May 2021 • Paper 2H • Q16**CH 1 • TOPIC 15** **6 marks**

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....

(b) Find the value of A when $r = 7.5A$

$$A = \text{.....}$$

(Total for Question 16 is 6 marks)

Q25 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 1H • Q15**CH 1 • TOPIC 15** **3 marks**

15 A is inversely proportional to C^2

$$A = 40 \text{ when } C = 1.5$$

Calculate the value of C when $A = 1000$

$$C = \text{.....}$$

(Total for Question 15 is 3 marks)

Q26**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q17

CH 1 • TOPIC 15 4 marks

17 M varies directly as the cube of h

$$M = 4 \text{ when } h = 0.5$$

Find the value of h when $M = 500$

(Total for Question 17 is 4 marks)

Q27**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q17

CH 1 • TOPIC 15 4 marks

17 F is inversely proportional to the square of r

$$F = 36 \text{ when } r = 4$$

(a) Find a formula for F in terms of r

(b) Work out the value of F when $r = 48$

(Total for Question 17 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q17

CH 1 • TOPIC 15 3 marks

17 Q is directly proportional to the square root of d

$$Q = 4.5 \text{ when } d = 324$$

Find a formula for Q in terms of d

(Total for Question 17 is 3 marks)

Q29**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q14

CH 1 • TOPIC 15 3 marks

14 B is inversely proportional to the square of d

$$B = 0.25 \text{ when } d = 12$$

Find a formula for B in terms of d

(Total for Question 14 is 3 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q18

CH 1 • TOPIC 15

5 marks

18 F is inversely proportional to the cube of r

$$F = 6 \text{ when } r = 2$$

(a) Find a formula for F in terms of r

.....

(b) Find the value of r when $F = 3072$

$$r = \text{.....}$$

(Total for Question 18 is 5 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q18

CH 1 • TOPIC 15

4 marks

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \text{.....}$$

(Total for Question 18 is 4 marks)

Q32**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q17

CH 1 • TOPIC 15 3 marks

17 T is inversely proportional to $\sqrt{m}$

$$T = 15 \text{ when } m = 36$$

Find a formula for T in terms of m

.....
(Total for Question 17 is 3 marks)

04 Graphs and Tables

Know the four shapes by heart. A table? Test $\frac{y}{x}$ (direct) or xy (inverse) — if it stays the same, you found the link.

WHEN YOU SEE → YOU DO

straight line through the origin

$$\rightarrow y \propto x$$

curve going down, never touching the axes

$$\rightarrow y \propto \frac{1}{x}$$

“the graph shows ... find a formula”

→ read one clear point, use it as the first pair

“complete the table” / “show the values fit”

→ find k from the full column, then use it

$$y \propto x$$

straight line through O

a line *not* through O
is not proportion

$$y \propto x^2$$

curve bending **up** from O

gets steeper

$$y \propto \sqrt{x}$$

curve bending **over** from O

gets flatter

$$y \propto \frac{1}{x}, \frac{1}{x^2}$$

curve coming down towards the x -axis

never meets the axes

WORKED MODEL • JUNE 2023 P2H

Q is directly proportional to $\sqrt{t}$. The graph passes through (4, 6). Find Q in terms of t .

1 $Q = k\sqrt{t}$

L

2 $6 = k\sqrt{4} = 2k \Rightarrow k = 3$

I • pick a grid point

ANSWER $Q = 3\sqrt{t}$

PICK THE POINT Choose a point where the curve crosses a grid corner exactly — and where $\sqrt{t}$ is nice ($t = 1, 4, 9$).

IDEA 04

Practise: Graphs and Tables

My questions first, then selected practice, then real past papers.

Q33 DR ESLAM Test the table

CH 1 • TOPIC 15 2 marks

x : 2, 4, 8 y : 12, 6, 3.

Is y directly or inversely proportional to x ? Give the formula.

SELECTED PRACTICE**Q34** PRACTICE Practice • Medium

CH 1 • TOPIC 15 2 marks

7 y is inversely proportional to x .

Complete the table.

x	12	6	
y		4	8

(2 marks)

Q35**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 4 marks

- 3 Given that $y \propto \frac{1}{x^2}$, complete this table of values.

x	1	2	5	10
y				1

(4 marks)

Q36**PRACTICE**

Practice • Hard

CH 1 • TOPIC 15 5 marks

- 12 (a) The following table gives values of x and y where y is inversely proportional to the square of x .

x	1.5	2	3	4
y	16	9	4	2.25

Find a formula for y in terms of x .

(3 marks)

- (b) Given that $x > 0$

find the value of x when $y = 144$

(2 marks)

Q37

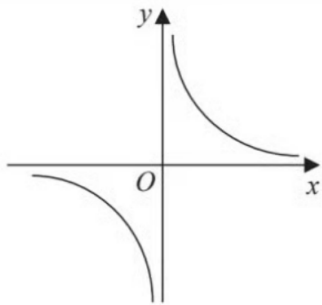
PRACTICE

Practice • Very Hard

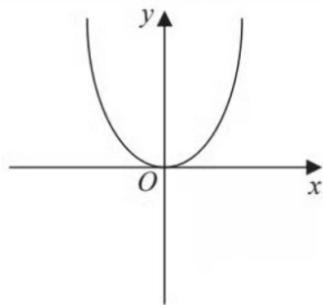
CH 1 • TOPIC 15

2 marks

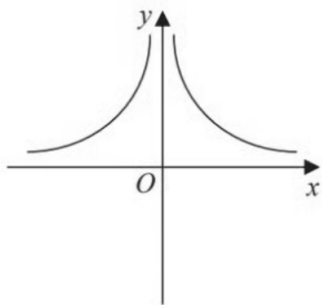
6 These graphs show four different proportionality relationships between y and x .



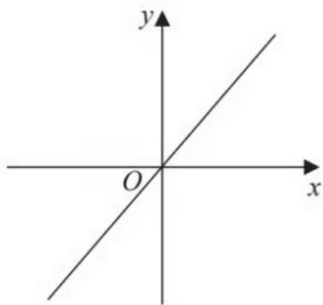
Graph A



Graph B



Graph C



Graph D

Match each graph with a statement in the table below.

Proportionality relationship	Graph letter
y is directly proportional to x	
y is inversely proportional to x	
y is proportional to the square of x	
y is inversely proportional to the square of x	

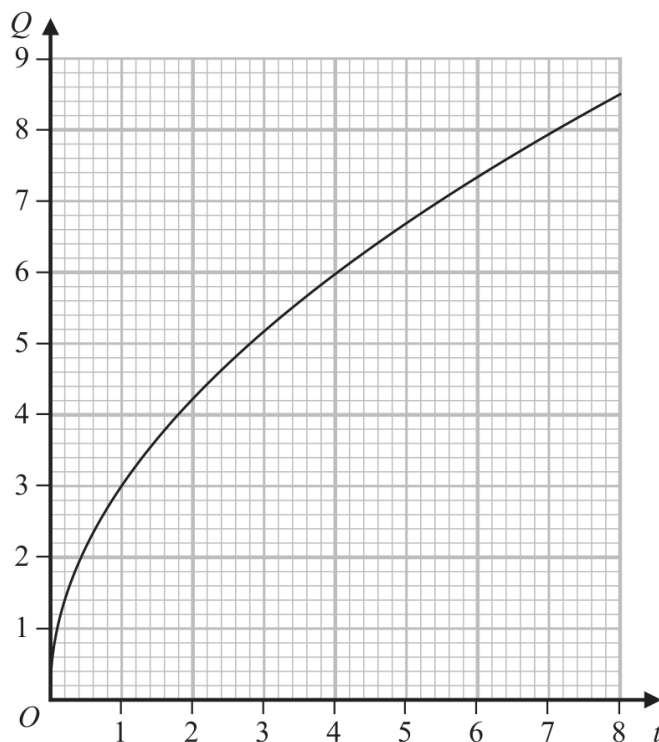
(2 marks)

PAST PAPERS

Q38**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q16

CH 1 • TOPIC 15 5 marks

16 Q is directly proportional to $\sqrt{t}$ The graph shows the relationship between Q and t for $0 < t < 8$ (a) Find a formula for Q in terms of t

.....

 Q is increased by 20%(b) Find the percentage increase in t

.....%

(Total for Question 16 is 5 marks)

05 Chained Proportion (Two Links)

Two sentences with $\propto$? Do LINK twice, then put Link 2 inside Link 1. Nothing new.

WHEN YOU SEE → YOU DO

" $y \propto \dots d$ and $d \propto \dots x$ "	→ Link 1 ($y-d$), Link 2 ($d-x$)
"find a formula for y in terms of x "	→ substitute Link 2 into Link 1
only one pair given for both links	→ use one constant K for the chain
"exact value"	→ keep fractions and roots — no decimals

1

Link 1

Find k for the first pair of letters.

2

Link 2

Find the second constant for the other pair.

3

Substitute

Replace the middle letter, then simplify. The middle letter must disappear.

WORKED MODEL • JANUARY 2022 P2HR

$y \propto \frac{1}{\sqrt{x}}$, $x \propto T^3$. $y = 8$ when $T = 25$. Find the exact value of T when $y = 27$.

$$1 \quad y = \frac{k}{\sqrt{cT^3}} = \frac{K}{T^{3/2}}$$

one chain constant

$$2 \quad 8 = \frac{K}{25^{3/2}} = \frac{K}{125} \Rightarrow K = 1000$$

I

$$3 \quad 27 = \frac{1000}{T^{3/2}} \Rightarrow T^{3/2} = \frac{1000}{27}$$

K

$$\text{ANSWER} \quad T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

× **TRAP** Do not use the numbers of Link 1 inside Link 2. Each link has its own pair of values.

IDEA 05

Practise: Chained Proportion (Two Links)

My questions first, then selected practice, then real past papers.

Q39 DR ESLAM Two links

CH 1 • TOPIC 15 4 marks

y is directly proportional to z ; $y = 12$ when $z = 4$. z is inversely proportional to x ; $z = 3$ when $x = 2$. Find a formula for y in terms of x .

SELECTED PRACTICE

Q40**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 5 marks

- 3 y is inversely proportional to d^2
When $d = 10$, $y = 4$

d is directly proportional to x^2
When $x = 2$, $d = 24$

Find a formula for y in terms of x .
Give your answer in its simplest form.

(5 marks)

Q41 **PRACTICE** Practice • Very Hard

CH 1 • TOPIC 15 4 marks

7 h is inversely proportional to p

p is directly proportional to $\sqrt{t}$

Given that $h = 10$ and $t = 144$ when $p = 6$
find a formula for h in terms of t

(4 marks)

Q42**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15 5 marks

13 P , Q and R have positive values.

P is directly proportional to the square of Q .

When $P = 1.25$, $Q = 0.5$

Q is inversely proportional to R .

When $Q = 0.5$, $R = 6$

Work out the value of R when $P = 0.8$

(5 marks)

Q43**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 15

4 marks

- 16** x is directly proportional to y .
 y is directly proportional to z .

When $x = 10$, $y = 60$.

When $y = 8$, $z = 1.6$.

Find a formula for z in terms of x .

(4 marks)

PAST PAPERS

Q44 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 2HR • Q20**CH 1 • TOPIC 15** **4 marks**

- 20 y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

06 Percentage Change: Use Multipliers

No numbers for k ? You do not need k . Turn the percentage into a multiplier and push it through the power.

WHEN YOU SEE → YOU DO

“ x is increased by 15%”

→ multiply by 1.15

“is doubled / multiplied by 4”

→ put 2 or 4 in for x

“find the percentage increase in y ”

→ $(\text{multiplier} - 1) \times 100$

“ Q is increased by 20%, find the % increase in t ”

→ work backwards: undo the power

$$y \propto x^2, x \times 1.15$$

$$y \times 1.15^2 = 1.3225$$

up 32.25%

$$D \propto n^3, n \times 2$$

$$D \times 2^3 = 8$$

not $\times 6$

$$m \propto \frac{1}{\sqrt{r}}, r \times 4$$

$$m \times \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$m \div 2$

$$V \propto \frac{1}{p}, V \times 1.25$$

$$p \times \frac{1}{1.25} = 0.8$$

down 20%

WORKED MODEL • JUNE 2023 P2H

$Q = 3\sqrt{t}$. Q is increased by 20%. Find the percentage increase in t .

1 $t = \left(\frac{Q}{3}\right)^2$, so $t \propto Q^2$

make t the subject

2 $t \times 1.2^2 = t \times 1.44$

multiplier through the power

ANSWER t increases by 44%

× **TRAP** A 20% increase in Q is **not** a 20% increase in t — the power changes the percentage.

IDEA 06

Practise: Percentage Change: Use Multipliers

My questions first, then selected practice.

Q45 DR ESLAM Multiplier

CH 1 • TOPIC 15 3 marks

y is directly proportional to x^3 . x is decreased by 10%. Find the percentage decrease in y .

SELECTED PRACTICE**Q46** PRACTICE Practice • Very Hard

CH 1 • TOPIC 15 3 marks

- 4 A pendulum of length L cm has time period T seconds.
 T is directly proportional to the square root of L .

The length of the pendulum is increased by 40%.

Work out the percentage increase in the time period.

(3 marks)

Q47**PRACTICE**

Practice · Very Hard

CH 1 • TOPIC 15 **1 mark**

- 5 D is directly proportional to the cube of n .

Mary says that when n is doubled, the value of D is multiplied by 6

Mary is wrong.

Explain why.

(1 mark)

Q48**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15

4 marks**12 (a)** y is directly proportional to x^3

$$y = 17 \text{ when } x = 4$$

Work out an equation connecting y and x .**(3 marks)****(b)** m is inversely proportional to $\sqrt{r}$.The value of r is multiplied by 4.Circle what happens to the value of m .

$\times 2$	$\times 16$	$\div 2$	$\div 16$
------------	-------------	----------	-----------

(1 mark)

Q49**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15 4 marks

14 y is directly proportional to the square of x .

Find the percentage increase in y when x is increased by 15%.

(4 marks)**Q50****PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 15 4 marks

15 At a constant temperature, the volume of a gas V is inversely proportional to its pressure p .
By what percentage will the pressure of a gas change if its volume increases by 25%?

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|----------------------------|--|
| 01 | Words | Direct: find one, then multiply. Inverse: total work first, then divide. |
| 02 | LINK | Layout ($y = kx$ or $\frac{k}{x}$) → Insert (find k) → New equation → Keep solving. |
| 03 | Powers & roots | Power on x <i>before</i> you find k ; undo it at the end. |
| 04 | Graphs & tables | Line through O = direct. Test $\frac{y}{x}$ or xy . Read an exact grid point. |
| 05 | Two links | LINK twice, then substitute; the middle letter disappears. |
| 06 | Percentages | Multiplier through the power: 1.15^2 , 2^3 , $\frac{1}{\sqrt{4}}$. |

BOTTOM LINE Layout → Insert (find k) → New equation → Keep solving.
 “Formula” = stop at N. “Value” = go to K.

Answers

Direct & Inverse Proportion · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • PROPORTION IN WORDS

- | | | |
|-----------|-----------------------|--|
| Q1 | Dr Eslam · Find one | £5.44 |
| Q2 | Dr Eslam · Total work | $6 \times 10 = 60$ worker-days; $60 \div 4 = 15$ days |
| Q3 | Practice · Medium | 22.5 cleaner-hours $\div 3 = 7.5$ h $\rightarrow$ paid 8 h:
£65.60 |
| Q4 | Practice · Medium | (a) 480 person-hours $\div 30 = 16$ days (b) if
some work slower it takes longer |
| Q5 | Practice · Medium | (a) 800 (b) $2.5 - 2 = 0.5$ h = 30 min longer
(c) they cannot keep the same rate all day;
overestimate |
| Q6 | Practice · Medium | (a)(i) 31 s per length; 3100 s = 51 min 40 s
< 55 min (ii) same speed every length (b) he
gets tired |
| Q7 | Practice · Hard | 8.5 hours |
| Q8 | Practice · Very Hard | $3\frac{1}{3}$ more days |

IDEA 02 • THE LINK METHOD: DIRECT AND INVERSE

Q9 Dr Eslam • Direct

$$y = 5x; y = 35$$

Q10 Dr Eslam • Inverse

$$p = \frac{24}{q}; p = 8$$

Q11 Practice • Medium

$$p = \frac{48}{t}; p = 8$$

Q12 Practice • Medium

$$d = \frac{7000}{c}; d = 20$$

Q13 Practice • Medium

$$y = \frac{54}{x}; y = 9$$

Q14 Practice • Medium

$$y = \frac{3.2}{x}; y = 0.1$$

Q15 Practice • Hard

V is inversely proportional to H ; V is directly proportional to $\frac{1}{H}$

Q16 Practice • Hard

y is directly proportional to $\frac{1}{x}$

IDEA 03 • SQUARES, CUBES AND ROOTS

- Q17** Dr Eslam • Square $y = 2x^2; y = 72$
- Q18** Dr Eslam • Inverse square $y = \frac{12}{x^2}; y = \frac{1}{3}$
- Q19** Practice • Medium (a) $T = 0.34r^3$ (b) 73.44
- Q20** Practice • Hard $I = \frac{120}{d^2}; I = 1.2$
- Q21** Practice • Hard (a) $y = \frac{12}{\sqrt{x}}$ (b) 2.4
- Q22** Practice • Very Hard (a) $y = \frac{20x^3}{h^2}$ (b) $x = \frac{3}{2}h$
- Q23** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q14 $F = \frac{104}{v^2}$
- Q24** Edexcel 4MA1 • May 2021 • Paper 2H • Q16 (a) $A = \frac{0.45}{r^2}$ (b) $A^3 = 0.008, A = 0.2$
- Q25** Edexcel 4MA1 • January 2022 • Paper 1H • Q15 $A = \frac{90}{C^2}; C = 0.3$
- Q26** Edexcel 4MA1 • June 2022 • Paper 2H • Q17 $M = 32h^3; h = 2.5$
- Q27** Edexcel 4MA1 • May 2023 • Paper 1HR • Q17 (a) $F = \frac{576}{r^2}$ (b) 0.25
- Q28** Edexcel 4MA1 • June 2024 • Paper 2H • Q17 $Q = 0.25\sqrt{d}$
- Q29** Edexcel 4MA1 • November 2024 • Paper 1H • Q14 $B = \frac{d^2}{576}$
- Q30** Edexcel 4MA1 • June 2025 • Paper 1HR • Q18 (a) $F = \frac{48}{r^3}$ (b) $r = 0.25$
- Q31** Edexcel 4MA1 • June 2025 • Paper 1H • Q18 $y = \frac{108}{x^3}; x = 0.5$
- Q32** Edexcel 4MA1 • November 2025 • Paper 2H • Q17 $T = \frac{90}{\sqrt{m}}$

IDEA 04 • GRAPHS AND TABLES

- Q33** Dr Eslam • Test the table $xy = 24$ every time: inverse, $y = \frac{24}{x}$
- Q34** Practice • Medium $x: 12, 6, 3 \cdot y: 2, 4, 8$
- Q35** Practice • Hard $y = \frac{100}{x^2}: 100, 25, 4, 1$
- Q36** Practice • Hard (a) $y = \frac{36}{x^2}$ (b) $x = 0.5$
- Q37** Practice • Very Hard $y \propto x: D \cdot \frac{1}{x}: A \cdot x^2: B \cdot \frac{1}{x^2}: C$
- Q38** Edexcel 4MA1 • June 2023 • Paper 2H • Q16 (a) $Q = 3\sqrt{t}$ (b) 44%

IDEA 05 • CHAINED PROPORTION (TWO LINKS)

- Q39** Dr Eslam • Two links $y = 3z, z = \frac{6}{x} \Rightarrow y = \frac{18}{x}$
- Q40** Practice • Very Hard $y = \frac{400}{d^2}, d = 6x^2 \Rightarrow y = \frac{100}{9x^4}$
- Q41** Practice • Very Hard $h = \frac{60}{p}, p = 0.5\sqrt{t} \Rightarrow h = \frac{120}{\sqrt{t}}$
- Q42** Practice • Very Hard $P = 5Q^2, Q = \frac{3}{R}; Q = 0.4, R = 7.5$
- Q43** Practice • Very Hard $y = 6x, y = 5z \Rightarrow z = 1.2x$
- Q44** Edexcel 4MA1 • January 2022 • Paper 2HR • Q20 $y = \frac{1000}{T^{3/2}}; T = \frac{100}{9}$

IDEA 06 • PERCENTAGE CHANGE: USE MULTIPLIERS

Q45 Dr Eslam • Multiplier

$$0.9^3 = 0.729: y \text{ decreases by } 27.1\%$$

Q46 Practice • Very Hard

$$\sqrt{1.4} = 1.183: 18.3\%$$

Q47 Practice • Very Hard

$$2^3 = 8: D \text{ is multiplied by } 8, \text{ not } 6$$

Q48 Practice • Very Hard

$$(a) y = \frac{17}{64}x^3 \quad (b) \div 2$$

Q49 Practice • Very Hard

$$1.15^2 = 1.3225: 32.25\%$$

Q50 Practice • Very Hard

$$\frac{1}{1.25} = 0.8: \text{pressure decreases by } 20\%$$